# SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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Abstract. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A finite classification routes every hypothetical cover to one of four mechanisms: trace-length deficit, propagation of boundary-reach lower bounds, area loss, or an equilateral-enclosure obstruction for nine forced points. The paper is organized as a single proof: the structural reduction is followed by one section for each mechanism, and repeated reader/verification layers have been removed. Long branchwise algebra remains in the propagation section, while the two mixed support-arc intersections are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

Contents



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## 1. Introduction and Proof Outline

### 1.1. The theorem and the canonical labeling.

Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \ldots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

Then

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1\left(\bigcup_{i=0}^{5} r_i\right) = 6, \qquad \mathcal{H}^1(S) = 12.$$

**Theorem 1.1** (Main theorem)**.** *The hexagon $H$ cannot be covered by seven open equilateral triangles of side length one.*

**Corollary 1.2** (Soifer's Hexagon Problem)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \ldots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \ldots, U_5$$

such that

$$O \in U_C, \qquad V_i \in U_i.$$

We call $U_C$ the *C triangle* and $U_i$ the *V triangle at $V_i$*. Set

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

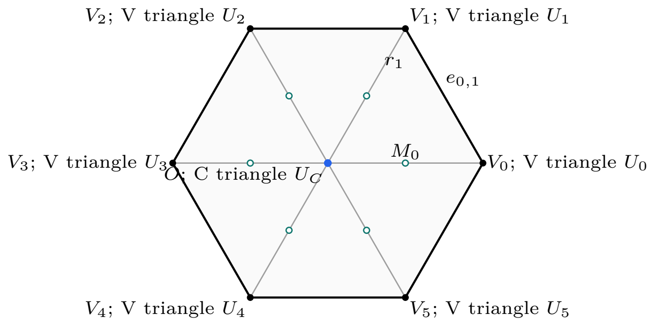
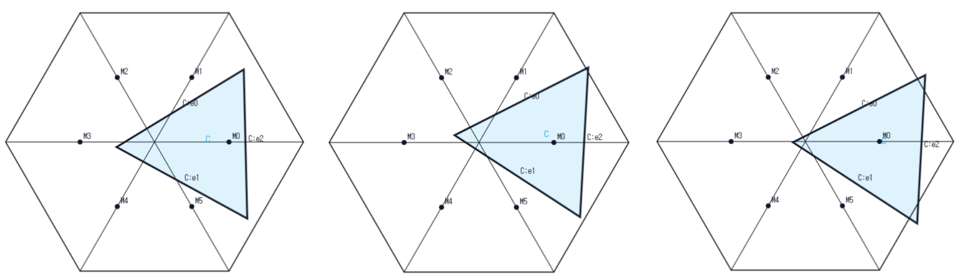



FIGURE 1. The center, vertices, radial segments, and radial midpoints of $H$.

1.2. **Finite classifications.** The center type is the number of boundary edges on which $T_C$ has a positive-length trace. The only possibilities are $0, 1, 2$, denoted by CE0, CE1, CE2.



(a) CE0 (b) CE1 (c) CE2

FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of $\partial H$; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle $T_i$, let $o(T_i)$ be the number of its triangle vertices outside $H$ and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|,$$

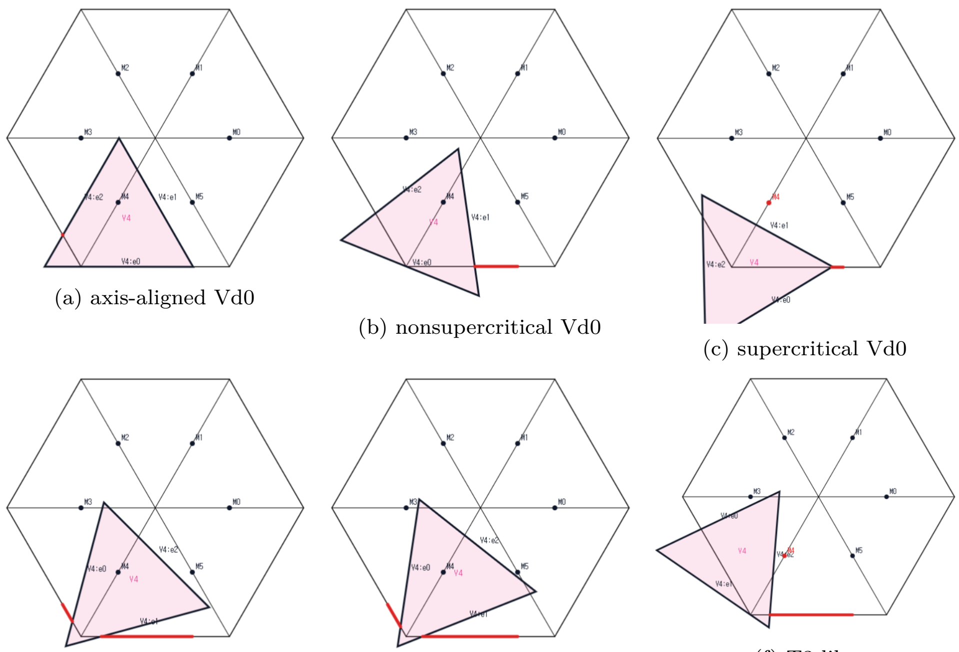
with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of $T_i$ lie outside $H$; it does not say that $T_i$ is disjoint from $H$, since $V_i \in \operatorname{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in $H$, still containing $V_i$ in its interior, and with at most two vertices outside $H$. We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

| type | $(o(T_i), n(T_i))$ |
|---|---|
| Vd0 | $(1, 0), (2, 0)$ |
| Vd1 | $(1, 1)$ |
| Vd2 | $(1, 2)$ |
| T3-like | $(2, 1)$. |

Let $A_i, B_i, C_i$ be the maximal reaches of $T_i$ toward $V_{i-1}, V_{i+1}, O$, respectively. The V triangle $T_i$ is *supercritical* when $A_i + B_i > 1$, and

$$N_+ = |\{i : A_i + B_i > 1\}| .$$

Lowercase $(a_i, b_i, c_i)$ always denotes selected lower bounds for these reaches and never the maxima defining $N_+$.



(a) axis-aligned Vd0

(b) nonsupercritical Vd0

(c) supercritical Vd0

(d) Vd1

(e) Vd2

(f) T3-like

FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \le 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

On $e_{i,i+1}$, parametrized from $V_i$ to $V_{i+1}$, the two incident open traces have the form $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their nonempty complement

$$[B_i, 1 - A_{i+1}]$$

is a *boundary gap*; equality gives a singleton gap, which remains uncovered. Let $N_{\mathrm{gap}}$ be the number of positive center traces that contain a boundary gap.

Let $N_{\mathrm{sp}}$ be the number of V triangles of type Vd1, Vd2, or T3-like. The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\mathrm{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\mathrm{sp}} \ge 3$ is impossible by a trace-length bound on $S$.

1.3. **Exhaustive routing.** Table 1 is read from top to bottom in the CE1/CE2 part. The first applicable row is the unique method for a hypothetical cover. The full classification proof is Proposition 2.14.

| center type and $N_+$ | exhaustive refinement | method |
|---|---|---|
| CE0, $N_+ = 0$ | all vertex types | 1 |
| CE0, $N_+ = 1$ | all six triangles Vd0 | 4 |
| CE0, $N_+ = 1$ | a Vd1 or Vd2 triangle | 1 |
| CE0, $N_+ = 1$ | no Vd1/Vd2 triangle and a T3-like triangle | 3 |
| CE0, $N_+ \geq 2$ | all vertex types | 3 |
| CE1/CE2 | $N_+ + N_{\mathrm{sp}} \geq 3$ | 1 |
| CE1/CE2, $N_+ = 0$ | all six triangles Vd0 | 2 |
| CE1/CE2, $N_+ = 0$ | a Vd1 or Vd2 triangle | 1 |
| CE1/CE2, $N_+ = 0$ | no Vd1/Vd2 triangle and a T3-like triangle | 2 |
| CE1/CE2, $N_+ = 1$ | all Vd0, no boundary gap | 4 |
| CE1/CE2, $N_+ = 1$ | all Vd0, a boundary gap | 2 |
| CE1/CE2, $N_+ = 1$ | exactly one T3-like, no Vd1/Vd2 | 2 |
| CE1/CE2, $N_+ = 1$ | exactly one Vd1/Vd2, no T3-like | 1 / 1+2 |

TABLE 1. The exhaustive routing. Method 1 is a trace-length argument, Method 2 propagates reach bounds, Method 3 sums area loss, and Method 4 uses a nine-point enclosure obstruction. In the last row, the first alternative is for CE1 and the second for CE2.

1.4. **Organization of the proof.** Section 2 proves the structural classification, strict handoff selection, midpoint facts, and the signed center normal form. Section 3 gives the complete trace-length argument. Section 4 develops the local admissible set and all boundary-reach propagation cases. Section 5 proves the local and cyclic area-loss estimates. Section 6 constructs the nine forced points and proves the equilateral-enclosure obstruction. Section 7 then applies Table 1 once and closes the proof.

## 2. Structural Reduction and Common Geometry

This section supplies the complete finite reduction used by every later method. It proves the open–closed scaling equivalence, the center and vertex classifications, strict boundary handoffs, midpoint facts, and exhaustive routing. It then records the signed center normal form used in both the trace and propagation arguments. Each result is stated and proved once.

2.1. **Structural Reduction to Finite Cases.** Throughout, indices are taken in $\mathbb{Z}/6\mathbb{Z}$. Put

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\}.$$

Thus $H$ is the regular hexagon of side length one. We use the notation

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and, when needed,

$$D = \bigcup_{i=0}^{5} r_i, \qquad S = \partial H \cup D, \qquad S_{1/2} = \partial H \cup \{O, M_0, \dots, M_5\}.$$

The arms $r_i$ meet only at $O$, up to sets of one-dimensional measure zero, and consequently

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1(D) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{1}$$

For $L > 0$, $H_L := LH$ denotes the concentric regular hexagon of side length $L$. All symmetries used below belong to the dihedral group $D_6$; applying a symmetry always transports the indices, orientations, and incoming and outgoing labels together.

2.1.1. *Open, closed, and scaled formulations.*

**Proposition 2.1** (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

(1) *The set $H$ is covered by seven open equilateral triangles of side length 1.*
(2) *For some $\varepsilon > 0$, the set $H$ is covered by seven closed equilateral triangles of side length $1-\varepsilon$.*
(3) *For some $L > 1$, the set $H_L$ is covered by seven closed equilateral triangles of side length 1.*

*Proof.* Suppose that open unit triangles $U_1, \dots, U_7$ cover $H$. Define

$$m_j(p) = \operatorname{dist}(p, \mathbb{R}^2 \setminus U_j), \qquad m(p) = \max_{1\le j\le 7} m_j(p).$$

The continuous function $m$ is positive on the compact set $H$, so $\eta := \min_{p\in H} m(p) > 0$. Choose

$$0 < \rho < \min\left\{\eta, \frac{\sqrt{3}}{6}\right\}.$$

Moving each side of $U_j$ inward through distance $\rho$ gives a closed equilateral triangle $K_j$ of side $1-2\sqrt{3}\rho$. Every $p \in H$ has $m_j(p) \ge \eta > \rho$ for some $j$, and hence belongs to $K_j$. This proves (1)⇒(2), with $\varepsilon = 2\sqrt{3}\rho$.

Conversely, enlarging a closed triangle of side $1-\varepsilon$ about its incenter to an open triangle of side 1 preserves containment, proving (2)⇒(1). Finally, dilation by $(1-\varepsilon)^{-1}$ identifies (2) with (3), with

$$L = (1-\varepsilon)^{-1} > 1.$$

□

We conduct the geometric argument in the open side-one model. Denote the original open triangles by $U_C$ and $U_i$ for $0 \le i \le 5$, and put

$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

The distinction between $U_i$ and $T_i$ will be maintained: closure may add endpoints or tangencies, although it does not change area or one-dimensional trace measure.

**Lemma 2.2** (Distinct triangles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \le i \le 5),$$

*and these seven triangles are distinct. Consequently*

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

*Proof.* The seven distinguished points $O, V_0, \dots, V_5$ are pairwise at distance at least 1. Any two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Thus one open triangle contains at most one distinguished point. Selecting a covering triangle for each of the seven points gives an injection from seven points to seven triangles, hence a bijection and the asserted labelling. □

2.1.2. *The center classification.* For a closed C triangle define

$$N_E(T_C) = |\{i : T_C \cap e_{i,i+1} \text{ contains a positive-length interval}\}|\,.$$

Point contacts, endpoint contacts, and tangencies do not contribute. We say that $T_C$ has type CE0, CE1, or CE2 according as $N_E(T_C)$ is 0, 1, or 2.

**Proposition 2.3** (Exhaustive center types)**.** *Every C triangle has exactly one of the types CE0, CE1, and CE2. Equivalently,*

$$N_E(T_C) \le 2.$$

*Proof.* Among any three edges of the six-cycle $\partial H$, two have no common endpoint. It is therefore enough to show that relative-interior points of two nonadjacent edges are more than unit distance apart. Up to symmetry there are two cases. For $0 < t, s < 1$, put

$$x = (1-t)V_0 + tV_1.$$

If $y = (1-s)V_2 + sV_3$, direct calculation gives

$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$

If instead $y = (1-s)V_3 + sV_4$ and $u = t+s$, then

$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$

A unit equilateral triangle has diameter 1. It therefore cannot contain relative-interior points on two nonadjacent hexagon edges. Three positive-length edge traces would supply such a pair, a contradiction. □

For the CE1/CE2 types, Proposition 2.12 below records the additional fact that $T_C$ contains exactly one radial midpoint.

2.1.3. *Vertex types and exact-trace normalizations.* For the closure $T_i$ of an original open V triangle $U_i$, let

$$o(T_i) = |\{\text{vertices of } T_i \text{ outside } H\}|$$

and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|.$$

**Lemma 2.4** (Local wedge)**.** *Every original V triangle has $1 \le o \le 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one of its vertices belongs to $H$; if both $\mathcal{H}^1(T_i \cap r_{i-1}) > 0$ and $\mathcal{H}^1(T_i \cap r_{i+1}) > 0$, at least two of its vertices belong to $H$.*

*Proof.* Normalize to $i = 0$ and write

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0).$$

Then $V_0 = (0,0)$,

$$\|(x,y)\|^2 = x^2 + y^2 - xy,$$

and, within the closed unit ball about $V_0$, membership in $H$ is equivalent to membership in the wedge

$$W = \{(x,y) : x \ge 0, y \ge 0\}.$$

Indeed,

$$x^2 + y^2 - xy = \left(y - \frac{x}{2}\right)^2 + \frac{3x^2}{4} = (x-y)^2 + xy$$

shows that $x, y \geq 0$ and norm at most one imply the remaining hexagon inequalities. The two relevant radial segments are

$$r_1 = \{(1,y) : 0 \leq y \leq 1\}, \qquad r_5 = \{(x,1) : 0 \leq x \leq 1\}.$$

Suppose $T_0 \cap r_1$ has positive length, and let $m$ be the largest $x$-coordinate of a vertex of $T_0$. If $m > 1$, a maximizing vertex $(m, y)$ lies in the unit ball and hence satisfies $0 < y < 1$; it belongs to $H$. If $m = 1$, positive-length contact with the supporting line $x = 1$ forces a side of $T_0$ to lie there, and both endpoints satisfy $1 + y^2 - y \leq 1$, hence belong to $H$. The same argument applies to $r_5$. If both $r_1$ and $r_5$ are met but only one triangle vertex belonged to $H$, that vertex would have to have both $x > 1$ and $y > 1$. This is impossible because $(x-y)^2 + xy > 1$.

Finally, if all three triangle vertices belonged to the convex set $H$, then $T_0 \subset H$. This is incompatible with the extreme point $V_0$ lying in the interior of $T_0$. Hence $o \geq 1$. $\square$

We use the following orientation normal forms for both translations below. Set

$$D_t = 1 - t + t^2, \qquad z = \sqrt{D_t}, \qquad 0 \leq t \leq 1.$$

After relabelling the vertices of the equilateral triangle, every orientation belongs to one of the two families

| | first side vector | second side vector |
|---|---|---|
| I | $z^{-1}(1,t)$ | $z^{-1}(1-t,1)$ |
| II | $z^{-1}(t,-(1-t))$ | $z^{-1}(1,t)$. |

In Type I write

$$U = \alpha + y - tx, \qquad V = \beta + x - (1-t)y, \qquad T = \{U \geq 0, V \geq 0, U + V \leq z\}, \tag{2}$$

and in Type II write

$$U = \alpha + (1-t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \geq 0, V \geq 0, U + V \leq z\}. \tag{3}$$

In either form, $V_0 \in \operatorname{int}(T)$ is equivalent to

$$\alpha > 0, \qquad \beta > 0, \qquad \alpha + \beta < z. \tag{4}$$

For later use, the Type-II vertex in the wedge is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z-\alpha) + (1-t)\beta}{D_t}\right). \tag{5}$$

Both coordinates are positive under (4).

**Proposition 2.5** (Exact-trace normalization of raw $(3,0)$ roles)**.** *Let $U$ be an original open $V_i$-triangle, put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3,0)$. There is a translate $\widehat{U}$ of $U$, with closure $\widehat{T}$, the same orientation, and the same side length, such that*

$$V_i \in \widehat{U}, \qquad (o(\widehat{T}), n(\widehat{T})) = (2,0), \tag{6}$$

*and both the closed and open traces on the hexagon are unchanged:*

$$T \cap H = \widehat{T} \cap H, \qquad U \cap H = \widehat{U} \cap H. \tag{7}$$

*Consequently the three maximal reaches on the two incident edges and the own radial segment are unchanged. In the notation introduced below, the values $A_i, B_i, C_i$, the inequality $A_i + B_i > 1$, and hence $N_+$ are all preserved.*

*Proof.* Normalize to $i = 0$ and use the common forms above. A Type-II triangle cannot have $o = 3$: the point $Q$ in (5) lies in $W$, and because $T$ contains the origin and has diameter one, the local wedge lemma puts $Q$ in $H$. Thus $T$ has Type I form.

The three Type-I vertices are

$$\begin{aligned} Q_0 &= \left(-\frac{(1-t)\alpha+\beta}{D_t}, -\frac{\alpha+t\beta}{D_t}\right), \\ Q_1 &= \left(\frac{z-(1-t)\alpha-\beta}{D_t}, \frac{tz-\alpha-t\beta}{D_t}\right), \\ Q_2 &= \left(\frac{(1-t)z-(1-t)\alpha-\beta}{D_t}, \frac{z-\alpha-t\beta}{D_t}\right). \end{aligned} \tag{8}$$

The vertex $Q_0$ has two negative coordinates, the first coordinate of $Q_1$ is positive, and the second coordinate of $Q_2$ is positive. Hence $o = 3$ is equivalent to

$$\alpha + t\beta > tz, \qquad (1-t)\alpha + \beta > (1-t)z. \tag{9}$$

These strict inequalities force $0 < t < 1$.

Put

$$s = \alpha + \beta, \qquad L = z - s > 0.$$

The two inequalities in (9) become

$$\alpha > \frac{tL}{1-t}, \qquad \beta > \frac{(1-t)L}{t}. \tag{10}$$

For every $(x, y) \in W$ satisfying

$$(1-t)x + ty \le L, \tag{11}$$

the minimum of $U$ is attained at $(L/(1-t), 0)$ and the minimum of $V$ at $(0, L/t)$. Thus (10) makes both $U$ and $V$ strictly positive throughout the triangle in (11). It follows that

$$T \cap W = \{(x, y) \in W : (1-t)x + ty \le L\}. \tag{12}$$

Keep $t$ and $s$ fixed and replace the affine constants by

$$\widehat{\alpha} = \frac{tL}{1-t}, \qquad \widehat{\beta} = s - \widehat{\alpha}. \tag{13}$$

The first inequality in (10) gives $0 < \widehat{\alpha} < \alpha$ and $\widehat{\beta} > \beta > 0$; also $\widehat{\alpha} + \widehat{\beta} = s < z$. Since the linear map defining $U, V$ in (2) is nonsingular, changing only these affine constants gives a translate $\widehat{T}$ of $T$. In particular, (4) shows that the origin remains in $\mathrm{int}(\widehat{T})$.

For the translated triangle, the inequality $\widehat{U} + \widehat{V} \le z$ is still exactly (11). On that triangle $\widehat{U} \ge 0$, with equality only at $(L/(1-t), 0)$, while $\widehat{V} > 0$ by the second inequality in (10) and $\widehat{\beta} > \beta$. Therefore

$$\widehat{T} \cap W = T \cap W,$$

and their interiors have the same trace on $W$: the only new equality point of $\widehat{U}$ already lies on the common boundary $(1-t)x+ty=L$. Both triangles contain the origin and have diameter one, so the local wedge lemma identifies their intersections with $H$ with their intersections with $W$. This proves both equalities in (7).

Equation (13) makes the second coordinate of $Q_1$ in (8) zero. Thus $Q_1 \in H$. The vertex $Q_0$ remains outside $H$, and the strict second inequality in (10), strengthened by $\widehat{\beta} > \beta$, keeps $Q_2$ outside $H$. Hence $o(\widehat{T}) = 2$. The local wedge lemma gives $n(T) = 0$ from $o(T) = 3$, and the exact trace equality preserves $n$, so $n(\widehat{T}) = 0$. Finally, all three reach segments lie in $H$, and (7) preserves their exact open and closed traces. This proves every assertion. $\square$

Apply Proposition 2.5 simultaneously to every raw $(3,0)$ vertex role and then reuse the symbols $U_i, T_i$ for the translated roles. The rolewise equalities (7) preserve the entire open cover, every closed trace, the condition $V_i \in U_i$, all actual reaches, and $N_+$. This simultaneous exact-trace normalization is the convention throughout the rest of the paper.

For the normalized roles, we use the names

| type | $(o,n)$ |
|---|---|
| Vd0 | $(1,0),(2,0)$ |
| Vd1 | $(1,1)$ |
| Vd2 | $(1,2)$ |
| T3-like | $(2,1)$. |

**Proposition 2.6** (Exhaustive normalized vertex types)**.** *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, and T3-like.*

*Proof.* We have $n \in \{0,1,2\}$. Lemma 2.4 gives $o \leq 2$ when $n \geq 1$, and $o \leq 1$ when $n = 2$. Together with $o \geq 1$, the possibilities for a raw role are

| $n$ | possible $o$ |
|---|---|
| 0 | $1,2,3$ |
| 1 | $1,2$ |
| 2 | $1.$ |

The exact-trace normalization replaces the sole extra possibility $(3,0)$ by $(2,0)$ and leaves all other pairs unchanged. The remaining pairs are precisely the four named classes. $\square$

The next normalization is useful for trace estimates, but its final warning is essential.

**Proposition 2.7** (T3-like closed-trace domination)**.** *Let $T$ be the closure of an original T3-like $V_i$-triangle. There is a translate $\widehat{T}$ of $T$, with the same orientation and side length, such that*

$$V_i \in \operatorname{relint}(\textit{a side of } \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

*and $\widehat{T}$ remains of type $(o,n) = (2,1)$. Every old open interior point in $H \setminus \{V_i\}$ remains an interior point of $\widehat{T}$. The point $V_i$ itself becomes a boundary point, so $\widehat{T}$ is a closed-trace majorant and is not a replacement open triangle.*

*Proof.* Use the wedge coordinates of Lemma 2.4 and the common orientation normal forms above. In Type I, direct inversion shows that if exactly two triangle vertices are outside $W$, the sole wedge vertex has both coordinates strictly below 1: in the two possible cases its coordinates are bounded respectively by $(1-t)/z$ and $t/z$, with the endpoint strictness supplied by $\alpha > 0$ or $\beta > 0$. The support argument in Lemma 2.4 then excludes positive-length contact with either $x = 1$ or $y = 1$. Thus Type I cannot be T3-like.

The triangle therefore has Type II form. Reflecting in $x = y$ if necessary, assume $\mathcal{H}^1(T_0 \cap r_1) > 0$. The two outside vertices have respectively a negative $x$- and a negative $y$-coordinate, while the unique wedge vertex is the point $Q$ in (5). The endpoint cases $t = 0, 1$ give no positive adjacent interval, so $0 < t < 1$. The exact axis reaches are

$$A = \beta, \qquad B = z - \alpha - \beta,$$

and T3-like support is equivalent to

$$Q_x > 1, \qquad Q_y \leq 1.$$

Define $\widehat{T}$ by replacing $\alpha$ with 0 and keeping $t, \beta$ fixed. Since the linear map $(x, y) \mapsto (U, V)$ is nonsingular, this changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so $V_0$ is in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \geq 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old $x$-axis reach is $B$ and the new one is $B + \alpha$; because $B < 1$ and $\alpha > 0$, the inclusion is strict inside $H$.

The two outside vertices remain outside, whereas $V_0$ and the new wedge vertex lie in $H$, so $o = 2$. Moreover $\widehat{Q}_x > Q_x > 1$. From $Q_x > 1$ one gets $t\beta < z - \alpha - D_t$, and hence

$$tD_t(1 - \widehat{Q}_y) > D_t(1 - z) + (1-t)\alpha > 0.$$

Thus $\widehat{Q}_y < 1$ and $n = 1$. Finally, for every nonzero point of $W$, $\widehat{U} > 0$; the displayed slack relations show that every old interior point other than the origin remains interior. $\square$

2.1.4. *Reach coordinates, supercritical V triangles, and gaps.* For the V triangle $U_i$ under the simultaneous normalization convention, define its maximal incoming, outgoing, and radial reaches by

$$A(U_i) = \sup_{\substack{t \in [0,1] \\ V_i + t(V_{i-1} - V_i) \in U_i}} t,$$

$$B(U_i) = \sup_{\substack{t \in [0,1] \\ V_i + t(V_{i+1} - V_i) \in U_i}} t,$$

$$C(U_i) = \sup_{\substack{t \in [0,1] \\ V_i + t(O - V_i) \in U_i}} t.$$

Closure does not change the suprema, so

$$A(U_i) = A(T_i), \qquad B(U_i) = B(T_i), \qquad C(U_i) = C(T_i).$$

For compact indexed formulas, put

$$A_i := A(U_i) = A(T_i), \qquad B_i := B(U_i) = B(T_i), \qquad C_i := C(U_i) = C(T_i).$$

The closure $T_i$ contains the corresponding endpoints. We reserve lowercase $(a_i, b_i, c_i)$ for selected or prescribed reach lower bounds whenever actual and selected values occur together. A realizing triangle then satisfies

$$A_i \ge a_i, \qquad B_i \ge b_i, \qquad C_i \ge c_i.$$

An actual V triangle is supercritical if $A_i + B_i > 1$, and

$$N_+ = \left|\{i : A_i + B_i > 1\}\right|. \tag{14}$$

This definition never counts a smaller reach pair in place of the actual V triangle.

Parametrize $e_{i,i+1}$ by

$$X_i(x) = V_i + x(V_{i+1} - V_i), \qquad 0 \le x \le 1.$$

The two incident open traces are

$$U_i \cap e_{i,i+1} = [0, B_i), \qquad U_{i+1} \cap e_{i,i+1} = (1 - A_{i+1}, 1]$$

in this coordinate. A *boundary gap* on this edge is the nonempty closed complement between these two traces,

$$[B_i, 1 - A_{i+1}] \qquad \text{when } B_i \le 1 - A_{i+1}.$$

In particular, equality gives a singleton gap. This convention retains the actual open traces; a point missing from both open triangles is not erased merely because both closures contain it.

**Lemma 2.8** (Exhaustiveness of the gap split)**.** *On each edge the complement of the two incident V triangle traces is either empty or one closed interval; a singleton interval counts as a gap. Every edge having a gap has positive-length intersection with the open center triangle. Consequently a* CE0 *center permits no gaps, a* CE1 *center permits at most one, and a* CE2 *center permits at most two.*

*Proof.* The two incident traces are intervals issuing from the two endpoints, so their complement is exactly the interval displayed above when nonempty. A nonincident V triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from the triangle's distinguished vertex. Thus every point of a gap lies in $U_C$. Because $U_C$ is open, its intersection with that edge contains a relative neighborhood and hence has positive length. Different gap edges therefore give different edges counted by $N_E(T_C)$. Proposition 2.3 gives the three stated bounds, including singleton gaps. ☐

**Lemma 2.9** (T3-like V triangles are nonsupercritical)**.** *Every original T3-like V triangle satisfies*

$$A_i + B_i \le 1. \tag{15}$$

*In particular, it is not counted by $N_+$.*

*Proof.* The proof of Proposition 2.7 showed that a T3-like triangle has Type II orientation with

$$A_i = \beta, \qquad B_i = z - \alpha - \beta, \qquad z = \sqrt{1 - t + t^2} \le 1.$$

Thus $A_i + B_i = z - \alpha \le 1$. The same conclusion holds after reflection. ☐

We state the precise passage from maximal reaches to strict handoff lower bounds. Its hypothesis that the six V triangles themselves cover the boundary is part of the statement.

**Proposition 2.10** (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover $\partial H$. On each edge choose*

$$\xi_i \in (1 - A_{i+1}, B_i)$$

*and define*

$$a_i = 1 - \xi_{i-1}, \qquad b_i = \xi_i.$$

*Then $0 < a_i < A_i$, $0 < b_i < B_i$, the two selected anchors lie in $U_i$, and*

$$a_i^2 + a_i b_i + b_i^2 < 1.$$

*Moreover:*

(1) *if exactly one actual V triangle, indexed by $p$, is supercritical, every strict selection has exactly one selected supercritical V triangle, also indexed by $p$;*
(2) *if at least two actual V triangles are supercritical, some simultaneous strict selection has at least two selected supercritical V triangles.*

*Proof.* Because $V_i$ is interior to a diameter-one closed triangle,

$$0 < A_i < 1, \qquad 0 < B_i < 1.$$

No nonincident V triangle contains a relative-interior point of $e_{i,i+1}$, since such a point is more than unit distance from its distinguished vertex. The two displayed incident traces therefore cover the edge and, being relatively open and nonempty, overlap. Hence each interval $(1 - A_{i+1}, B_i)$ is nonempty. The anchor and distance assertions follow from the choice and from the fact that two points of $U_i$ are at distance strictly less than one.

Put $\ell_i = 1 - A_{i+1}$ and $u_i = B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i. \tag{16}$$

If actual V triangle $i$ is nonsupercritical, then

$$\xi_i < u_i \le \ell_{i-1} < \xi_{i-1},$$

so every selected version of it is strictly subcritical. Also

$$\sum_{i=0}^{5} (a_i + b_i) = \sum_{i=0}^{5} (1 - \xi_{i-1} + \xi_i) = 6. \tag{17}$$

If $p$ is the sole actual supercritical index, the other five selected sums are strictly below one; equation (17) forces the $p$th sum above one. This proves (1).

For (2), first take two nonadjacent supercritical indices $p, q$. For each $r \in \{p, q\}$ choose

$$\ell_{r-1} < z_r < u_r,$$

then choose

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \quad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The two pairs of variables are disjoint and give ascents at $p, q$. Any set of at least three indices on a six-cycle contains two nonadjacent indices. It remains to consider exactly two adjacent supercritical indices $p, p+1$. For the other four nonsupercritical V triangles, successive inequalities $\ell_i < u_i \le \ell_{i-1}$ give

$$\ell_{p-1} < \ell_{p+1}.$$

Together with supercriticality at $p, p+1$, this implies

$$\max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose $\xi_p$ between these bounds and then choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their respective overlap intervals. By (16), both $T_p$ and $T_{p+1}$ are selected supercritical. $\square$

2.1.5. *Midpoint forcing.* We first need a local fact for V triangles. Take unit vectors

$$u = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \qquad v = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

**Lemma 2.11** (Self-midpoint obstruction)**.** *Let a closed unit triangle contain*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2}.$$

*Then $a+b \le 1$. Consequently, for every V triangle,*

$$M_i \in T_i \implies A_i + B_i \le 1.$$

*If the two demanded boundary points and the midpoint are contained with a positive margin, the conclusion is strict.*

*Proof.* Suppose $a+b>1$ and set

$$K = \operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}.$$

For outward unit normals $n_0, n_1, n_2$ separated by $120°$, the least equilateral triangle with those normals and containing $K$ has side length

$$\frac{2}{\sqrt{3}} \sum_{j=0}^{2} h_K(n_j).$$

The finite-caliper theorem, Theorem 6.1, reduces its minimum to an outward normal of one of the four edges of $K$.

For the edge from 0 to $au$, direct projection gives

$$S_{0A} = \max\left\{\frac{\sqrt{3}}{4}, \frac{\sqrt{3}a}{2}\right\} + \frac{\sqrt{3}b}{2} > \frac{\sqrt{3}}{2};$$

if $a \ge 1/2$ this is $(\sqrt{3}/2)(a+b)$, and if $a < 1/2$ then $b > 1/2$. The edge through $bv$ is symmetric. For the edge from $au$ to $(u+v)/2$, put $D_a = \sqrt{4a^2 - 2a + 1}$. Projection gives

$$S_{AM} = \begin{cases} \dfrac{\sqrt{3}(a+b)}{2D_a}, & a \le \frac{1}{2}, \\ \dfrac{\sqrt{3}(2a^2+b)}{2D_a}, & a > \frac{1}{2}. \end{cases}$$

The first value is strictly above $\sqrt{3}/2$ because $D_a \le 1$. In the second case, $b > 1-a$ and

$$(2a^2 - a + 1)^2 - D_a^2 = a^2(2a-1)^2 \ge 0,$$

so again $S_{AM} > \sqrt{3}/2$. The fourth edge is symmetric. Every enclosing equilateral triangle therefore has side greater than one, a contradiction.

If equality $a+b=1$ occurred while all relevant points had positive margin, both boundary reach lower bounds could be increased slightly; the closed conclusion would then be violated. $\square$

The corresponding center fact is exact and uses the interior condition at $O$.

**Proposition 2.12** (CE1/CE2 exactly-one-midpoint property)**.** *Let $T_C$ be a closed unit equilateral triangle with $O \in \mathrm{int}(T_C)$. If $T_C$ is CE1 or CE2, equivalently if it has positive-length intersection with a boundary edge of $H$, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

*Proof.* Normalize the positive trace by a dihedral symmetry. Proposition 2.15 gives a single signed side-slack model for both CE1 and CE2 and proves directly that its midpoint set is $\{M_0\}$. Undoing the normalization proves the assertion. □

*Remark* 2.13. The hypothesis $O \in \mathrm{int}(T_C)$ cannot be weakened to $O \in T_C$: the triangle $\mathrm{conv}\{O, V_0, V_1\}$ has a boundary-edge overlap and contains both $M_0$ and $M_1$.

**Proposition 2.14** (Exhaustive structural reduction)**.** *Every hypothetical cover by seven open unit equilateral triangles admits the distinct center and V triangles used in Table 1, and its maximal reaches determine exactly one row of that table.*

*Proof.* Lemma 2.2 gives the seven triangles. Propositions 2.3 and 2.6 give the exhaustive and mutually exclusive center and vertex types. The maximal reaches determine exactly one of $N_+ = 0$, $N_+ = 1$, and $N_+ \geq 2$. For CE0 these data and the vertex classification give the five displayed rows.

Now suppose the center is CE1 or CE2. It contains exactly one radial midpoint by Proposition 2.12. The direct skeleton theorem, Theorem 3.14, removes every state with $N_+ + N_{\mathrm{sp}} \geq 3$. If $N_+ \geq 2$, Lemma 3.5 supplies a distinct positive-support V triangle, so $N_{\mathrm{sp}} \geq 1$ and that theorem applies. Hence every surviving branch has $N_+ \leq 1$ and $N_+ + N_{\mathrm{sp}} \leq 2$. If $N_+ = 0$, the vertex classification gives, in order, all Vd0, at least one Vd1/Vd2, or no such triangle and at least one T3-like. If $N_+ = 1$, then $N_{\mathrm{sp}} \leq 1$: the triangles are all Vd0, exactly one T3-like, or exactly one Vd1/Vd2. In the all-Vd0 case, Lemma 2.8 supplies the exhaustive zero-gap/nonempty-gap split. These alternatives are disjoint, proving the claim. □

2.2. **Signed CE1/CE2 Reductions.** This section gives the signed center model used by both later methods. The propagation interface depending on that model is deferred until after the transfer calculus in Section 4.1.

2.2.1. *One signed center normal form.*

**Proposition 2.15** (Signed CE1/CE2 center normal form)**.** *Let $T_C$ be a closed unit equilateral triangle with $O \in \mathrm{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

*There are parameters*

$$0 < R < 1, \qquad W = 1 - R,$$
$$E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E),$$

*and nonnegative center slacks $\alpha, \delta$ such that, with*

$$k = \eta + \alpha + \delta,$$

*the C triangle is*

$$\boxed{\begin{aligned}F_0 &= R+\alpha-a+Wb,\\ F_1 &= Rb+Wa-k,\\ F_2 &= W+\delta-b+Ra,\end{aligned}} \qquad T_C = \{F_0, F_1, F_2 \geq 0\}. \tag{18}$$

*Define*

$$\Delta_R = P-\alpha-W\delta, \qquad \Delta_L = P-R\alpha-\delta. \tag{19}$$

*Then*

$$T_C \cap e_{0,1} = \left[\frac{k}{R}, W+\delta\right], \qquad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R}, \tag{20}$$

*and the possible companion trace is*

$$T_C \cap e_{5,0} = \left[\frac{k}{W}, R+\alpha\right] \quad \textit{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{21}$$

*The normalization requires $\Delta_R > 0$, and*

$$T_C \textit{ is CE1} \iff \Delta_L \leq 0, \qquad T_C \textit{ is CE2} \iff \Delta_L > 0. \tag{22}$$

*The six center exits, measured from $O$, are*

$$\boxed{\begin{aligned}&d_0^C = E-\alpha-\delta, && d_1^C = \frac{\delta}{R}, && d_2^C = \delta,\\ &d_3^C = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\}, && d_4^C = \alpha, && d_5^C = \frac{\alpha}{W}.\end{aligned}} \tag{23}$$

*Moreover,*

$$T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}. \tag{24}$$

*For closures of original open C triangles one has $\alpha, \delta > 0$.*

*Proof.* Write

$$T_C \cap e_{0,1} = [s,t], \qquad 0 \leq s < t \leq 1.$$

The two sides through these endpoints have a positive slope ratio. Reflect across the perpendicular bisector of $e_{0,1}$ if necessary and call the ratio $R \leq 1$; put $W = 1-R$. Direct equations for the three side lines give

$$F_1 = Rb+Wa-Rs, \qquad F_2 = -b+Ra+t, \qquad F_0 = Wb-a+E+Rs-t,$$

where $E = \sqrt{1-RW}$ is forced by the unit side-length relation. If $R = 1$, then $F_0(O) = s-t < 0$, contrary to $O \in \mathrm{int}(T_C)$; hence $0 < R < 1$.

Set

$$\alpha = F_0(O), \qquad \delta = F_2(O).$$

Then

$$t = W+\delta, \qquad Rs = \eta+\alpha+\delta = k,$$

and substitution gives (18). The gradients are the three unit-equilateral side-normal directions and

$$F_0+F_1+F_2 = E,$$

so the three inequalities define the same unit equilateral triangle.

On $e_{0,1}$ one has $a = 0$. The active inequalities are

$$Rb \geq k, \qquad b \leq W+\delta,$$

which gives (20); its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$ one has $b = 0$, and the active inequalities are

$$Wa \geq k, \qquad a \leq R + \alpha.$$

Their signed interval length is

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W},$$

which proves (21).

The positive right trace gives

$$\delta < \frac{P}{W} = \frac{ER}{1+E} < \frac{R}{2}.$$

On $e_{1,2}$, parameterized by $b = 1 + q$, $a = q$, the slack $F_2$ is

$$\delta - R - Wq < 0.$$

Thus a second positive trace cannot occur on $e_{1,2}$. The CE classification says that a C triangle has at most two positive boundary traces and that two such traces are adjacent. Hence $e_{5,0}$ is the only possible companion edge, and its signed length proves (22). Equality $\Delta_L = 0$ is only a point contact and is therefore CE1.

For the radial exits, substitute the six ray parametrizations in (18). On $r_1$ the decreasing slack is $\delta - Rq$; on $r_2$ it is $\delta - q$; on $r_3$ the two decreasing slacks are $\alpha - Rq$ and $\delta - Wq$. Reflection gives $r_4, r_5$, and on $r_0$ the decreasing slack is $E - \alpha - \delta - q$. This proves (23).

Finally,

$$W(\alpha + \delta) \leq \alpha + W\delta < P,$$

so

$$E - \alpha - \delta > E - \frac{P}{W} = \frac{E(E-R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $2E(E-R) - W = (E-R)^2 > 0$. Together with

$$\alpha < P < \min\left\{\frac{R}{2}, \frac{W}{2}\right\}, \qquad \delta < \frac{R}{2},$$

the exits in (23) satisfy

$$d_0^C > \frac{1}{2}, \qquad d_i^C < \frac{1}{2} \quad (i = 1, \dots, 5).$$

Thus the exact midpoint set is $\{M_0\}$. For the closure of an original open C triangle, all three center slacks are positive; hence $\alpha, \delta > 0$. □

*Remark* 2.16 (Affine-chart variables). If $\kappa_j = F_j(O)$ denotes the auxiliary side slack in the CE1 affine chart, take

$$\lambda = R, \qquad s = \frac{k}{R}, \qquad t = W + \delta, \qquad \kappa_0 = \alpha, \qquad \kappa_2 = \delta.$$

For CE2, take

$$x = \frac{k}{W}, \qquad \nu = R + \alpha, \qquad y = \frac{k}{R}, \qquad v = W + \delta.$$

If $S = x + y$ and $D = \sqrt{x^2 + xy + y^2}$, then

$$S = \frac{k}{RW}, \qquad D = \frac{Ek}{RW},$$

and the old coupling equation is

$$(\nu + v)S - xy = D.$$

Thus the CE1 and CE2 formula lists are two coordinate readings of Proposition 2.15.

2.2.2. *Boundary contribution and diameter transfer.*

**Corollary 2.17** (Common center-boundary formula)**.** *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{25}$$

*In CE1,*

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}.$$

*In CE2,*

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

*Proof.* Equation (25) is the sum of (20) and (21). If $\Delta_L \le 0$, then $R\alpha + \delta \ge P$. Multiplying by $W$ and using $\alpha \ge RW\alpha$ gives

$$\alpha + W\delta \ge WP.$$

Thus $\Delta_R \le RP$, so the contribution is at most $P$. Since $E \ge \sqrt{3}/2$ and $E(1-E)$ decreases on this interval, the CE1 cap follows.

If $\Delta_L > 0$, direct addition gives

$$\begin{aligned} L_{\partial H}(T_C) &= \frac{P - \alpha - W\delta}{R} + \frac{P - R\alpha - \delta}{W} \\ &= \frac{P - E^2(\alpha + \delta)}{RW}. \end{aligned}$$

Here $W + R^2 = W^2 + R = E^2$ and $P/(RW) = E/(1+E)$, so the last expression is strictly below $1/2$. ☐

**Lemma 2.18** (Adjacent-edge diameter transfer)**.** *For $0 \le q \le 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

*Here the argument-free symbol $M_0$ continues to denote the radial midpoint; $M_0(q)$ denotes the envelope map. If a unit equilateral triangle reaches parameters $q, b$ on two adjacent boundary edges, then*

$$b \le M_0(q).$$

*The map $M_0$ is strictly decreasing. For $q > 0$,*

$$M_0(q) < 1 - \frac{q}{2}, \qquad 1 - M_0(q) > \frac{q}{2}.$$

*Proof.* The squared distance between the two boundary points is $q^2 + qb + b^2$, so it is at most one. Solving the quadratic in $b$ gives the first assertion. Differentiating the displayed formula gives $M_0'(q) = (-1 - 3q/\sqrt{4-3q^2})/2 < 0$. For $q > 0$, $\sqrt{4-3q^2} < 2$, which gives both strict linear comparisons. □

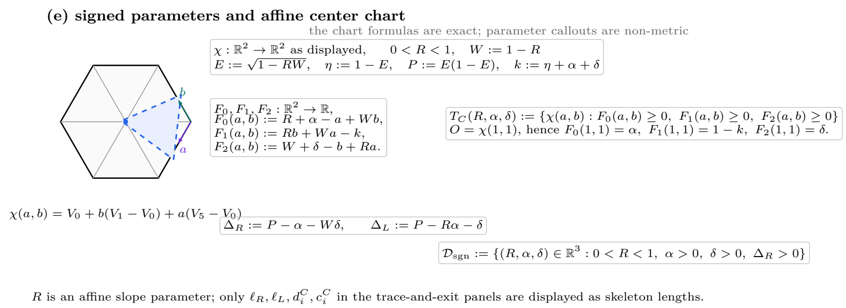



FIGURE 4. The affine center chart and the exact dependency structure of the signed parameters.

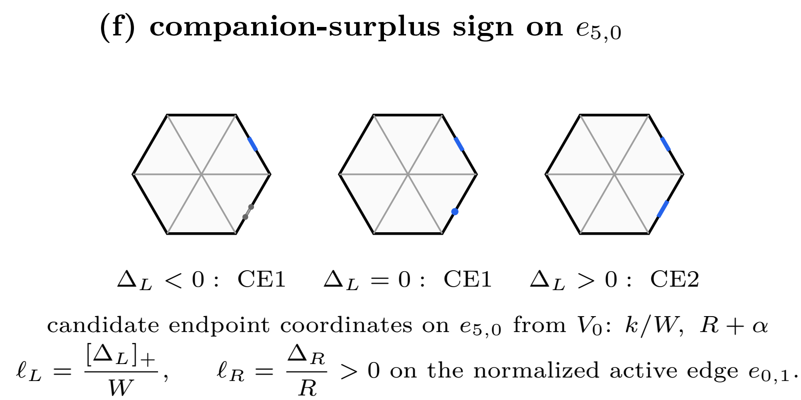



FIGURE 5. The exact sign test for the candidate companion trace, including the CE1 point-contact case.

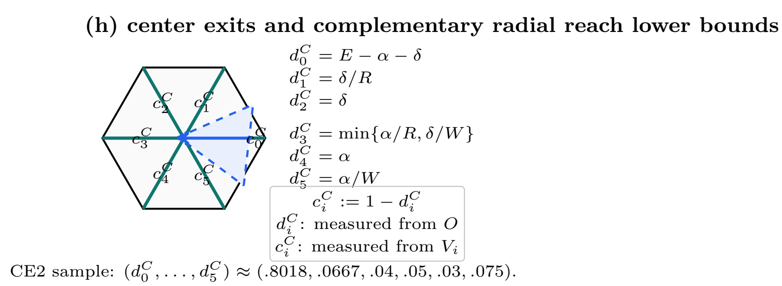
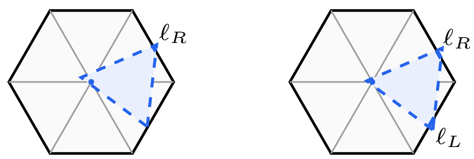



FIGURE 6. The six exact center exits $d_i^C$ and complementary radial lower bounds $c_i^C = 1 - d_i^C$.

**(i) boundary-contribution bounds on the skeleton**




$$\text{CE1}: \quad L_{\partial H}(T_C) = \ell_R \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4}$$
$$\text{CE2}: \quad L_{\partial H}(T_C) = \ell_R + \ell_L < \frac{1}{2}$$

FIGURE 7. The exact CE1 and CE2 center-boundary contributions and their general bounds.

## 3. Trace-Length Bounds

The first mechanism compares the total one-dimensional trace available from the seven triangles with the length of either the perimeter $\partial H$ or the full skeleton $S$. The local trace estimates, the master perimeter deficit, the conditional skeleton theorem, and every routed application are proved in this section.

3.1. **Detailed Trace-Length Estimates.** This section develops the boundary and conditional-skeleton trace bounds used by the length mechanism.

For a closed triangle $T$ and a rectifiable target $E$, write

$$L_E(T) = \mathcal{H}^1(T \cap E).$$

The passage from an original open triangle $U$ to its closure $T = \overline{U}$ can only enlarge the trace. Hence a cover of $E$ by the open triangles implies

$$\mathcal{H}^1(E) \le L_E(T_C) + \sum_{i=0}^{5} L_E(T_i). \tag{26}$$

All estimates in this section concern full traces. They therefore remain valid for any assigned measurable subsets of those traces.

The strategy uses only the perimeter $\partial H$ and the full skeleton $S$, whose measures were recorded in (1). Every skeleton estimate below is conditional on its stated center and V triangle hypotheses; no unconditional noncoverage assertion for $S$ is made.

$\partial H$: **length** 6 $S$: **length** 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 8. The two one-dimensional targets. The full skeleton is used only under the hypotheses of the corresponding trace budget.

3.1.1. *Conditional skeleton trace caps.* We now develop the skeleton estimate used only when the center is CE1 or CE2. The first lemma is the center contribution.

**Lemma 3.1** (Center skeleton cap)**.** *If $T_C$ is a CE1 or CE2 C triangle and contains exactly one radial midpoint, then*

$$L_S(T_C) < \frac{3}{2}.$$

*Proof.* Normalize the unique midpoint as $M_0$ and the positive boundary trace to $e_{0,1}$. Use the signed variables of Proposition 2.15; in particular

$$W = 1 - R, \qquad E = \sqrt{1 - RW}, \qquad \eta = 1 - E, \qquad P = E(1 - E).$$

The positive trace condition is

$$\alpha + W\delta < P. \tag{27}$$

The two possible boundary contributions are

$$\ell_{01} = W + \delta - \frac{\eta + \alpha + \delta}{R}, \qquad \ell_{50} = \frac{[P - (R\alpha + \delta)]_+}{W}.$$

The six radial contributions are bounded by

| $i$ | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| $\mathcal{H}^1(T_C \cap r_i)$ | $E - \alpha - \delta$ | $\delta/R$ | $\delta$ | $\min\{\alpha/R, \delta/W\}$ | $\alpha$ | $\alpha/W.$ |

Adding gives

$$L_S(T_C) \le K_R + \mathcal{E}(\alpha, \delta),$$

where

$$K_R = W + E - \frac{\eta}{R}$$

and

$$\mathcal{E}(\alpha, \delta) = -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting only according to the minimum and the positive part gives

$$\mathcal{E}(\alpha, \delta) \le \frac{P}{W}. \tag{28}$$

For example, if $\alpha/R \le \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when the positive part is active, its excess over $P/W$ is $\alpha - R\delta/W \le 0$. The reflected order gives excess $\delta - W\alpha/R \le 0$. If the positive part vanishes, (27) gives the same bound strictly.

Put $A = 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Thus $K_R + P/W < 3/2$. □

**Lemma 3.2** (Positive-support skeleton cap)**.** *Let $T_i$ be the closure of an original open V triangle, so $V_i \in \operatorname{int}(T_i)$. If*

$$\mathcal{H}^1(T_i \cap r_j) > 0 \quad \textit{for some } j \in \{i-1, i+1\},$$

*then*

$$L_S(T_i) < \frac{3}{2}.$$

*Proof.* Normalize to $i = 0$ and translate the hexagon by $V_0$:

$$\widetilde{H} = V_0 + H, \qquad \widetilde{O} = V_0, \qquad W_j = V_0 + V_j.$$

Then $W_3 = O$, $W_2 = V_1$, and $W_4 = V_5$. The five original skeleton components on which a unit triangle containing $V_0$ can have positive-length trace are

$$[V_0, V_1],\ [V_0, V_5],\ [O, V_1],\ [O, V_0],\ [O, V_5].$$

All five belong to the skeleton $\widetilde{S}$ of $\widetilde{H}$. Since $V_0 = \widetilde{O}$ is interior to $T_0$, the triangle is a legitimate center triangle for $\widetilde{H}$. Its positive trace on $r_1$ or $r_5$ is a positive boundary-edge trace of $\widetilde{H}$, so it is CE1 or CE2 by Proposition 2.3. Proposition 2.12 and Lemma 3.1 therefore give

$$L_{\widetilde{S}}(T_0) < \frac{3}{2}.$$

A nonlocal component of $S$ is more than unit distance from the interior point $V_0$, apart from zero-measure endpoints. Hence $L_S(T_0) \le L_{\widetilde{S}}(T_0)$. □

**Lemma 3.3** (No-support skeleton cap)**.** *If a V triangle has $A_i + B_i \le 1$ and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

*then*

$$L_S(T_i) \le 2.$$

*Proof.* Its boundary contribution is $A_i + B_i \le 1$ by (31). For $P = sV_j \in r_j \setminus \{O\}$,

$$\|V_i - P\|^2 = 1 + s^2 - 2s\cos\frac{(j-i)\pi}{3} > 1$$

when $j \notin \{i-1, i, i+1\}$. The traces on $r_{i-1}$ and $r_{i+1}$ have zero length by hypothesis, so only $r_i$ can contribute positive length. This adds at most one. □

For the remaining cap we use one specialized part of the exact local equilateral enclosure criterion. In local directions $u, v$ meeting at 120°, suppose a closed unit triangle contains

$$0, \quad au, \quad bv, \quad c(u+v),$$

with $0 \le a \le b \le 1$ and $a + b > 1$. The support-line calculation gives the selected supercritical cell

$$\begin{aligned} &a^2 + ab + b^2 \le 1, \qquad c \le \frac{1}{2}, \\ &(a^2 - 1)c^2 + (2ab^2 + b)c + (b^4 - b^2) \le 0, \end{aligned} \tag{29}$$

where $c$ lies on the connected component containing $c = 0$. To see the selector directly, the quadratic is negative at $c = 0$, positive at $c = 1/2$, and opens downward. For the middle assertion, four times its value at $1/2$ is increasing in $a$ and, at $a = 1 - b$, equals $b^2(2b-1)^2$; it is therefore positive when $a + b > 1$. Hence $c \le 1/2$ selects its smaller-root component. Formula (29) follows by fixing in the support sum the three active contacts $\{B\}$, $\{B, c(u+v)\}$, and $\{A\}$, and squaring the positive unit-side equation.

**Lemma 3.4** (Supercritical skeleton cap)**.** *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

*Proof.* By Proposition 2.6, a V triangle is Vd0, Vd1, Vd2, or T3-like. Proposition 3.6 makes every Vd1/Vd2 V triangle nonsupercritical, and Lemma 2.9 does the same for T3-like V triangles. Hence a supercritical triangle is Vd0 and, by definition, $\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0$. Thus, after normalizing and writing $a = A_i$, $b = B_i$, the skeleton trace has length $a + b + c$, where $c$ is the reach on $r_i$. Reflect so that $a \le b$, and put

$$s = a + b, \qquad \delta = b - a, \qquad c_0 = \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \le \frac{2}{\sqrt{3}}, \qquad 0 \le \delta \le \sqrt{4 - 3s^2}.$$

Let $P(c)$ denote the quadratic in (29). Substitution and multiplication by 16 give

$$\begin{aligned} 16P(c_0) = Q(s,\delta) = {} & \delta^4 + 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ & - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ & + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s-3) + (s-1)(2-s)\big) \ge 0.$$

Therefore

$$Q(s,\delta) \ge Q(s,0) = (s-1)(s^3 - 5s^2 - 24s + 36) > 0.$$

Indeed, the cubic is decreasing on $[1, 2/\sqrt{3}]$ and at its right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(c_0) > 0$. Since (29) selects the smaller-root component, $P(c) \le 0$ forces $c < c_0$. Thus $a + b + c < 3/2$. □

**Lemma 3.5** (A positive-support adjacent exceptional triangle is forced)**.** *Suppose $T_C$ is CE1 or CE2 and, after symmetry, $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. If at least two V triangles are supercritical, then for a supercritical index $s \ne 0$ there is some $j \in \{s-1, s+1\}$ such that*

$$M_s \in U_j, \qquad \mathcal{H}^1(T_j \cap r_s) > 0.$$

*Proof.* Choose a supercritical index $s \ne 0$. Lemma 2.11 gives $M_s \notin T_s$, and $M_s \notin T_C$ by the unique-midpoint property. In the original open cover, $M_s$ is therefore contained by some other open V triangle $U_j$. Diameter one restricts $j$ to $\{s-1, s+1\}$. Because $U_j$ is open and $M_s$ is an interior point of $r_s$, one has $\mathcal{H}^1(T_j \cap r_s) > 0$. □

3.1.2. *Vd1/Vd2 corner estimates.* We next isolate the much stronger trace loss caused by Vd1 and Vd2 geometry. For the following two results use the temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0).$$

Thus $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

**Proposition 3.6** (Vd1/Vd2 corner normal form)**.** *Let $T$ be the closure of an original Vd1 or Vd2 triangle at $V_0$, and let $a, b$ be its exact reaches toward $V_5, V_1$. There is a unique $t > 0$ such that, with $d = \sqrt{t^2+t+1}$,*

$$T = \left\{ (x,y) : \begin{array}{r} x - (t+1)y \le a, \\ ty - (t+1)x \le tb, \\ tx + y \le d - a - tb \end{array} \right\}.$$

*The raw traces, in coordinates measured from the indicated outer vertex, are*

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d-a-tb}{t+1} \\ r_1 & \dfrac{t(1-b)}{t+1} \le s \le \dfrac{d-a-tb-1}{t} \\ r_5 & \dfrac{1-a}{t+1} \le s \le d - a - tb - t. \end{array}$$

*Intersecting with $[0,1]$ gives the actual traces. Moreover,*

$$a + b < \frac{1}{2}. \tag{30}$$

*Proof.* A Vd1/Vd2 triangle has one vertex $W$ outside $H$ and two vertices $U, Z$ in $H$. Since $V_0$ is an interior convex combination of them, both coordinates of $W$ are negative. The two axis endpoints $(a,0), (0,b)$ lie on distinct sides incident with $W$. Put $p = U - W$, $q = Z - W$. Both vectors have positive coordinates and satisfy

$$\|p\| = \|q\| = \|p - q\| = 1.$$

The order in which the two sides cross the positive axes selects the $60°$ rotation

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \qquad q = \frac{(t, t+1)}{d}.$$

The two side lines through the axis endpoints and the parallel third side are exactly those in (3.6). Substitution of $(s,s), (s,1), (1,s)$ gives (3.6).

If the $r_1$ trace has positive length, its lower endpoint is smaller than its upper endpoint. Clearing denominators gives

$$(t+1)a + tb < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(a+b)$,

$$a + b < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

For the last inequality, both sides are positive and

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2+t+1) = \frac{t^2}{4} > 0.$$

Reflection proves the same conclusion when the supported arm is $r_5$. $\square$

**Lemma 3.7** (Vd2 adjacent-midpoint cap)**.** *If an open role $U_i$ has Vd2 closure $T_i = \overline{U_i}$ and $T_i$ contains $M_{i-1}$ or $M_{i+1}$, then its exact boundary reaches satisfy*

$$a + b < \frac{1}{3}.$$

*Proof.* In Proposition 3.6, reflection sends $(t, a, b, M_1, M_5)$ to $(1/t, b, a, M_5, M_1)$, so assume $t \geq 1$ and put $U = a + tb$.

If $M_5 = (1, 1/2)$ is contained, the third half-plane in (3.6) gives

$$a + b \leq U \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3};$$

the middle expression decreases for $t \geq 1$.

If $M_1 = (1/2, 1)$ is contained, the second and third half-planes give

$$b \geq \frac{t-1}{2t}, \qquad a + b \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with $r_5$ gives, by (3.6),

$$a + b < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows that $S$ is increasing and $R$ decreasing on $[1, \infty)$. For example, the sign calculation for $S'$ reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ yields

$$1 \leq t \leq \frac{4}{3}: \quad a + b \leq S(4/3) = \frac{8\sqrt{37} - 41}{24} < \frac{1}{3},$$

$$t \geq \frac{4}{3}: \quad a + b < R(4/3) = \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}.$$

The final inequalities follow respectively from $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. $\square$

3.1.3. *Boundary trace caps.* The signed center normal form gives the CE1 and CE2 center contributions from one formula; no separate maximal-interval calculation is needed here.

**Theorem 3.8** (Boundary trace table)**.** *For closures of original open triangles, the following bounds hold:*

| *triangle* | *hypothesis* | $L_{\partial H}$ |
|---|---|---|
| $T_C$ | CE0 | $0$ |
| $T_C$ | CE1 | $\leq \frac{\sqrt{3}}{2} - \frac{3}{4}$ |
| $T_C$ | CE2 | $< \frac{1}{2}$ |
| $T_i$ | Vd0, $A_i + B_i \leq 1$ | $\leq 1$ |
| $T_i$ | Vd0, $A_i + B_i > 1$ | $\leq \frac{2}{\sqrt{3}}$ |
| $T_i$ | Vd1/Vd2 | $< \frac{1}{2}$ |
| $T_i$ | *T3-like* | $< 1.$ |

*Proof.* For every V triangle,

$$L_{\partial H}(T_i) = A_i + B_i. \tag{31}$$

Indeed, the two incident traces are initial intervals of these lengths. A relative-interior point of any nonincident edge is more than unit distance from $V_i$, whereas $V_i$ is interior to the diameter-one triangle $T_i$. This excludes any other positive-length boundary trace.

The CE0 C-triangle row is the definition. The other two C-triangle rows are the class specializations of the common signed formula in Corollary 2.17.

For a nonsupercritical Vd0 V triangle, (31) is enough. For a supercritical V triangle, its incident endpoints are at squared distance $A_i^2 + A_iB_i + B_i^2 \leq 1$, and hence

$$\frac{3}{4}(A_i + B_i)^2 \leq 1.$$

This proves the $2/\sqrt{3}$ cap. The Vd1/Vd2 V triangle is (30). Finally, the Type II calculation in Proposition 2.7 gives for an original T3-like triangle

$$A_i + B_i = z - \alpha < z < 1$$

because $0 < t < 1$ and $\alpha > 0$. □

3.1.4. *Final length contradictions.*

**Proposition 3.9** (Vd2 adjacent-midpoint branch)**.** *Assume a CE2, $N_+ = 1$ configuration has exactly one Vd1/Vd2 triangle, that triangle is $T_i$ of type Vd2 and contains $M_{i-1}$ or $M_{i+1}$, and all other V triangles are Vd0. Then the boundary cannot be covered.*

*Proof.* This is item (5) of Corollary 3.15, using Lemma 3.7. □

**Proposition 3.10** (Branches closed by Method 1)**.** *Direct length-sum obstructions close the following routing entries and hybrid subcase:*

(1) *CE0, $N_+ = 0$;*
(2) *CE0, $N_+ = 1$, with a Vd1/Vd2 triangle;*
(3) *CE1/CE2 with $N_+ + N_{\mathrm{sp}} \geq 3$;*
(4) *CE1/CE2, $N_+ = 0$, with a Vd1/Vd2 triangle;*
(5) *CE1, $N_+ = 1$, with exactly one Vd1/Vd2 triangle and no T3-like triangle;*
(6) *the CE2 hybrid subcase in which the Vd2 triangle $T_i$ contains $M_{i-1}$ or $M_{i+1}$.*

*Proof.* For item (1), the CE0 open C triangle and every nonincident V triangle miss the relative interior of each boundary edge. The two incident open V traces therefore overlap, so $1 < B_i + A_{i+1}$ on every edge. Summing contradicts $N_+ = 0$.

Items (2), (4), and (5) are items (1), (2), and (3) of Corollary 3.15; each is one application of the master perimeter deficit. Item (3) is Theorem 3.14. Item (6) is Proposition 3.9. This is only the indicated subcase of the exceptional CE2 V triangle; its remaining placements are handled by the boundary-reach argument. □

3.2. **Common perimeter and skeleton budgets.** The signed center formula also removes repeated length arithmetic. We first state a generic perimeter deficit, then give the direct CE1/CE2 skeleton count used by the routing table.

**Lemma 3.11** (Master perimeter deficit)**.** *Put*

$$\theta = \frac{2}{\sqrt{3}} - 1.$$

*Suppose a branch has a center contribution $L_C$, $n$ supercritical Vd0 V triangles, $m \geq 1$ distinguished nonsupercritical V triangles with strict boundary caps $q_1, \dots, q_m < 1$, and $6 - n - m$ remaining V triangles whose boundary contributions are at most 1. If*

$$L_C + n\theta \leq \sum_{j=1}^{m} (1 - q_j), \tag{32}$$

*then the seven triangles do not cover $\partial H$.*

*Proof.* The supercritical Vd0 cap is $2/\sqrt{3}$. Hence the total boundary contribution is strictly less than

$$\begin{aligned} L_C + n\frac{2}{\sqrt{3}} + \sum_{j=1}^{m} q_j + (6 - n - m) &= 6 + L_C + n\theta - \sum_{j=1}^{m}(1 - q_j) \\ &\leq 6. \end{aligned}$$

The strictness follows from $m \geq 1$ and the strict distinguished-V triangle caps. A cover of the perimeter would give the opposite weak inequality by subadditivity. $\square$

**Lemma 3.12** (Small center slacks in the one-Vd1/Vd2 branch)**.** *Assume there is exactly one supercritical Vd0 V triangle, exactly one Vd1/Vd2 V triangle, and four nonsupercritical Vd0 V triangles. Then every candidate surviving the perimeter budget is CE2 and satisfies*

$$\boxed{\alpha + \delta < \frac{1}{24}, \qquad \alpha + \delta < \frac{\min\{R, W\}}{6}.} \tag{33}$$

*Proof.* The Vd1/Vd2 boundary cap is strict and equals 1/2. If the perimeter were covered, Lemma 3.11 would force

$$L_C + \theta > \frac{1}{2}.$$

Put

$$\kappa = \frac{1}{2} - \theta = \frac{3}{2} - \frac{2}{\sqrt{3}}.$$

Thus $L_C > \kappa$. The CE1 cap in Corollary 2.17 gives

$$L_C + \theta \leq \left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \left(\frac{2}{\sqrt{3}} - 1\right) < \frac{1}{2},$$

so every survivor is CE2.

Set $T = \alpha + \delta$. The CE2 formula gives

$$T < M(E) := \frac{RW}{E^2}\left(\frac{E}{1+E} - \kappa\right).$$

Since $RW = 1 - E^2$,

$$M(E) = \frac{1-E}{E^2}\big(E - \kappa(1+E)\big),$$

and

$$M'(E) = -\frac{E - 2\kappa}{E^3} < 0 \qquad \left(\frac{\sqrt{3}}{2} \le E < 1\right).$$

Here $2\kappa < \sqrt{3}/2$, because it is equivalent to $6\sqrt{3} < 11$, whose square is $108 < 121$. Therefore

$$T < M\left(\frac{\sqrt{3}}{2}\right) = \frac{8\sqrt{3}}{9} - \frac{3}{2} < \frac{1}{24}.$$

The last inequality is equivalent to $64\sqrt{3} < 111$, and follows after squaring from $12288 < 12321$.

For the second estimate, use $E/(1+E) < 1/2$:

$$T < \frac{RW}{E^2}\left(\frac{1}{2} - \kappa\right) = \frac{RW}{E^2}\theta.$$

Since

$$E^2 = W + R^2 = R + W^2,$$

one has $RW/E^2 \le R$ and $RW/E^2 \le W$. Also $\theta < 1/6$, equivalently $12 < 7\sqrt{3}$, whose square is $144 < 147$. This proves (33). ☐

**Lemma 3.13** (CE2 initial endpoints dominate the total slack)**.** *Assume* $\Delta_R > 0$ *and* $\Delta_L > 0$. *Put*

$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$

*Then*

$$\boxed{T < \frac{1}{2}\min\{x, y\}.} \tag{34}$$

*Proof.* Equation (71) gives $T < \eta/E$. Moreover

$$E^2 - (2R-1)^2 = 3RW > 0,$$

so $|2R-1| < E$ and therefore $|2R-1|T < \eta$. Hence

$$x - 2T = \frac{\eta - (1-2R)T}{W} > 0,$$

and

$$y - 2T = \frac{\eta - (2R-1)T}{R} > 0.$$

This is (34). ☐

**Theorem 3.14** (Direct CE1/CE2 skeleton count)**.** *If a CE1/CE2 branch satisfies* $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$ *and*

$$N_+ + N_{\rm sp} \ge 3, \tag{35}$$

*then the seven triangles do not cover the full skeleton* $S$.

*Proof.* Every Vd1, Vd2, or T3-like V triangle has positive support on an adjacent radial segment and is nonsupercritical: use Proposition 3.6 and Lemma 2.9. Conversely, every supercritical V triangle is Vd0 and has no such support. The two classes counted by $N_+$ and $N_{\rm sp}$ are therefore disjoint.

The center skeleton cap is strictly below $3/2$. Each of the $N_+ + N_{\mathrm{sp}}$ distinguished V triangles also contributes strictly less than $3/2$, by the supercritical or positive-support cap. Every remaining V triangle is nonsupercritical Vd0 and contributes at most 2. Consequently

$$\begin{aligned} L_S(T_C) + \sum_{i=0}^{5} L_S(T_i) &< \frac{3}{2} + \frac{3}{2}(N_+ + N_{\mathrm{sp}}) + 2(6 - N_+ - N_{\mathrm{sp}}) \\ &= \frac{27 - N_+ - N_{\mathrm{sp}}}{2} \leq 12. \end{aligned}$$

This contradicts subadditivity for a cover of the length-12 skeleton. ☐

**Corollary 3.15** (Length branches closed by the common budgets)**.** *The following branches are impossible.*

(1) *CE0, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;*
(2) *$N_+ = 0$ and at least one Vd1/Vd2 V triangle, for either CE1 or CE2;*
(3) *CE1, $N_+ = 1$, and at least one Vd1/Vd2 V triangle;*
(4) *CE2, $N_+ = 1$, and at least two Vd1/Vd2 V triangles;*
(5) *CE2, $N_+ = 1$, with a Vd2 triangle $T_i$ containing $M_{i-1}$ or $M_{i+1}$;*
(6) *CE1 or CE2 with $N_+ + N_{\mathrm{sp}} \geq 3$;*
(7) *CE1 or CE2 with $N_+ \geq 2$.*

*Proof.* For item 1, take $L_C = 0$, one supercritical cap $2/\sqrt{3}$, and one strict Vd1/Vd2 cap $1/2$ in Lemma 3.11; the needed inequality is

$$\frac{2}{\sqrt{3}} - 1 < \frac{1}{2}.$$

For item 2, the center contribution is strictly less than $1/2$ and the chosen Vd1/Vd2 V triangle has strict cap $1/2$; the other five V triangles contribute at most 1 each. This is Lemma 3.11 with $n = 0$ and $q_1 = 1/2$.

For item 3, the unique supercritical V triangle is Vd0, while the CE1 center cap, the Vd1/Vd2 cap, and the four remaining unit caps give

$$\left(\frac{\sqrt{3}}{2} - \frac{3}{4}\right) + \frac{1}{2} + \frac{2}{\sqrt{3}} + 4 < 6.$$

For item 4, the corresponding CE2 sum is strictly less than

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{2}{\sqrt{3}} + 3 < 6.$$

For item 5, the $M_{i-1}/M_{i+1}$ Vd2 cap is $1/3$, and the total is strictly less than

$$\frac{1}{2} + \frac{1}{3} + \frac{2}{\sqrt{3}} + 4 < 6.$$

Item 6 is Theorem 3.14.

Finally suppose $N_+ \geq 2$. By Proposition 2.12, normalize so that

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

Choose a supercritical index $s \neq 0$. Its own midpoint $M_s$ is not covered by $T_s$, and it is not covered by $T_C$. Diameter locality leaves only $U_{s-1}, U_s, U_{s+1}$ as possible V triangles containing $M_s$. Thus $M_s \in U_j$ for some $j \in \{s-1, s+1\}$ and, by openness, $\mathcal{H}^1(T_j \cap r_s) > 0$. Thus $T_j$ is counted by $N_{\mathrm{sp}}$ and is distinct from the two supercritical V triangles. Hence $N_+ + N_{\mathrm{sp}} \geq 3$, and Theorem 3.14 applies. ☐

## 4. Boundary-Reach Propagation

The second mechanism converts a backward boundary reach and a radial reach into an upper bound for the forward reach of the same V triangle. Coverage of the next edge then produces a lower bound for the next backward reach. This section develops the common equilateral-enclosure calculus, the center-assisted path budget, the exceptional T3-like and Vd placements, and the complete branchwise propagation proof. Formalization-only optimization registries are kept in the repository but are not repeated in the printed article.

### 4.1. **Universal Enclosure and Propagation Calculus.**

This section collects the geometry shared by the propagation and direct-witness arguments. The same equilateral enclosure functional governs the local admissible set of Method 2 and the forced witness set of Method 4. The transfer statements below distinguish three situations that must not be conflated: a center-free edge, an edge with a specified center interval, and a free strict-supercritical outgoing envelope.

#### 4.1.1. *The equilateral enclosure gauge.*

Let $\mathsf{R}$ denote counterclockwise rotation through $2\pi/3$. For a nonempty compact set $K \subset \mathbb{R}^2$, put

$$h_K(n) = \max_{x\in K}\langle x, n\rangle$$

and

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{36}$$

**Proposition 4.1** (Equilateral enclosure gauge)**.** *The number $\Lambda(K)$ is the least side length of a closed equilateral triangle containing $K$. It is monotone under inclusion, translation invariant, and positively homogeneous. If $K$ is a finite polygonal hull, a minimizing normal may be chosen perpendicular to one of its hull edges.*

*Proof.* For a fixed unit normal $n$, the three supporting half-planes with outward normals $n, \mathsf{R}n, \mathsf{R}^2 n$ form an equilateral triangle. Its side is $2/\sqrt{3}$ times the sum of the three support numbers, proving (36). Monotonicity and homogeneity are immediate, while translation invariance follows from

$$n + \mathsf{R}n + \mathsf{R}^2 n = 0.$$

On an open normal-direction cell with three fixed support points, the support sum has the form $A\cos\theta + B\sin\theta$ and has second derivative equal to its negative. In the nondegenerate case an interior critical point is a maximum. Hence a minimum occurs at a cell boundary, where one support line contains a hull edge. □

For the anchor hull

$$K(a, b, c) = \operatorname{conv}\{0, au, bv, c(u + v)\},$$

where $u$ and $v$ are the two unit boundary directions meeting at $120°$, the admissible set is

$$\mathcal{A} = \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \le 1\}. \tag{37}$$

The complete support calculation is given in Subsection 4.6 below.

4.1.2. *One equilateral radical.*

For $0 \le x \le 1$, define

$$\omega(x) = \sqrt{1 - x + x^2}, \qquad \sigma(x) = 1 - \omega(x). \tag{38}$$

Then

$$\omega(1 - x) = \omega(x), \qquad \sigma(x) = \frac{x(1-x)}{1 + \omega(x)}, \tag{39}$$

and direct differentiation gives

$$\sigma''(x) = -\frac{3}{4\omega(x)^3} < 0. \tag{40}$$

Thus $\sigma$ is strictly concave. In the signed center calculus,

$$E = \omega(R), \qquad \eta = \sigma(R).$$

For $0 < x < 1$, put $z = \sigma(x)/x$. Eliminating the square root gives

$$x = \frac{1-2z}{1-z^2}, \qquad \sigma(x) = \frac{z(1-2z)}{1-z^2}, \qquad 0 < z < \frac{1}{2}. \tag{41}$$

4.1.3. *Forward envelope and nonsupercritical propagation.*

For $0 \le a, c \le 1$, define the raw forward envelope

$$M_c(a) = \max\{b \in [0,1] : (a,b,c) \in \mathcal{A}\}. \tag{42}$$

Its zero-radial specialization, introduced geometrically in Lemma 2.18, is

$$M_0(a) = \frac{-a + \sqrt{4 - 3a^2}}{2}. \tag{43}$$

The exact nonsupercritical forward cap and the following-edge propagation map are

$$\overline{M}_c(a) = \begin{cases} 1 - a, & 0 \le c \le 1/2, \\ M_c(a), & 1/2 < c \le 1, \end{cases} \qquad \Phi_c(a) = 1 - \overline{M}_c(a). \tag{44}$$

Equivalently,

$$\overline{M}_c(a) = \min\{M_c(a), 1 - a\}.$$

**Lemma 4.2** (Midpoint threshold)**.** *For $0 \le a \le 1$,*

$$M_c(a) \le 1 - a \qquad \left(\frac{1}{2} \le c \le 1\right), \qquad M_{1/2}(a) = 1 - a.$$

*Consequently $\Phi_c(a) \ge a$ whenever $c \ge 1/2$.*

*Proof.* If $(a,b,c) \in \mathcal{A}$ and $c \ge 1/2$, the containing triangle also contains the radial midpoint $(u+v)/2$. Lemma 2.11 gives $a + b \le 1$, hence $b \le 1 - a$.

For the reverse inequality at $c = 1/2$, set $b = 1 - a$. In the exact cell description of Proposition 4.23, one has $s = a + b = 1$, $d = a^2 + ab + b^2 \le 1$, $q = ab \ge 0$, and, with $M = \max\{a, b\} \ge 1/2$,

$$F_T(a, b, 1/2) = M(1/2 - M) \le 0, \qquad 1/2 \le 2M.$$

Thus $(a, b, 1/2) \in \mathcal{A}$, proving equality. $\square$

**Lemma 4.3** (Raw envelope and nonsupercritical cap)**.** *Let $T$ be an actual V triangle satisfying $A(T) \geq a$ and $C(T) \geq c$, and let $T_{\mathrm{next}}$ be the next incident actual V triangle. Then*

$$B(T) \leq M_c(a).$$

*If no center interval intervenes on the next edge, then*

$$A(T_{\mathrm{next}}) \geq 1 - M_c(a).$$

*If $T$ is nonsupercritical, then*

$$B(T) \leq \overline{M}_c(a) \tag{45}$$

*and, on a center-free next edge,*

$$A(T_{\mathrm{next}}) \geq \Phi_c(a) \geq a. \tag{46}$$

*Proof.* The closure of $T$ realizes $(a, B(T), c)$, so the definition of $M_c$ gives the raw outgoing bound. On a center-free edge, coverage gives $A(T_{\mathrm{next}}) \geq 1 - B(T)$. Under nonsupercriticality, $B(T) \leq 1 - A(T) \leq 1 - a$; combining this with the raw bound gives $B(T) \leq \overline{M}_c(a)$. Taking complements proves the propagation bound. □

**Proposition 4.4** (Zero-radial warning)**.** *For $0 < a < 1$,*

$$M_0(a) > 1 - a, \qquad 1 - M_0(a) < a.$$

*Thus a nonsupercritical zero-radial handoff uses the cap $\overline{M}_0(a) = 1 - a$, not the raw envelope $M_0(a)$.*

*Proof.* By (43), the inequality $M_0(a) > 1 - a$ is equivalent, after squaring positive quantities, to $4a(1 - a) > 0$. □

4.1.4. *Center-assisted propagation and the free envelope.*

Let $J \subseteq [0, 1]$ be empty or a closed interval. For $0 \leq p \leq 1$, put

$$e_J(p) = \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup J\}, \qquad \mathcal{R}_J(p) = 1 - e_J(p). \tag{47}$$

If $J = [L, U]$, then

$$\mathcal{R}_{[L,U]}(p) = \begin{cases} 1 - p, & p < L, \\ 1 - \max\{p, U\}, & p \geq L, \end{cases} \qquad \mathcal{R}_\varnothing(p) = 1 - p. \tag{48}$$

The map is nonincreasing in $p$ and under enlargement of $J$.

**Lemma 4.5** (Generalized boundary handoff)**.** *Suppose a unit boundary edge is covered by a left trace contained in $[0, p]$, a C triangle trace $J$, and a right trace contained in $[1 - q, 1]$. Then*

$$q \geq \mathcal{R}_J(p).$$

*For an actual V triangle satisfying $A(T) \geq a$ and $C(T) \geq c$,*

$$A(T_{\mathrm{next}}) \geq \mathcal{R}_J(M_c(a)). \tag{49}$$

*If $T$ is nonsupercritical, then*

$$A(T_{\mathrm{next}}) \geq \mathcal{R}_J(\overline{M}_c(a)). \tag{50}$$

*For $J = \varnothing$, the latter is exactly $A(T_{\mathrm{next}}) \geq \Phi_c(a)$.*

*Proof.* The component of $[0, p] \cup J$ containing 0 is $[0, e_J(p)]$. If $1 - q > e_J(p)$, the intervening open interval is uncovered. Apply this with the raw outgoing bound $B(T) \leq M_c(a)$, or with the nonsupercritical bound $B(T) \leq \overline{M}_c(a)$, and use monotonicity of $\mathcal{R}_J$. □

For $0 \le c < 1/2$, define

$$M_c^{\sup} = \sup\{M_c(a) : M_c(a) > 1 - a\} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}. \tag{51}$$

The supremum is not attained. Hence an actual strict-supercritical V triangle $T$ with $C(T) \ge c$ satisfies

$$B(T) < M_c^{\sup}. \tag{52}$$

If, in addition, its outgoing edge is center-free and $T_{\text{next}}$ is the next incident actual V triangle, then coverage gives

$$A(T_{\text{next}}) > 1 - M_c^{\sup}. \tag{53}$$

At $c = 1/2$ the strict feasible set is empty. When convenient we write $M_{1/2}^{\sup} = 1/2$ only for the continuous extension of the displayed formula, never for an attained strict supremum.

4.1.5. *Relaxed composition and path budgets.*

For maps listed in geometric V triangle order, write

$$[\Psi_1 \mid \cdots \mid \Psi_r](x) = (\Psi_r \circ \cdots \circ \Psi_1)(x). \tag{54}$$

The leftmost slot acts first. Write $\mathrm{I}(x) = x$.

**Lemma 4.6** (Relaxed composition)**.** *Suppose propagated reach lower bounds satisfy*

$$x_j \ge \Psi_j(x_{j-1}) \qquad (1 \le j \le r),$$

*where every $\Psi_j$ is nondecreasing. If $\underline{\Psi}_j \le \Psi_j$, then*

$$x_r \ge [\underline{\Psi}_1 \mid \cdots \mid \underline{\Psi}_r](x_0).$$

*Proof.* Set $y_0 = x_0$ and $y_j = \underline{\Psi}_j(y_{j-1})$. Monotonicity gives $x_j \ge \Psi_j(x_{j-1}) \ge \Psi_j(y_{j-1}) \ge y_j$ inductively. □

**Lemma 4.7** (Boundary-path budget)**.** *Let $E_0, \ldots, E_k$ be consecutive boundary edges and let $T_1, \ldots, T_k$ be the triangles at their intermediate vertices. Suppose:*

1. *on $E_0$, the component covered from the endpoint opposite $T_1$ by all triangles other than $T_1$ has extent at most $u$;*
2. *on $E_k$, the analogous component opposite $T_k$ has extent at most $v$;*
3. *on every middle edge, no triangle other than the two incident path triangles has positive-length trace;*
4. *the chain is covered.*

*Then*

$$\sum_{j=1}^{k} (A_j + B_j) \ge k + 1 - u - v. \tag{55}$$

*Consequently, if all $k$ internal V triangles are nonsupercritical, the chain is impossible whenever $u + v < 1$.*

*Proof.* The first and last component hypotheses give $A_1 \ge 1-u$ and $B_k \ge 1-v$. On every middle edge the C triangle and all nonincident triangles have been excluded, so $B_j + A_{j+1} \ge 1$. Adding proves the formula. □

4.1.6. *Low roots, selected $Q_+$ curves, and thresholds.*
For the high-radial maps define

$$e(d) = \frac{1-d}{2}\left(1 - \sqrt{4(1-d)^2 - 3}\right) \qquad \left(0 < d < 1 - \frac{\sqrt{3}}{2}\right), \tag{56}$$

and, for the low-radial order transition,

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}} \qquad \left(\frac{1}{2} < c \le \frac{\sqrt{3}}{2}\right). \tag{57}$$

**Proposition 4.8** (Universal selected-$Q_+$ curve)**.** *Fix $0 < d < 1/2$, put $c = 1 - d$, and suppose an input $p$ belongs to a selected $Q_+$ arc of $\Phi_c$. Write*

$$q = \Phi_c(p), \qquad \nu = q - p, \qquad x = \frac{q-d}{1-d}.$$

*Then*

$$\nu = \sigma(x) = 1 - \sqrt{1 - x + x^2}, \tag{58}$$

*and*

$$q = d + (1-d)x, \qquad p = d + (1-d)x - \sigma(x). \tag{59}$$

*Consequently the selected arc is increasing and strictly concave.*

*Proof.* The selected quadratic is

$$(1-q)(q-d) = (1-d)^2\nu(2-\nu).$$

Substitution gives $x(1-x) = \nu(2-\nu)$, and the selected component uses the smaller root $\nu = \sigma(x)$. The input function is increasing and strictly convex because $\sigma$ is strictly concave; inversion gives the assertion. $\square$

The corresponding chord bounds are

$$\Phi_{1-d}(p) \ge p + \frac{d}{e(d) - d}(p - d), \tag{60}$$

and, with $t = h(1-d)$,

$$\Phi_{1-d}(p) \ge p + \frac{1-2t}{t-d}(p - d). \tag{61}$$

For every coefficient $\lambda$ proved valid on the relevant selected arc, write

$$\Phi^{\lambda}_{1-d}(p) := p + \lambda(p - d) \le \Phi_{1-d}(p). \tag{62}$$

The rational form (41) gives, in the historical notation $x = 1 - \beta$,

$$\beta = \frac{z(2-z)}{1-z^2}, \qquad m_\beta = \frac{1 - z + z^2}{1 - z^2}. \tag{63}$$

For $0 < d < 1 - \sqrt{3}/2$, the exact high-reach lower-bound threshold is

$$z > e(d) \implies \Phi_{1-d}(z) \ge 1 - e(d). \tag{64}$$

Define

$$\Phi^{\mathrm{th}}_{1-d}(z) = \begin{cases} z, & z \le e(d), \\ 1 - e(d), & z > e(d), \end{cases} \qquad \Phi^{\mathrm{th}}_{1-d} \le \Phi_{1-d}. \tag{65}$$

**Lemma 4.9** (Threshold routing)**.** *Let* $z_j = \Psi_j(z_{j-1})$*, where every* $\Psi_j$ *is nondecreasing and extensive. If* $\Psi_k = \Phi_{1-d}$ *and* $z_{k-1} > e(d)$*, then* $z_j \geq 1 - e(d)$ *for all* $j \geq k$*. If a chain contains later occurrences of* $\Phi_{1-d_1}$ *and* $\Phi_{1-d_2}$ *and*

$$z_0 > e(d_1), \qquad \min\{e(d_1), e(d_2)\} < Q,$$

*then at least one of those two V triangles forces every subsequent output above* $1-Q$*.*

*Proof.* The first assertion is (64) followed by extensivity. If $e(d_1) < Q$, use the first threshold. Otherwise $z_0 > e(d_1) \geq Q > e(d_2)$, so the second threshold fires. The conclusion is "at least one", not uniqueness; both thresholds may be available. ☐

4.2. **The local propagation interface.** For the high-radial parameters used below, $M_c$ is the raw outgoing envelope and $\Phi_c = 1 - \overline{M}_c$ is the following-edge propagation map. The latter is nondecreasing and extensive:

$$\Phi_c(z) \geq z. \tag{66}$$

The exact reflection duality is

$$\Phi_c(a) \leq z \iff \Phi_c(1-z) \leq 1-a. \tag{67}$$

Let a nonsupercritical V triangle have maximal reaches $(A_i, B_i, C_i)$ satisfying

$$A_i \geq z, \qquad C_i \geq c_i > \frac{1}{2}.$$

Then its forward reach satisfies

$$B_i \leq M_{c_i}(z).$$

If the next boundary handoff gives $A_{i+1} \geq 1 - B_i$, then

$$\begin{aligned} A_{i+1} &\geq 1 - B_i \\ &\geq 1 - M_{c_i}(z) \\ &= \Phi_{c_i}(z). \end{aligned} \tag{68}$$

Every composition below is shorthand for repeated applications of this inequality to specified actual V triangles.

4.3. **The common one-gap interface.**

**Proposition 4.10** (Common five-V triangle one-gap reduction)**.** *Assume every V triangle is Vd0,* $N_+ = 1$*, and the normalized right center trace contains a boundary gap, possibly a singleton. In the CE2 case assume the companion trace contains no boundary gap. Put*

$$X = R - \delta, \qquad h_0 = \frac{k}{2R}, \qquad m_3 = \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\},$$

*and*

$$\begin{aligned} c_1 &= 1 - \frac{\delta}{R}, \quad c_2 = 1 - \delta, \qquad c_3 = 1 - m_3, \\ c_4 &= 1 - \alpha, \quad c_5 = 1 - \frac{\alpha}{W}, \quad c_0 = k. \end{aligned}$$

*Let*

$$z_0 = X, \qquad z_i = \Phi_{c_i}(z_{i-1}) \quad (1 \leq i \leq 5),$$

*and* $Z = z_5$*. Then the maximal reaches satisfy*

$$A_0 \geq Z, \qquad A_0 \leq M_k\left(\frac{k}{R}\right) < 1 - h_0. \tag{69}$$

*Consequently it is enough to prove*

$$[\Phi_{1-\alpha} \mid \Phi_{1-m_3} \mid \Phi_{1-\delta}](h_0) > W + \delta. \tag{70}$$

*Proof.* Equation (24), together with Lemma 2.11 and diameter locality, makes V triangles $1, \dots, 5$ nonsupercritical and V triangle 0 uniquely supercritical. Gap containment gives

$$B_0 \geq \frac{k}{R}, \qquad A_1 \geq X.$$

Suppose inductively that $A_i \geq z_{i-1}$. Radial coverage gives actual radial reach at least $c_i > 1/2$, and the raw high-radial bound gives

$$B_i \leq M_{c_i}(z_{i-1}).$$

The next boundary handoff therefore gives

$$A_{i+1} \geq 1 - B_i \geq \Phi_{c_i}(z_{i-1}) = z_i.$$

For the final handoff, the CE1 companion trace is empty and the CE2 companion trace is gap-free, so $A_0 \geq z_5 = Z$.

The same V triangle has boundary reach at least $k/R$ and radial reach at least $k$. Reflection and down-closedness of $\mathcal{A}$ give

$$\left(\frac{k}{R}, A_0, k\right) \in \mathcal{A},$$

so

$$A_0 \leq M_k\left(\frac{k}{R}\right).$$

Lemma 2.18 gives

$$M_k\left(\frac{k}{R}\right) \leq M_0\left(\frac{k}{R}\right) < 1 - \frac{k}{2R} = 1 - h_0,$$

which proves (69).

The first and fifth maps are extensive and the middle maps are nondecreasing, so it is enough to prove

$$[\Phi_{c_2} \mid \Phi_{c_3} \mid \Phi_{c_4}](X) > 1 - h_0.$$

Three applications of (67) show that its negation is equivalent to the negation of (70). This proves the reduction. □

The CE1 sign $\Delta_L \leq 0$ is completed by the selected-$Q_+$ relaxation in Proposition 4.26. We now give the shorter CE2 scalar clause.

**Lemma 4.11** (CE2 total slack and two thresholds)**.** *Assume $\Delta_R > 0$ and $\Delta_L > 0$, and put*

$$T = \alpha + \delta, \qquad Q = \frac{\eta + T}{2R}.$$

*Then*

$$T < \frac{\eta}{E}, \tag{71}$$

*and*

$$\min\{e(\alpha), e(\delta)\} < Q. \tag{72}$$

*Proof.* The CE2 inequalities are

$$\alpha + W\delta < P, \qquad R\alpha + \delta < P.$$

Multiply the first by $W$, the second by $R$, and add. Since $W + R^2 = W^2 + R = E^2$, this gives

$$E^2 T < P = E\eta,$$

which is (71).

Let $d = \min\{\alpha, \delta\}$. Since $d \le T/2$ and the rational low-root bound is increasing,

$$\min\{e(\alpha), e(\delta)\} < \frac{2d}{1-2d} \le \frac{T}{1-T}.$$

It remains to prove

$$\frac{T}{1-T} < \frac{\eta + T}{2R}.$$

Define

$$\chi(t) = (\eta + t)(1-t) - 2Rt.$$

This function is concave, so its minimum on $[0, \eta/E]$ is at an endpoint. Clearly $\chi(0) = \eta > 0$. Direct simplification using $E^2 = 1 - R + R^2$ gives

$$\chi\left(\frac{\eta}{E}\right) = \frac{R(E-R)^2}{E^2} > 0.$$

Thus $\chi(T) > 0$, proving (72). $\square$

**Proposition 4.12** (Short CE2 one-gap completion)**.** *Under the CE2 hypotheses of Proposition 4.10, the one-gap state is impossible in either orientation.*

*Proof.* The right-gap initial input is $z_0 = R - \delta$. The strict CE2 domain gives

$$Wz_0 = RW - W\delta = \eta + P - W\delta > \eta + \alpha.$$

For

$$\pi(q) = (\eta + q)(1 - 2q) - 2qW,$$

one has

$$\pi(0) = \eta > 0, \qquad \pi(P) = \eta(\eta + 2ER^2) > 0.$$

Concavity therefore gives $\pi(\alpha) > 0$, and the low-root bound yields

$$z_0 > \frac{\eta + \alpha}{W} > \frac{2\alpha}{1 - 2\alpha} > e(\alpha). \tag{73}$$

If $e(\alpha) < Q$, extensivity carries (73) to the V triangle-4 map $\Phi_{1-\alpha}$, which gives an output at least $1 - e(\alpha) > 1 - Q$. All later maps preserve it. If $e(\alpha) \ge Q$, Lemma 4.11 gives $e(\delta) < Q$, and

$$z_0 > e(\alpha) \ge Q > e(\delta).$$

The V triangle-2 map $\Phi_{1-\delta}$ then gives an output above $1 - Q$, again preserved by all later maps. Thus $Z > 1 - Q = 1 - h_0$, contradicting (69).

Reflection exchanges

$$R \leftrightarrow W, \qquad \alpha \leftrightarrow \delta$$

and reverses the V triangle order. The signed CE2 domain and the proof above are invariant under this exchange, so the left-gap orientation is impossible as well. $\square$

*Remark* 4.13 (Why the CE1 scalar clause remains separate). The geometric one-gap interface is common, but (72) is not valid on the full CE1 sign domain. Thus the merger stops at Proposition 4.10; the CE1 selected-$Q_+$ clause and the CE2 threshold clause are genuinely different scalar proofs of the same target (70).

4.4. **Exceptional Local Profiles and Vd Placements.** This subsection proves the local profile estimates used by the one-special-triangle branches. A T3-like adjacent exceptional triangle and a Vd1 adjacent exceptional triangle satisfy the same two scalar inequalities and therefore share one center-hiding theorem. The remaining Vd1/Vd2 placements use the interval residuals from Section 4.1, one global radial envelope, and two immediate consequences of the corner normal form.

Use the strict-supercritical envelope $M_c^{\mathrm{sup}}$ from (51); its explicit formula is strictly decreasing for $0 \le c < 1/2$. At $c = 1/2$ we use only the continuous extension $M_{1/2}^{\mathrm{sup}} = 1/2$. The strict feasible set is empty there, so this endpoint value is never described as an attained supremum.

4.4.1. *A global quarter radial envelope.*

For $(p, h) \in [0, 1]^2$, let $c_{\max}(p, h)$ be the maximal radial coordinate such that $(p, h, c)$ belongs to the local admissible set.

**Lemma 4.14** (Quarter radial envelope)**.** *If*

$$0 \le h \le p, \qquad p + h < 1,$$

*then*

$$c_{\max}(p, h) \le 1 - \frac{h}{4}. \tag{74}$$

*The inequality is strict on the selected $\mathcal{A}_T$ component.*

*Proof.* Put $s = p+h$ and $q = s^4 - s^2 + ph$. Since $s < 1$, only the two nonsupercritical cells of Proposition 4.23 occur.

On $\mathcal{A}_L$, the smaller boundary coordinate is $h$, and the selected radial frontier is the unique root in $[\sqrt{3}/2, 1]$ of

$$P_h(c) = c^4 - c^2 + hc - h^2 = 0.$$

At $c_0 = 1 - h/4$,

$$P_h(c_0) = \frac{h}{256}(h^3 - 16h^2 - 240h + 128).$$

The cubic decreases on $[0, 1/2]$ and has value $33/8$ at $1/2$, so this is positive for $h > 0$. Moreover

$$P_h\left(\frac{\sqrt{3}}{2}\right) = -\left(h - \frac{\sqrt{3}}{4}\right)^2 \le 0, \qquad P_h'(c) > 0 \quad (c \ge \sqrt{3}/2).$$

Hence the selected root is below $c_0$; for $h = 0$ equality gives $c_{\max} = 1$.

On $\mathcal{A}_T$, put

$$r = \sqrt{4s^2 - 3}, \qquad \gamma = \frac{p - h}{s}.$$

The exact identity

$$\frac{q}{s^2} = \frac{r^2 - \gamma^2}{4}$$

gives $0 \le \gamma < r$. Write $\gamma = wr$, $0 \le w < 1$. Then

$$p = \frac{s(1+wr)}{2}, \qquad h = \frac{s(1-wr)}{2}, \qquad c_{\max} = \frac{s(1+wr)}{1+r}.$$

For

$$\Psi(w) = c_{\max} + \frac{h}{4}$$

one has

$$\Psi'(w) = \frac{sr(7-r)}{8(1+r)} > 0,$$

so

$$\Psi(w) < \Psi(1) = \frac{s(9-r)}{8}.$$

Since $4s^2 = r^2 + 3$,

$$256 - (r^2+3)(9-r)^2 = (1-r)(r^3 - 17r^2 + 67r + 13) > 0$$

for $0 \le r < 1$; the cubic is increasing and positive on this interval. Thus $s(9-r) < 8$ and $c_{\max} + h/4 < 1$. □

4.4.2. *The rational T3-like adjacent-exception profile.*

**Lemma 4.15** (T3-like adjacent-exception profile)**.** *Let a translated T3-like triangle at $V_0$ contain $M_1$, have boundary trace $[0,a]$ on $e_{5,0}$, and have trace $[c,u]$ on $r_1$ measured from $V_1$ toward $O$. Then*

$$a \le 1 - M_c^{\sup}, \qquad \frac{a}{a+1-u} \le 1 - M_c^{\sup}. \tag{75}$$

*Proof.* The translated T3-like normal form supplies

$$1 \le D \le \frac{2}{\sqrt{3}}, \qquad E = \sqrt{4-3D^2}, \qquad \varrho = \frac{D+E}{2},$$

and

$$\frac{E}{2D} \le a \le \frac{4-3D+E}{4D}.$$

Its radial interval is

$$c = \frac{D(1+a)-1}{\varrho}, \qquad u = 1 - \frac{\varrho}{D} + a.$$

Put

$$x = \frac{aD}{\varrho}, \qquad \theta = \frac{D-1}{\varrho}.$$

The relation $\varrho^2 - D\varrho + D^2 = 1$ gives

$$\varrho = \frac{1-2\theta}{1-\theta+\theta^2}, \qquad D = \frac{1-\theta^2}{1-\theta+\theta^2}, \qquad E = \frac{1-4\theta+\theta^2}{1-\theta+\theta^2},$$

with $0 \le \theta \le 2-\sqrt{3}$. Moreover

$$c = x + \theta, \qquad \frac{1-4\theta+\theta^2}{2(1-2\theta)} \le x \le \frac{1}{2} - \theta. \tag{76}$$

Let

$$\psi(z) = \frac{z(2-z)}{1+z}.$$

It is increasing on $[0, 1/2]$ and $\psi(1 - M_c^{\sup}) = c$. Hence it is enough to prove $\psi(x) \le x + \theta$, or equivalently

$$Q_\theta(x) = 2x^2 + (\theta - 1)x + \theta \ge 0.$$

The vertex is $x_* = (1-\theta)/4$. If $0 \le \theta \le 1/5$, the lower endpoint $x_L$ in (76) satisfies

$$x_L - x_* = \frac{1 - 5\theta}{4(1 - 2\theta)} \ge 0,$$

and

$$Q_\theta(x_L) = \frac{\theta(1 - 5\theta + 11\theta^2 - \theta^3)}{2(1 - 2\theta)^2} \ge 0.$$

If $1/5 \le \theta \le 2 - \sqrt{3}$, the vertex lies in the feasible interval and

$$Q_\theta(x) \ge Q_\theta(x_*) = \frac{10\theta - 1 - \theta^2}{8} > 0.$$

Thus $x \le 1 - M_c^{\sup}$. Since $D \ge \varrho$, one has $a \le x$, and

$$\frac{a}{a + 1 - u} = \frac{aD}{\varrho} = x.$$

This proves both inequalities. □

4.4.3. *The Vd1 adjacent-exception profile.*

**Lemma 4.16** (Vd1 adjacent-exception profile)**.** *Let a Vd1 triangle at $V_0$ contain $M_1$, have boundary trace $[0, a]$ on $e_{5,0}$, and have trace $[c, u]$ on $r_1$. Assume it has no positive support on $r_5$. Then*

$$a \le 1 - M_c^{\sup}, \qquad \frac{a}{a + 1 - u} \le 1 - M_c^{\sup}. \tag{77}$$

*Proof.* The Vd corner normal form gives $t > 0$, $d = \sqrt{t^2 + t + 1}$, and

$$c = \frac{t(1 - b)}{t + 1}, \qquad u = \frac{d - a - tb - 1}{t}.$$

The midpoint condition implies

$$tb = t - c(t + 1), \qquad a + tb \le d - 1 - \frac{t}{2}.$$

One-sided support forces $t \ge 1$. Indeed, if $t < 1$, then

$$d - a - tb - t \ge 1 - \frac{t}{2} > \frac{1}{t + 1} > \frac{1 - a}{t + 1},$$

so the raw $r_5$ interval has positive length, contrary to the Vd1 type.

Put

$$F(t, c) = \sqrt{t^2 + t + 1} - 1 - \frac{3t}{2} + c(t + 1).$$

Then $a \le F(t, c)$. Since $c \le 1/2$,

$$\frac{\partial F}{\partial t} = \frac{2t + 1}{2\sqrt{t^2 + t + 1}} - \frac{3}{2} + c \le \frac{2t + 1}{2\sqrt{t^2 + t + 1}} - 1 < 0,$$

where the last inequality follows from $(2t + 1)^2 < 4(t^2 + t + 1)$. Thus $F$ decreases for $t \ge 1$, and

$$a \le F(t, c) \le L(c) := \sqrt{3} - \frac{5}{2} + 2c.$$

Every feasible V triangle has $L(c) \geq 0$. The inequality $2L(c) \leq 1 - M_c^{\sup}$ is equivalent to

$$\sqrt{c^2 - 8c + 4} \leq 12 - 4\sqrt{3} - 9c.$$

The right side is positive on $[0, 1/2]$. Squaring is therefore legitimate, and the squared difference is $4Q(c)$, where

$$Q(c) = 20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3}.$$

The polynomial is decreasing on $[0, 1/2]$ and $Q(1/2) = 26 - 15\sqrt{3} > 0$. Hence $2L(c) \leq 1 - M_c^{\sup}$, and therefore $a \leq 1 - M_c^{\sup}$.

Put $\varepsilon = 1 - u > 0$. The endpoint formula gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \qquad \varepsilon_0 = \frac{2t + 1 - c(t+1) - d}{t}.$$

Since $c \leq 1/2$ and $t \geq 1$,

$$2t + 1 - c(t+1) - d \geq \frac{3t+1}{2} - d > 0;$$

the last inequality follows after squaring from $(3t+1)^2/4 - d^2 = (5t^2 + 2t - 3)/4 > 0$. Thus $\varepsilon_0 > 0$. Moreover,

$$\varepsilon_0 + \frac{F(t,c)}{t} = \frac{1}{2}.$$

Since $z/(z + \varepsilon_0 + z/t)$ increases in $z$,

$$\frac{a}{a+\varepsilon} \leq \frac{F(t,c)}{F(t,c) + 1/2} \leq \frac{L(c)}{L(c) + 1/2} \leq 2L(c) \leq 1 - M_c^{\sup}.$$

□

4.4.4. *A common adjacent-exception theorem.*

**Lemma 4.17** (Common adjacent-exception obstruction)**.** *Let a local adjacent exceptional triangle $T_0$ have trace $[0, a]$ on $e_{5,0}$ and trace $[c, u]$ on $r_1$, measured from $V_1$ toward $O$. Put*

$$\varepsilon = 1 - u > 0, \qquad \Theta = \frac{a}{a+\varepsilon}.$$

*Assume*

$$a \leq 1 - M_c^{\sup}, \qquad \Theta \leq 1 - M_c^{\sup}, \tag{78}$$

*and assume radial isolation forces the center to cover the $O$-side gap of length $\varepsilon$. If $T_1$ is the unique strict supercritical V triangle and $T_2, T_3, T_4, T_5$ are nonsupercritical, then the perimeter and $r_1$ are not covered.*

*Proof.* The center exit on $r_1$ is $\delta/R$, so radial isolation gives

$$\delta \geq R\varepsilon.$$

Let $h$ be the reach forced on $T_5$ along $e_{5,0}$. If the companion center trace does not hide the gap after coordinate $a$, then

$$h \geq 1 - a \geq M_c^{\sup}.$$

In the only hiding order,

$$\frac{k}{W} \leq a < R + \alpha.$$

The left endpoint inequality and radial isolation give

$$k \leq Wa, \qquad R\varepsilon \leq Wa,$$

so $R(a+\varepsilon) \le a$ and $R \le \Theta$. Also

$$\alpha \le Wa - R\varepsilon - \eta,$$

whence

$$R + \alpha \le a + R(u-a).$$

The hiding order forces $u > a$, and therefore

$$R + \alpha \le a + \Theta(u-a) = \Theta \le 1 - M_c^{\sup}.$$

Thus in every ordering

$$h \ge M_c^{\sup}.$$

Coverage of $r_1$ gives the supercritical V triangle radial lower bound at least $c$, so its actual forward reach satisfies

$$B_1 < M_c^{\sup} \le h.$$

Lemma 4.7, applied to $T_2, T_3, T_4, T_5$, gives

$$\sum_{i=2}^{5}(A_i + B_i) \ge 4 + h - B_1 > 4,$$

contrary to their four nonsupercritical caps. □

**Corollary 4.18** (Adjacent T3-like/Vd1 rescue)**.** *The common adjacent-exception lemma applies both to the exactly-one-T3-like branch and to a normalized Vd1 triangle $T_i$ containing $M_{i-1}$ or $M_{i+1}$.*

*Proof.* Lemmas 4.15 and 4.16 give the two local inequalities. In both branches, midpoint and support isolation force the center to cover the $O$-side radial gap before the adjacent exceptional triangle begins. □

4.4.5. *Adjacent Vd1/Vd2 placement.*

**Lemma 4.19** (Adjacent Vd1/Vd2 placement)**.** *Assume a reduced CE2 candidate with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$ has $N_+ = 1$, V triangle $T_0$ uniquely supercritical, V triangle $T_1$ the unique Vd1/Vd2 V triangle, and V triangles $T_2, T_3, T_4, T_5$ nonsupercritical Vd0. Then the perimeter together with $r_2$ is not covered. The reflected $T_5$ placement is also impossible.*

*Proof.* Let

$$I_L = \left[\frac{k}{W}, R + \alpha\right], \qquad I_R = \left[\frac{k}{R}, W + \delta\right].$$

For the actual supercritical reaches $(A_0, B_0)$, put

$$A = \mathcal{R}_{I_R}(B_0), \qquad h_0 = \mathcal{R}_{I_L}(A_0).$$

The endpoint-distance inequality is

$$A_0^2 + A_0 B_0 + B_0^2 \le 1.$$

It implies $A_0, B_0 < 1$. The signed center bounds give $R + \alpha < 1$, so $h_0 > 0$. Choose selected lower bounds

$$0 \le a_i \le A_i, \qquad 0 \le b_i \le B_i \qquad (1 \le i \le 5)$$

so that boundary propagation gives

$$a_1 \ge A, \qquad b_i \ge h_0 \quad (1 \le i \le 5).$$

The Vd cap gives

$$A + h_0 < \frac{1}{2}. \tag{79}$$

The common perimeter budget gives

$$T := \alpha + \delta < \frac{1}{24}, \qquad T < \frac{\min\{R, W\}}{6}.$$

We claim

$$\delta < \frac{h_0}{4}. \tag{80}$$

If $h_0 = W - \alpha$, then

$$4\delta + \alpha \le 4T < \frac{2W}{3} < W.$$

Suppose $h_0 = 1 - A_0$. The other residual cannot be $1 - B_0$, since otherwise (79) would give $A_0 + B_0 > 3/2$, whereas the endpoint-distance inequality gives $A_0 + B_0 \le 2/\sqrt{3} < 3/2$. Hence

$$A = 1 - (W + \delta), \qquad B_0 \ge y := \frac{k}{R},$$

and $W + \delta > 1/2 + h_0$. Diameter transfer gives

$$h_0 > \frac{y}{2} > \frac{\eta}{2R}.$$

Thus

$$\delta > \vartheta(R) := R - \frac{1}{2} + \frac{\eta}{2R}.$$

Writing $\eta/R = W/(1+E)$, exact differentiation gives

$$4E(1+E)^2\vartheta'(R) = 2E(1+E)(1+2E) - W(2R-1).$$

If $R \le 1/2$, the second term is nonnegative after subtraction. If $R \ge 1/2$, then $W(2R-1) \le 1/8$, while the first term is larger than $1/8$. Thus $\vartheta'(R) > 0$. Since $\vartheta(3/8) = 1/24$, the bound $\delta < 1/24$ forces $R < 3/8$. Then

$$(2-3R)^2 - E^2 = (3-8R)(1-R) > 0,$$

so $\eta/R > 1/3$ and $h_0 > 1/6$. Hence $4\delta < 1/6 < h_0$, proving (80).

The center interval on $r_2$ begins from the vertex side at $1 - \delta$. If $T_1$ supports $r_2$, its exact endpoint satisfies

$$u_2 < 1 - b_1 - \frac{a_1}{t} \le 1 - h_0 < 1 - \delta.$$

Thus it does not bridge to the center. For $T_2$, boundary handoff gives

$$a_2 > \frac{1}{2} + A, \qquad b_2 \ge h_0.$$

Put $p = 1/2 + A$. Equation (79) gives $0 \le h_0 \le p$ and $p + h_0 < 1$. Lemma 4.14 gives

$$c_2 \le c_{\max}(p, h_0) \le 1 - \frac{h_0}{4} < 1 - \delta.$$

Thus neither possible radial bridge reaches the center interval, and $r_2$ is uncovered. $\square$

4.4.6. *Nonadjacent Vd1/Vd2 placements.*

**Lemma 4.20** (Nonadjacent Vd1/Vd2 placements)**.** *Under the same reduced CE2 branch, let the unique Vd1/Vd2 V triangle be $T_\tau$ with $\tau \in \{2,3,4\}$. Then the perimeter together with $r_\tau$ is not covered.*

*Proof.* Put
$$T = \alpha + \delta, \qquad x = \frac{\eta + T}{W}, \qquad y = \frac{\eta + T}{R}.$$
Lemma 3.13 gives
$$T < \frac{1}{2}\min\{x, y\}.$$
Let $(A_0, B_0)$ be the actual reaches of the unique supercritical V triangle. Then
$$A_0 + B_0 > 1, \qquad A_0^2 + A_0 B_0 + B_0^2 \le 1.$$
Let
$$B = e_{[y,W+\delta]}(B_0), \qquad U = e_{[x,R+\alpha]}(A_0), \qquad A = 1 - B, \qquad h_0 = 1 - U.$$
Choose $a_\tau \le A_\tau$ and $b_\tau \le B_\tau$. Boundary propagation and the Vd cap permit the choices
$$a_\tau \ge A, \qquad b_\tau \ge h_0, \qquad A + h_0 < \frac{1}{2}.$$

The component endpoint $B$ is either $B_0$ or $W + \delta$. In the latter case the center diameter gives $B \le M_0(x)$ by Lemma 2.18. In the former case, if $A_0 < x$, then $U = A_0$ and $A + h_0 < 1/2$ would force $A_0 + B_0 > 3/2$, contrary to $A_0 + B_0 \le 2/\sqrt{3}$. Hence $A_0 \ge x$, and $B = B_0 \le M_0(A_0) \le M_0(x)$, using the decreasingness in that lemma. Reflection gives $U \le M_0(y)$. Consequently
$$A > \frac{x}{2} > T, \qquad h_0 > \frac{y}{2} > T.$$
Thus
$$d_\tau^C < \min\{A, h_0\} \qquad (\tau = 2, 3, 4),$$
because the three exits are $\delta$, $\alpha$, and $\min\{\alpha/R, \delta/W\} \le T$.

The Vd corner normal form gives the radial margin
$$c_\tau < 1 - \min\{A, h_0\}.$$
But the center interval begins from the vertex side at
$$1 - d_\tau^C > 1 - \min\{A, h_0\}.$$
The two radial traces are disjoint, and all other triangles are excluded by Vd0 locality and diameter locality. □

4.4.7. *Placement interfaces.*

The local profile and radial-separation lemmas above are assembled exactly once in Section 4.10. This subsection contains no independent CE2 one-Vd placement assembly.

4.5. **Propagation roadmap and the T3 endpoint state.** The proof below has eight geometric outputs: the one- and two-gap endpoint inequalities, the CE1 and CE2 return inequalities, the support-isolated T3 endpoint inequality, the common adjacent-exception obstruction, and the two Vd separation statements. Their proof-bearing locations carry the labels 4.33, 4.34, 4.26, 4.12, 4.31, 4.18, 4.19, and 4.20, respectively. This roadmap introduces no second registry or universal optimization layer.

**Definition 4.21** (Support-isolated T3-like endpoint state)**.** Assume $T_C$ is CE1 or CE2, $N_+ = 0$, no V triangle is Vd1 or Vd2, and, after normalization and reflection, $T_0$ is T3-like with $\mathcal{H}^1(T_0 \cap r_1) > 0$. Apply the closed-trace domination of Proposition 2.7 and retain the symbol $T_0$ for the resulting trace-dominating majorant. The support-isolation hypotheses are

$$\mathcal{H}^1(T_j \cap r_1) = 0 \quad (j \notin \{0, 1\}), \qquad \mathcal{H}^1(T_j \cap r_5) = 0 \quad (j \neq 5).$$

Parameterize $e_{0,1}$ and $e_{5,0}$ from $V_0$, and $r_1, r_5$ from $O$. Write

$$T_0 \cap e_{0,1} = [0, p_1], \qquad T_0 \cap e_{5,0} = [0, p_5],$$

$$J_1 = T_C \cap e_{0,1}, \qquad J_5 = \begin{cases} T_C \cap e_{5,0}, & T_C \text{ is CE2}, \\ \varnothing, & T_C \text{ is CE1}. \end{cases}$$

In CE1 a possible closure-only point contact is discarded: it has zero trace, and the original open C triangle supplies neither open coverage nor a positive boundary interval there. Write

$$T_C \cap r_i = [0, \gamma_i] \quad (i \in \{1, 5\}), \qquad T_0 \cap r_1 = [1 - u, 1 - c].$$

With the residual operator $\mathcal{R}_J$ of (47), define

$$(a_5^*, c_5^*) = (\mathcal{R}_{J_5}(p_5), 1 - \gamma_5)\,,$$

$$(a_1^*, c_1^*) = \begin{cases} (\mathcal{R}_{J_1}(p_1), c)\,, & 1 - u \le \gamma_1 \le 1 - c, \\ (\mathcal{R}_{J_1}(p_1), 1 - \gamma_1)\,, & \text{otherwise}. \end{cases}$$

For $i \in \{1, 5\}$ the endpoint capacity is $\overline{M}_{c_i^*}(a_i^*)$; it is the direct nonsupercritical value $1 - a_i^*$ when $c_i^* \le 1/2$. If $A_i, C_i$ denote the actual maximal boundary and radial reaches of $T_i$, then the displayed residual quantities are the selected lower bounds

$$a_i^* \le A_i, \qquad c_i^* \le C_i \qquad (i \in \{1, 5\}).$$

**Proposition 4.22** (Common CE2 two-gap application)**.** *Assume both CE2 center traces contain boundary gaps and $T_1, \ldots, T_5$ are nonsupercritical Vd0 triangles. Then the perimeter is not covered. No hypothesis on the criticality of $T_0$ is required.*

*Proof.* The two-endpoint inequality, Theorem 4.34, makes the endpoint capacities sum to less than one. The three intervening V triangles form the center-free path required by Lemma 4.7, giving the contradiction. □

4.6. **Detailed Local Reach Calculus.** This section develops the local calculus for configurations in which a coarse length sum does not by itself retain enough information. The mechanism is always the same. A center trace fixes boundary inputs and complementary radial lower bounds; an exact local map propagates these data through nonsupercritical V triangles; and the returning reach lower bound exceeds the remaining boundary or radial capacity.

Actual maximal reaches continue to be denoted by $(A_i, B_i, C_i)$, while lowercase letters denote prescribed reach lower bounds. Every estimate in this section is proved analytically. In particular, no empirical branch catalogue, machine-assisted certificate, or formal root from a discarded algebraic component is used.

4.6.1. *The exact local feasible set.* Retain $K(a,b,c)$, the admissible set $\mathcal{A}$, and its raw envelope $M_c(a)$ from (37) and (42). The variables $a,b,c$ are selected reach lower bounds; actual maximal reaches remain uppercase.

**Proposition 4.23** (Exact admissible set and radial envelope)**.** *Put*

$$s = a+b, \qquad d = a^2+ab+b^2, \qquad m = \min\{a,b\}, \qquad M = \max\{a,b\},$$

*and*

$$q = s^4 - s^2 + ab.$$

*Define*

$$\begin{aligned} F_L(a,b,c) &= c^4 - c^2 + mc - m^2, \\ F_T(a,b,c) &= (s^2-1)c^2 + Mc - M^2, \\ F_S(a,b,c) &= (m^2-1)c^2 + (2mM^2+M)c + M^4 - M^2. \end{aligned}$$

*Then* $\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S$, *where*

$$\begin{aligned} \mathcal{A}_L &= \{d \le 1, s \le 1, q \le 0, F_L \le 0\}, \\ \mathcal{A}_T &= \{d \le 1, s \le 1, q \ge 0, F_T \le 0, c \le 2M\}, \\ \mathcal{A}_S &= \{d \le 1, s > 1, F_S \le 0, c \le \tfrac{1}{2}\}. \end{aligned}$$

*All variables in these three sets are also constrained to* $[0,1]$.

*For* $d \le 1$*, the maximal radial lower bound is*

$$c_{\max}(a,b) = \begin{cases} 1, & a=b=0, \\ C_L(m), & s \le 1, q \le 0, (a,b) \ne (0,0), \\ C_T(m,M), & s \le 1, q > 0, \\ C_S(m,M), & s > 1, \end{cases} \tag{81}$$

*where* $C_L(m)$ *is the unique root in* $[\sqrt{3}/2, 1]$ *of*

$$z^4 - z^2 + mz - m^2 = 0,$$

*and*

$$C_T(m,M) = \frac{2M}{1+\sqrt{4s^2-3}}, \qquad C_S(m,M) = \frac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}.$$

*If* $d > 1$*, the radial envelope is undefined. At* $q = 0$ *the two subcritical formulas agree and equal* $s$.

*Proof.* If $s > 0$ and $cs \le ab$, then $c/a + c/b \le 1$ (with the zero-coordinate cases obtained by continuity), so $c(u+v)$ lies in $\operatorname{conv}\{0, au, bv\}$. Its minimum enclosing equilateral side is the diameter

$$\sqrt{a^2 + ab + b^2}.$$

Assume now that $cs > ab$. The four points occur cyclically as $0, bv, c(u+v), au$. For three outward unit normals separated by $120°$, the required side equals $2/\sqrt{3}$ times the sum of the three support values. On an angular cell with fixed support points this sum has second derivative equal to its negative. Hence an angular minimum occurs when a normal is perpendicular to one of the four hull edges. The four resulting side lengths are

$$L_{0A} = b + \max\{a, c\}, \qquad L_{B0} = a + \max\{b, c\},$$

$$L_{AC} = \begin{cases} (a^2 + bc)/\sqrt{a^2 - ac + c^2}, & a \ge c, \\ c(a+b)/\sqrt{a^2 - ac + c^2}, & a < c \le a + b, \\ c^2/\sqrt{a^2 - ac + c^2}, & c > a + b, \end{cases}$$

and $L_{CB}$ is obtained by interchanging $a, b$. Thus the minimum side is the minimum of these four expressions.

Ordering $a, b$ as $m \le M$, squaring the positive support equations gives $F_L = 0, F_T = 0, F_S = 0$ in the three respective contact regimes; the inactive support inequalities give $q \le 0, q \ge 0, s > 1$. Squaring creates one spurious component in each of the last two cells. In the $T$ cell the two roots are

$$\frac{2M}{1 + \sqrt{4s^2 - 3}}, \qquad \frac{2M}{1 - \sqrt{4s^2 - 3}},$$

and $c \le 2M$ selects the first. In the $S$ cell the smaller root lies below $1/2$ and the other above $1/2$; hence $c \le 1/2$ is exactly the geometric selector. The constant-branch fiber is the interval from zero to its unique selected root. Solving the selected equations gives (81). This also proves that $\mathcal{A}$ is coordinatewise down-closed. $\square$

4.6.2. *Branchwise nonsupercritical output.*

**Proposition 4.24** (Exact midpoint split)**.** *For every $a, c \in [0, 1]$,*

$$\overline{M}_c(a) = \min\{M_c(a), 1 - a\}.$$

*If an actual triangle has reaches $(A, B, C)$ with*

$$A \ge a, \qquad C \ge c, \qquad A + B \le 1,$$

*then*

$$B \le \overline{M}_c(a). \tag{82}$$

*Moreover,*

$$\overline{M}_c(a) = 1 - a \quad (c \le 1/2), \qquad \overline{M}_c(a) = M_c(a) \quad (c > 1/2).$$

*Proof.* This is Lemmas 4.2 and 4.3. The equality-boundary assignment at $c = 1/2$ is retained in (44). $\square$

4.6.3. *The exact high-radial map.* For $1/2 < c < 1$ put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c)$$

when $c \geq \sqrt{3}/2$, and use the order-transition function $h(c)$ from (57). Also define

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$

$$Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

**Proposition 4.25** (Four-branch high-radial map)**.** *For $1/2 < c \leq \sqrt{3}/2$,*

$$\overline{M}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases} \tag{83}$$

*For $\sqrt{3}/2 < c < 1$,*

$$\overline{M}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases} \tag{84}$$

*Thus there are exactly four branches*

$$\text{Lin}, \qquad \text{Const}, \qquad Q_-, \qquad Q_+.$$

*At $a = \ell(c)$ in* (84)*, $\overline{M}_c(a) = r(c)$; immediately to its right the value is $\ell(c)$. This downward jump is genuine. The formal other root of the $Q_+$ quadratic is not feasible because it violates $c \leq 2\max\{a, \overline{M}_c(a)\}$.*

*The maps $\overline{M}_c$ are nonincreasing, the maps $\Phi_c = 1 - \overline{M}_c$ are nondecreasing and extensive, and*

$$\Phi_c(a) \leq z \quad \Longleftrightarrow \quad \Phi_c(1 - z) \leq 1 - a. \tag{85}$$

*Proof.* Because $c > 1/2$, the midpoint threshold forces $a + b \leq 1$ and eliminates the $S$ cell of Proposition 4.23. On the cap $b = 1 - a$, the $T$ inequality reduces to $M(c - M) \leq 0$; hence the linear face is feasible exactly for $a \leq 1 - c$ or $a \geq c$. Off that face a maximal point lies on $F_L = 0$ or $F_T = 0$. The constant-branch equation has roots $\ell(c), r(c)$ and the selector $q \leq 0$ keeps only the smaller coordinate. The ordered $T$ halves give respectively the displayed lower quadratic root $Q_+$ and the root $Q_-$. Their order boundary is $a = b = h(c)$, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$. These observations give (83)–(84), including all equality assignments.

Down-closedness gives monotonicity, while $\overline{M}_c(a) \leq 1 - a$ gives $\Phi_c(a) \geq a$. Finally,

$$\Phi_c(a) \leq z \iff \overline{M}_c(a) \geq 1 - z \iff (a, 1 - z, c) \in \mathcal{A}, \quad a + 1 - z \leq 1.$$

Reflection $a \leftrightarrow b$ gives (85). $\square$

Put

$$\mu = \frac{2\sqrt{3}-3}{4},$$

and, for $0 < X < 1 - \sqrt{3}/2$, write

$$e(X) = \ell(1-X).$$

The root equation gives

$$\begin{aligned} X &< e(X) && \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right), \\ e(X) &< \frac{2X}{1-2X} && (0 < X \le \mu), \\ e(X) &\le 2X + 5X^2 && (0 < X \le 1/8). \end{aligned} \tag{86}$$

Moreover,

$$a > e(X) \implies \Phi_{1-X}(a) \ge 1 - e(X) \qquad \left(0 < X < 1 - \frac{\sqrt{3}}{2}\right). \tag{87}$$

Indeed, for

$$p_X(z) = z^2 - (1-X)z + (1-X)^2 - (1-X)^4,$$

one has

$$p_X(X) = X(1 - 3X + 4X^2 - X^3) > 0.$$

Since $X < (1-X)/2$, this puts $X$ strictly below the smaller root $e(X)$. For $U = 2X/(1-2X)$, direct substitution gives

$$p_X(U) = -\frac{X^2}{(1-2X)^2}(4X^4 - 20X^3 + 37X^2 - 28X + 3).$$

The polynomial in parentheses is decreasing on $[0,\mu]$ and at the right endpoint equals $8217/64 - 74\sqrt{3} > 0$. Indeed, its derivative is at most $16/512 + 74/8 - 28 < 0$, and $(8217/64)^2 - 3\cdot 74^2 = 230001/4096 > 0$. Hence $p_X(U) < 0$, which puts $U$ strictly between the roots. Finally,

$$p_X(2X + 5X^2) = X^2(3X+4)(8X-1) \le 0,$$

and $X < 2X + 5X^2 < (1-X)/2$ on the stated range. Thus the trial point lies between the two roots, proving the third comparison. The implication (87) is read directly from (84). We shall also use $e(X) < 2X + 3X^2$ for $0 < X \le 1/10$, obtained from $p_X(2X+3X^2) = X^2(8X^2 + 19X - 2) < 0$; here too the trial point lies strictly between $X$ and $(1-X)/2$.

### 4.7. **Cyclic Propagation in the All-**Vd0 **Cases.**

4.7.1. *The exactly-one-supercritical all-*Vd0 *branch.* For a center interval, call the nonempty closed set missed by its two open endpoint triangles a *boundary gap*. It may be a singleton. If the maximal closed center trace is $[s,t]$, open coverage gives the weak bounds $B_i \ge s$ and $A_{i+1} \ge 1-t$; thus none of the arguments below discards a singleton gap.

**Proposition 4.26** (CE1 one-gap five-map obstruction)**.** *Assume that every V triangle is* Vd0*,* $N_+ = 1$*, and the C triangle is CE1. If its boundary trace contains a boundary gap, the triangles do not cover* $H$*.*

*Proof.* Use the signed center variables of Proposition 2.15, and put

$$w = W, \qquad A = \alpha, \qquad D = \delta, \qquad X = R - D, \qquad h_0 = \frac{\eta + A + D}{2R}.$$

The CE1 sign gives $D + RA \geq P$ and $RD > wA$, so

$$m_3 = \min\left\{\frac{A}{R}, \frac{D}{w}\right\} = \frac{A}{R}.$$

The exact strict domain needed below is

$$\begin{gathered} A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P, \\ A < \frac{w}{2}, \quad D < \frac{R}{2}, \quad RD - wA > 0, \end{gathered} \tag{88}$$

Proposition 4.10 contains the five actual-V triangle handoffs, the three duality steps, and the final diameter comparison. It remains only to prove its scalar target

$$\left(\Phi_{1-D} \circ \Phi_{1-m_3} \circ \Phi_{1-A}\right)(h_0) > 1 - X. \tag{89}$$

Here $A < P \leq \mu$, because $P = E(1-E)$ and $E \geq \sqrt{3}/2$. We first prove the two input comparisons used at V triangle 4. Since $D = R - X$, the inequality $A + wD < P$ and the identity $Rw = \eta + P$ give

$$A < wX - \eta.$$

Put

$$\varphi(q) = wq - \frac{q}{2(1+q)}.$$

The bound $0 < D < R/2$ gives $R/2 < X < R$, and $\varphi$ is convex. At the endpoints,

$$\varphi(R/2) < \frac{Rw}{2} < \eta,$$

while $\varphi(R) < \eta$ is equivalent to

$$E(1 - 2R^2) < 1,$$

which follows from $E < 1$. Convexity therefore gives $\varphi(X) < \eta$, and hence

$$A < \frac{X}{2(1+X)}.$$

Using the rational low-root bound in its valid range,

$$e(A) < \frac{2A}{1-2A} < X. \tag{90}$$

We also have

$$2R(h_0 - A) = \eta + D + (1-2R)A.$$

This is positive if $R \leq 1/2$. If $R > 1/2$, then $A < P = \eta E$ gives

$$2R(h_0 - A) > \eta\big(1 - (2R-1)E\big) > 0.$$

Thus $h_0 > A$. The positive signed right trace gives

$$\frac{\eta + A + D}{R} < w + D < 1,$$

so $h_0 < 1/2 < 1 - A$.

Now consult the exact high-radial catalog for $c_4 = 1 - A$. The lower and upper linear ranges would require $h_0 \le A$ and $h_0 \ge 1 - A$, respectively, so both are impossible. On the constant or $Q_-$ branch range,

$$\overline{M}_{1-A}(h_0) \le e(A),$$

and therefore

$$\Phi_{1-A}(h_0) \ge 1 - e(A) > 1 - X$$

by (90). Thus only the $Q_+$ range remains. Its selector gives

$$h_0 \le e(A) \le 2A + 5A^2.$$

We claim that this range forces $X > 1/2$. Indeed, if $X \le 1/2$, the selector and the lower center inequality give

$$2Rh_0 = \eta + A + D \ge \eta + P + wA = w(R + A).$$

Together with $h_0 < 2A/(1 - 2A)$, this implies

$$f_R(A) < 0, \qquad f_R(z) := w(R + z)(1 - 2z) - 4Rz.$$

But $f_R''(z) = -4w < 0$ and $f_R(0) = Rw > 0$. We check the other endpoint of the possible interval for $A$.

If $R \le 1/2$, then $0 < A < P$. Direct simplification gives

$$f_R(P) = (1 - E)\left(4E^3 - 2E^2 + 1 - R(2E^3 - 2E^2 + 5E)\right).$$

The coefficient of $R$ is positive, so

$$f_R(P) \ge \frac{1 - E}{2}\left(6E^3 - 2E^2 - 5E + 2\right).$$

For $\phi(E) = 6E^3 - 2E^2 - 5E + 2$ on $[\sqrt{3}/2, 1)$,

$$\phi'(E) = 18E^2 - 4E - 5 \ge \frac{17}{2} - 2\sqrt{3} > 0,$$

and

$$\phi\left(\frac{\sqrt{3}}{2}\right) = \frac{2 - \sqrt{3}}{4} > 0.$$

Hence $f_R(P) > 0$. Strict concavity puts $f_R$ above its endpoint chord, so $f_R(A) > 0$ for $0 < A < P$, a contradiction.

If $R > 1/2$, the assumption $X \le 1/2$ gives $D \ge R - 1/2$, and therefore

$$0 < A < A_*, \qquad A_* := P - w\left(R - \frac{1}{2}\right) = \frac{w}{2} - \eta.$$

Direct simplification gives

$$f_R(A_*) = \frac{1 - E}{2}(E + 11R - 3ER - 5).$$

Since $11 - 3E > 0$ and $R > 1/2$,

$$E + 11R - 3ER - 5 > \frac{11 - 3E}{2} + E - 5 = \frac{1 - E}{2} > 0.$$

Thus $f_R(A_*) > 0$. Concavity again gives $f_R(A) > 0$ on $(0, A_*)$, a contradiction. Both ranges are impossible, proving $X > 1/2$.

Put $p_1 = \Phi_{1-A}(h_0)$. If V triangle 3 is not on the $Q_+$ branch, then $p_1 \le A + e(A) < 3A + 5A^2 < 29/64 < 1/2$, and $2R(h_0 - m_3) = \eta + D - A > \eta - P > 0$.

Extensivity gives $p_1 \geq h_0 > m_3$, while $p_1 < 1/2 < 1 - m_3$; thus both linear ranges are impossible. On the constant, $Q_-$, or the low-$c$ tie one has $\overline{M}_{1-m_3}(p_1) \leq p_1$, so

$$\Phi_{1-m_3}(p_1) \geq 1 - p_1 > 1 - X.$$

Extensivity of the final reverse map $\Phi_{1-D}$ preserves this strict bound and proves (89). It remains to treat the selected V triangle-3 $Q_+$ branch. Proposition 4.8 proves once for all that every selected $Q_+$ arc is increasing and strictly concave.

When $d < 1-\sqrt{3}/2$, the arc endpoints are $(d, d)$ and $(e(d), d+e(d))$, so concavity gives

$$q \geq p + \frac{d}{e(d) - d}(p - d). \tag{91}$$

For $0 < d \leq \mu$, the valid rational low-root bound gives

$$\frac{d}{e(d) - d} > \frac{1 - 2d}{1 + 2d} > 1 - 4d > 1 - 5d.$$

This covers V triangle 4, because its parameter is $d = A < P \leq \mu$. For the V triangle-3 parameter $d = m_3$, which need not be at most $\mu$, the remaining high-radial range is handled without the rational estimate. If $\mu < d \leq 1/8$, then

$$\frac{d}{e(d) - d} \geq \frac{1}{1 + 5d} > 1 - 5d$$

by $e(d) \leq 2d + 5d^2$. If $1/8 < d < 1 - \sqrt{3}/2$, then $e(d) < (1 - d)/2$ and hence

$$\frac{d}{e(d) - d} > \frac{2d}{1 - 3d} > 1 - 5d.$$

The last comparison is equivalent to $15d^2 - 10d + 1 < 0$; its smaller root $(5-\sqrt{10})/15$ is less than $1/8$. In the low-$c$ V triangle-3 range $1 - \sqrt{3}/2 \leq d < 1/5$, the other endpoint is $(h(1 - d), 1 - h(1 - d))$, and the chord coefficient is

$$\frac{1 - 2h(1 - d)}{h(1 - d) - d}.$$

On $3/4 \leq c \leq \sqrt{3}/2$, direct differentiation of the formula for $h$ shows that $h(c)$ is nonincreasing. Moreover,

$$h\left(\frac{3}{4}\right) = \frac{3}{2(\sqrt{6} + 1)} < \frac{7}{16}.$$

Here low $c$ gives $d \geq 1 - \sqrt{3}/2 > 2/15 > 1/16$, and hence

$$h(1 - d) < \frac{7}{16} < \frac{3 + d}{7}.$$

Thus the displayed chord coefficient is greater than $1/3$, while $1 - 5d < 1/3$. For $d \geq 1/5$, extensivity alone is stronger than the V triangle-3 estimate.

Applying these chord bounds first with $d = A$ and then with $d = m_3$ proves, without an omitted selected branch,

$$p_1 > L_1 := (2 - 4A)h_0 - (1 - 4A)A,$$
$$p_2 := \Phi_{1-m_3}(p_1) > L_2 := (2 - 5m_3)L_1 - (1 - 5m_3)m_3.$$

Finally let

$$D_0 = P - RA, \qquad D_h = A(4R - 1) + 10RA^2 - \eta, \qquad x = Rw.$$

Feasibility puts $D_0 \le D \le D_h$. We first prove that the actual feasible value satisfies $D < 1/10$. Since

$$P = \frac{xE}{1+E} < \frac{x}{2}, \qquad \eta = \frac{x}{1+E} > \frac{x}{2},$$

the contrary assumption $D \ge 1/10$ and $A + wD < P$ give

$$A < \overline{A} := \frac{w(5R-1)}{10}.$$

The function $D_h(A)$ is increasing, so

$$D \le D_h(A) < D_h(\overline{A}) < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{x}{2}.$$

Direct expansion yields

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R), \qquad P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R-1 \in [0,1]$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \le -101y^2 + 124y - 39 < 0.$$

The last quadratic has discriminant $-380$. Hence $U(R) < 1/10$, a contradiction, and therefore $D < 1/10$.

It remains to compare $L_2$ with $2D + 3D^2$. Put

$$\chi(D) = L_2(D) - \frac{23}{10}D.$$

Since

$$L_2 - 2D - 3D^2 - \chi(D) = 3D\left(\frac{1}{10} - D\right) > 0,$$

it is enough to prove $\chi(D) > 0$. Exact differentiation gives

$$\chi'(D) = \frac{S(R,A)}{10R^2}, \qquad S(R,A) = 100A^2 - (40R+50)A + R(20-23R).$$

The inequality $D_0 \le D_h$ is equivalent to

$$x \le A(5R - 1 + 10RA).$$

Because $A < P < x/2 \le 1/8$,

$$A > L := \frac{4x}{25R-4}.$$

Also $\partial_A S = 200A - 40R - 50 < -45$, and direct substitution gives

$$S(R,L) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R-1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 \ge 812 > 0.$$

Thus $\chi'(D) < 0$ and $\chi(D) \ge \chi(D_h)$.

At $D = D_h$ one has $h_0 = 2A + 5A^2$. Write $L_{2,h} = L_2(D_h)$. Expansion gives

$$L_{2,h} - A(1+4R) = \frac{A}{R^2}\mathcal{Q}(R,A),$$

where

$$\begin{aligned}\mathcal{Q}(R,A) = 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A\\ - 4R^3 + 5R^2 - R.\end{aligned}$$

On $0 \le A \le x/2$,

$$\partial_A^2 \mathcal{Q} = 20R(30A - 4R - 3) < 0.$$

Its two endpoint values are

$$\mathcal{Q}(R,0) = x(4R-1) > 0, \qquad \mathcal{Q}\left(R, \frac{x}{2}\right) = \frac{x}{2}p(R),$$

where

$$p(R) = 25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3.$$

For $y = 2R - 1$,

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0,$$

because the terms of degree at least three are nonnegative and $106y^2 - 35y + 5$ has discriminant $-895$. Concavity therefore gives $L_{2,h} > A(1 + 4R)$.

On the other hand, $A < P = \eta E$ and $A < P < x/2$ imply

$$\frac{D_h}{A} = 4R - 1 + 10RA - \frac{\eta}{A} < C_* := 4R - 2 + 5Rx - \frac{x}{2}.$$

Here we used

$$\frac{1}{E} > 1 + \frac{x}{2}, \qquad (1-x)\left(1 + \frac{x}{2}\right)^2 = 1 - \frac{x^2(3+x)}{4} < 1.$$

Finally,

$$1 + 4R - \frac{23}{10}C_* = \frac{h_3(R)}{20}, \qquad h_3(R) = 230R^3 - 253R^2 - 81R + 112.$$

Since $h_3''(R) = 46(30R - 11) \ge 184$, strong convexity at $R_0 = 20/23$ gives

$$h_3(R) \ge \frac{788}{529} - \frac{(17/23)^2}{368} = \frac{289695}{194672} > 0.$$

Consequently

$$\chi(D_h) > A\left(1 + 4R - \frac{23}{10}C_*\right) > 0,$$

and hence

$$p_2 > L_2 > 2D + 3D^2.$$

The low root $e(D)$ is the smaller root of

$$P_D(z) = z^2 - (1-D)z + (1-D)^2 - (1-D)^4,$$

and direct substitution gives $P_D(2D + 3D^2) = D^2(8D^2 + 19D - 2) < 0$. Hence $e(D) < 2D + 3D^2 < p_2$. The one-hit part of Lemma 4.9, applied to the V triangle-2 map, gives

$$\Phi_{1-D}(p_2) \ge 1 - e(D) > 1 - X,$$

because

$$e(D) < 2D + 3D^2 < \frac{23}{100} < \frac{1}{2} < X.$$

This is (89). The actual-V triangle and final inequalities in Proposition 4.10 now give the contradiction. $\square$

**Proposition 4.27** (nonempty gap all-Vd0, $N_+ = 1$)**.** *If the C triangle is CE1 or CE2, every V triangle is* Vd0*, and $N_+ = 1$, then no cover of $H$ can have a boundary gap in the actual open V triangle traces.*

*Proof.* The midpoint argument used above makes $T_0$ uniquely supercritical. In an assumed cover, every boundary gap in the V triangle traces must lie in a center trace. CE1 therefore has the one-gap state excluded by Proposition 4.26. For CE2, the exactly-one-gap state is Proposition 4.12; the two-gap state is the common application in Proposition 4.22. This exhausts the one- and two-gap states, including singleton gaps. The zero-gap state is intentionally owned by the center-independent obstruction of Method 4. □

## 4.8. **Nonsupercritical, T3-like, and Vd1/Vd2 Reach Branches.**

### 4.8.1. *The exactly-one-$T$3 branch.*

**Proposition 4.28** (Exactly one T3-like V triangle)**.** *Assume the C triangle is CE1 or CE2, $N_+ = 1$, exactly one V triangle is T3-like, and no triangle is Vd1 or Vd2. Then the triangles do not cover $H$.*

*Proof.* By Proposition 2.12, normalize so that $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. A T3-like V triangle misses its own midpoint, a supercritical V triangle misses its local midpoint, and an open role $U_i$ whose closure $T_i$ is Vd0 cannot contain $M_{i-1}$ or $M_{i+1}$. Consequently, after reflection,

$$T_0 \text{ is T3-like}, \qquad M_1 \in T_0, \qquad T_1 \text{ is uniquely supercritical},$$

and $T_2, \dots, T_5$ are nonsupercritical Vd0 V triangles. The rational local calculation is Lemma 4.15. It verifies the two local inequalities in (78), while midpoint and support isolation force the center to cover the $O$-side radial gap. The common adjacent-exception obstruction, Corollary 4.18, therefore gives the contradiction. □

4.8.2. *The nonsupercritical CE1/CE2 boundary packages.* Figure 9 gives a common schematic for the package proved below.

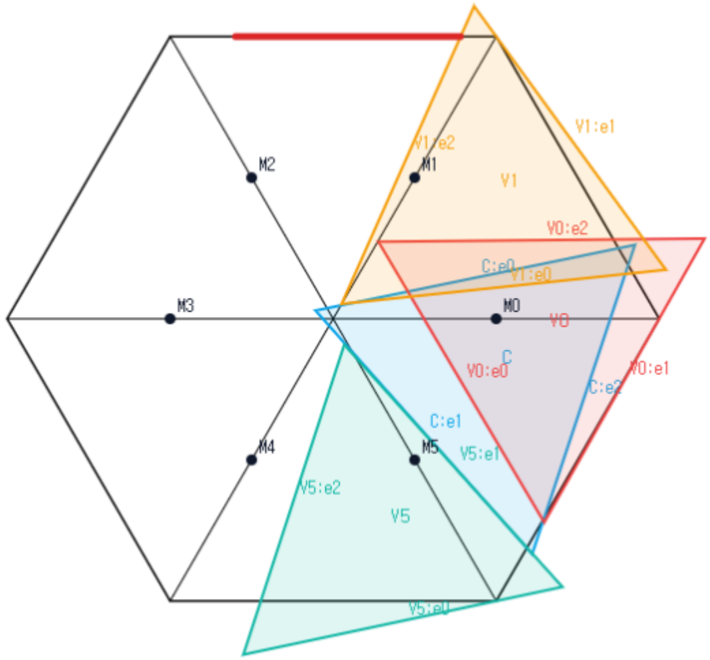



FIGURE 9. Schematic view of the common CE1/CE2, $N_+ = 0$, all-Vd0 boundary package, not to scale. The C triangle and three representative Vd0 vertex triangles are shown; the remaining three triangles are omitted for readability. The simultaneous placement is not a candidate cover, and no proof depends on visual inspection.

**Proposition 4.29** (All-Vd0 skeleton-data obstruction)**.** *Let the C triangle be CE1 or CE2. Assume all six V triangles are Vd0,*

$$A_i + B_i \le 1 \qquad (i = 0, \dots, 5),$$

*and the seven open triangles cover the full skeleton*

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

*Then no such configuration exists. Consequently the same distinguished-index data cannot cover all of $H$.*

*Proof.* Let $d_i^C$ be the center reach on $r_i$, measured from $O$, and set $c_i^C = 1 - d_i^C$. Since every V triangle is Vd0, $\mathcal{H}^1(T_{i-1} \cap r_i) = \mathcal{H}^1(T_{i+1} \cap r_i) = 0$; diameter locality excludes every $T_j$ with $j \notin \{i-1, i, i+1\}$. Skeleton coverage therefore forces

$$(*) \qquad C_i \ge c_i^C \qquad (i = 0, \dots, 5).$$

Let $N_{\mathrm{gap}}$ be the number of positive center traces containing a boundary gap. Singleton boundary gaps are included. The center classification gives $N_{\mathrm{gap}} \in \{0, 1, 2\}$, with $N_{\mathrm{gap}} \le 1$ in CE1.

If $N_{\mathrm{gap}} = 0$, the six open vertex traces cover every boundary edge. Thus

$$B_i > 1 - A_{i+1}.$$

Nonsupercriticality gives $1 - B_i \geq A_i$, while open boundary coverage gives

$$A_{i+1} > 1 - B_i \geq A_i,$$

which produces the impossible strict cycle

$$A_0 < A_1 < \cdots < A_5 < A_0.$$

Suppose $N_{\text{gap}} = 1$. Normalize the active trace to

$$I_R = \left[\frac{k}{R}, W + \delta\right] \subset e_{0,1}$$

and put

$$s = \frac{k}{R}, \qquad q = R - \delta.$$

Gap containment gives $B_0 \geq s$ and $A_1 \geq q$. The companion edge is absent or gap-free, so $A_0 + B_5 \geq 1$; since $A_0 + B_0 \leq 1$, one has $B_5 \geq s$. By $(*)$,

$$C_1 \geq 1 - \frac{\delta}{R}, \qquad C_5 \geq 1 - \frac{\alpha}{W}.$$

The exact one-side endpoint theorem gives

$$M_{1-\alpha/W}(s) + M_{1-\delta/R}(q) < 1.$$

The three internal V triangles $T_2, T_3, T_4$ lie on a center-free boundary path. The path budget forces their total boundary contribution above three, contrary to their three nonsupercritical caps.

If $N_{\text{gap}} = 2$, the center is CE2. The paired endpoint theorem, together with $(*)$, gives

$$p = W - \alpha, \qquad q = R - \delta, \qquad \overline{M}_{p/W}(p) + \overline{M}_{q/R}(q) < 1.$$

The same center-free three-V triangle path gives the contradiction.

The three ranks are exhaustive. □

We next treat the additional T3-like V triangles in the $N_+ = 0$ branch. The following local facts also reappear later.

**Lemma 4.30** (T3-like midpoint and side tradeoff)**.** *Let $T_i$ be T3-like. Then $M_i \notin T_i$. After applying the closed-trace domination of Proposition 2.7, suppose $\mathcal{H}^1(T_i \cap r_{i+1}) > 0$ and $M_{i+1}$ belongs to the dominated trace. If $b$ is its boundary reach toward $V_{i+1}$, $c$ its exit on $r_i$, and $\ell$ its entry on $r_{i+1}$, all measured from the relevant vertex, then*

$$b + c < 1, \qquad b + \ell > 1. \tag{92}$$

*Consequently two T3-like traces $T_i, T_{i+1}$ that cover each other's midpoint, with positive traces on $r_{i+1}$ and $r_i$, respectively, cannot cover both segments $[V_i, M_i]$ and $[V_{i+1}, M_{i+1}]$ if no other triangle has positive trace there.*

*Proof.* Apply Proposition 2.7 and reflect if necessary. Its translated Type-II majorant has parameters

$$0 < t < 1, \qquad z = \sqrt{1 - t + t^2}, \qquad \alpha = 0, \qquad 0 < \beta < z.$$

Its wedge vertex has

$$Q_x = \frac{z - t\beta}{1 - t + t^2} > 1.$$

Hence
$$t\beta < z - z^2.$$
The majorant's radial exit is $c = \beta/(1-t)$. Since
$$(1-z)(1+z) = t(1-t),$$
we obtain
$$c < \frac{z-z^2}{t(1-t)} = \frac{z}{1+z} < \frac{1}{2}.$$
Here $z < 1$ because $0 < t < 1$. Thus the translated majorant misses $M_0$, and its closed-trace domination implies that the original T3-like triangle misses $M_0$ as well.

For the tradeoff, the translated T3-like trace has the exhaustive Type-II form
$$\begin{aligned} (1-t)x + ty &\ge 0, \\ \beta + tx - y &\ge 0, \qquad 0 < t < 1, \quad z = \sqrt{1-t+t^2}, \quad 0 < \beta < z. \\ \beta + x - (1-t)y &\le z, \end{aligned}$$
For the selected support on $r_1$, substitution in the Type-II half-planes gives
$$b = z - \beta, \qquad c = \frac{\beta}{1-t}, \qquad \ell = \frac{\beta + 1 - z}{1-t}.$$
The condition $M_1 \in T$ is $\beta \le z - (1+t)/2$. Therefore
$$b + c \le z + \frac{t}{1-t}\left(z - \frac{1+t}{2}\right) = 1 - \frac{(1-z)^2}{2(1-t)} < 1,$$
whereas
$$b + \ell - 1 = \frac{t(1-z+\beta)}{1-t} > 0.$$
For a crossed adjacent pair, write $(b, c, \ell)$ for the first triangle's boundary reach, own-arm exit, and opposite-arm entry, and write $(\widetilde{b}, \widetilde{c}, \widetilde{\ell})$ for the reflected quantities of the second. Equation (92) and its reflection give
$$b + c < 1, \quad b + \ell > 1, \qquad \widetilde{b} + \widetilde{c} < 1, \quad \widetilde{b} + \widetilde{\ell} > 1.$$
Coverage forces $c \ge \widetilde{\ell}$ and $\widetilde{c} \ge \ell$. The first inequality gives $\widetilde{b} + c > 1 > b + c$, hence $\widetilde{b} > b$; the second gives $b + \widetilde{c} > 1 > \widetilde{b} + \widetilde{c}$, hence $b > \widetilde{b}$, a contradiction. □

**Lemma 4.31** (The support-isolated four-branch inequality)**.** *Assume the support-isolated T3-like endpoint state of Definition 4.21. Then the two endpoint V triangles satisfy*
$$\overline{M}_{c_5^*}(a_5^*) + \overline{M}_{c_1^*}(a_1^*) < 1, \tag{93}$$
*where $a_1^*, a_5^*$ are the exact boundary inputs and $c_1^*, c_5^*$ the exact radial lower bounds defined there.*

*Proof.* Use the CE1 side variables $0 < \lambda < 1$, $\rho = \sqrt{1-\lambda+\lambda^2}$ and center slacks
$$X = F_0(O), \qquad Y = F_2(O).$$
Then
$$a_C = \lambda - Y, \qquad \lambda s = X + Y + 1 - \rho,$$
and Proposition 2.15 gives
$$\rho - X - Y > \frac{1}{2}, \qquad \frac{Y}{\lambda} < \frac{1}{2}, \qquad \frac{X}{1-\lambda} < \frac{1}{2}.$$

Consequently the apparent minima in the two radial exits collapse to

$$\gamma_1 = \frac{Y}{\lambda}, \qquad \gamma_5 = \frac{X}{1-\lambda}.$$

For CE2 the second boundary interval is $[S, T]$, where

$$S = \frac{X+Y+1-\rho}{1-\lambda}, \qquad T = X + \lambda.$$

The T3-like normal form supplies $D > 1$, $0 < R < 1$, and $\alpha, p \geq 0$ with

$$R^2 - DR + D^2 = 1, \qquad \alpha + p = 1/D, \qquad q = 1 - p = \alpha + (D-1)/D < 1/2.$$

Its interval on $r_1$, measured from $V_1$, is

$$[c, u], \qquad c = Dq/R, \qquad u = q + (1-R)/D.$$

Using the residual operator (47), the exact boundary inputs are, with $J_1 = [s, t]$ and with $J_5 = [S, T]$ in CE2 but $J_5 = \varnothing$ in CE1,

$$a_1^* = \mathcal{R}_{J_1}(p), \qquad a_5^* = \mathcal{R}_{J_5}(\alpha),$$

and

$$c_5^* = 1 - \gamma_5, \qquad c_1^* = \begin{cases} c, & 1-u \leq \gamma_1 \leq 1-c, \\ 1-\gamma_1, & \text{otherwise}. \end{cases}$$

If $a_1^* + a_5^* > 1$, the two caps already imply $\overline{M}_{c_5^*}(a_5^*) + \overline{M}_{c_1^*}(a_1^*) < 1$. Assume the hard region $a_1^* + a_5^* \leq 1$. The following table is an exhaustive case analysis of the four possible first branches; the middle column states the exact decisive inequality.

| first branch | right branches | conclusion |
|---|---|---|
| Const | Lin | $e(\gamma_5) < a_1^*$ |
| Const | Const, $Q_-$ | $\overline{M}_{c_5^*}(a_5^*) + \overline{M}_{c_1^*}(a_1^*) < 1$ |
| Const | $Q_+$ | $e(\gamma_5) < a_1^*$ |
| $Q_-$ | Const, $Q_-$ | $\overline{M}_{c_5^*}(a_5^*) + \overline{M}_{c_1^*}(a_1^*) < 1$ |
| $Q_-$ | Lin, $Q_+$ | $\overline{M}_{c_5^*}(a_5^*) < a_1^*$ |
| Lin | all | impossible |
| $Q_+$ | Lin, $Q_+$ | impossible |
| $Q_+$ | Const, $Q_-$ | $\overline{M}_{c_5^*}(a_5^*) + \overline{M}_{c_1^*}(a_1^*) < 1.$ |

We verify the nonformal entries. Put $\kappa = (1-\rho)/(1-\lambda)$. From $X + (1-\lambda)Y < \rho(1-\rho)$ one obtains

$$\gamma_1 \geq a_C \Longrightarrow e(\gamma_5) < a_C, \tag{94}$$

$$S \leq Y/\lambda \Longrightarrow e(\gamma_5) < S. \tag{95}$$

For (94), one has $\gamma_5 < a_C - \kappa$ and $a_C - \kappa < 1/8$; then (86) applies, and the remaining comparison is

$$(a_C - \kappa) + 5(a_C - \kappa)^2 \leq \kappa.$$

After squaring its positive sides, the difference is $\lambda(23\lambda^3 + 58\lambda^2 + 7\lambda + 72) > 0$. For (95), combine the premise with the center inequality and put $x = \lambda$. This gives $0 < x < 3/8$ and

$$X < X_* = \frac{(1-\rho)(\rho(1-2x) - x(1-x))}{1-x-x^2}.$$

Set
$$\delta = \frac{1-\rho}{1-x} = \frac{x}{1+\rho}, \qquad \eta = \frac{X}{1-x}, \qquad f = \rho(1-2x) - x(1-x), \qquad d = 1-x-x^2.$$
Then $\gamma_5 \le \eta < \delta f/d$. Moreover
$$\frac{f}{d} < 1-2x,$$
because, after multiplication by positive denominators, this is just
$$\rho < d + \frac{x(1-x)}{1-2x} = 1 + \frac{2x^3}{1-2x}.$$
Hence $\eta < \delta(1-2x) < 1/8$. If $\eta = 0$, the assertion is immediate. Otherwise,
$$S \ge \frac{X+1-\rho}{1-2x} = \frac{1-x}{1-2x}(\eta+\delta) > 2\eta\left(\frac{1-x}{1-2x}\right)^2.$$
On the other hand,
$$5\eta < 5\delta(1-2x) < 2\left[\left(\frac{1-x}{1-2x}\right)^2 - 1\right].$$
For the middle sign use $\rho > 1-x$ and
$$2(2-3x)(2-x) - 5(1-2x)^3 = (4x-3)(10x^2-6x-1) > 0 \qquad (0 < x < 3/8).$$
Indeed both factors are negative; for the quadratic use $10x^2 \le 15x/4 < 6x+1$. The low-root estimate (86) now gives
$$e(\gamma_5) \le 2\gamma_5 + 5\gamma_5^2 \le 2\eta + 5\eta^2 < S,$$
proving (95) without any polynomial coefficient list. The constant-branch separation follows from its exact polynomial: if its output is $z = e(\eta)$ at input $1-\beta$, then
$$\beta \ge z + \eta.$$
Indeed the selector polynomial $P(x,z) = x^4 - 4x^3 + 5x^2 - (2+z)x + z - z^2$ is strictly decreasing on the low range and vanishes at $x = \eta$. Equations (94)–(4.8.2), with the three residual cases in $\mathcal{R}_J$, prove every first-constant entry. For a first $Q_-$ branch the exact root test
$$Q_-(a,c) < z \iff c^2(a+z)^2 - (a^2 - ac + c^2) > 0$$
is increasing in $z$. Substitution of $a_1^*$, followed by the center inequality, proves $\overline{M}_{c_5^*}(a_5^*) < a_1^*$ except in the low–low entries, which are immediate from $Q_-(a,c) \le \min\{a, 1-a\}$ and the constant-branch output is below $1/2$. First-coordinate linear would force either $X \ge (1-\lambda)\alpha$ and $Y \ge \lambda - \alpha$, contradicting $s \le p$, or, in the CE2 overlap case, $S \ge 1/2$, contradicting $S \le \alpha < 1/2$.

For a first $Q_+$ branch put $r = 1-\lambda$, $y = Y/\lambda$ and choose $\beta$ so that
$$m = \sqrt{\beta^2 - \beta + 1}, \quad 1 - \gamma_5 = \frac{m}{r+\beta}, \quad \overline{M}_{c_5^*}(a_5^*) = \frac{\beta m}{r+\beta}.$$
The exact selector is
$$\beta \ge \max\left\{r, \frac{1}{2}, \frac{1-r^2}{1+2r}\right\}, \qquad r \ge (1-\beta)(r+\beta)^2. \tag{96}$$

The center inequalities give $y < 1/10$ and, with

$$y_* = \frac{r}{1-r}\left(\frac{m}{r+\beta} - \frac{1}{2} - \frac{1-r}{1+\sqrt{r^2-r+1}}\right),$$

give $y < y_*$ and

$$3y_* \le 1 - \frac{\beta m}{r+\beta}.$$

This proves the right-constant entry by $e(y) < 3y$. It excludes right linear and right high by the exact high-input filter. For right $Q_-$ with input $a_C$, direct substitution gives a sum below one. For input $q$, put $c = 1-y$, $q = tc$, and let $\widehat{b}_1 = \overline{M}_{c_1^*}(q)$. The selected $Q_-$ equations give

$$\widehat{b}_1 \le \kappa := \frac{\sqrt{13}-1}{6}, \qquad q \le \tau < \frac{93}{200},$$

where $\tau \in (0, 1/2)$ is the unique root of $\tau^4 - 3\tau^3 + 3\tau^2 - 3\tau + 1 = 0$. In this CE2 overlap subcase the support ordering gives $q > S$. The common left-high identities are

$$S = S_0 + \frac{1-r}{r}y, \qquad S_0 = 1 - \frac{m}{r+\beta} + \frac{1-r}{1+\sqrt{r^2-r+1}}.$$

Since $y > 0$, we have the complete chain

$$S_0 < S < q \le \tau < \frac{93}{200}.$$

We now replace the former interval subdivision by a one-variable comparison. Write $s = r+\beta$ and $\rho_r = \sqrt{r^2-r+1}$. The selected Cell 2 inequality factors as

$$r - (1-\beta)(r+\beta)^2 = (s-1)(\beta^2 + r\beta - r).$$

The second factor is nonnegative because $r \le \beta$ and $\beta \ge 1/2$; if it vanishes, then $r = \beta = 1/2$. Hence $s \ge 1$. It follows that $\rho_r \le m$, since $\rho_r^2 - m^2 = (r-\beta)(s-1) \le 0$.

Define the one-variable lower bound

$$\overline{S}(\beta) = 1 - m + \frac{\beta}{1+m}.$$

Then

$$S_0 \ge 1 - \frac{m}{s} + \frac{1-r}{1+m} \ge \overline{S}(\beta).$$

Indeed, the second difference is

$$(s-1)\left(\frac{m}{s} - \frac{1}{1+m}\right) \ge 0.$$

For the last sign use $s \le 2\beta \le m(1+m)$: if $\gamma_\beta = 3\beta - \beta^2 - 1$, then $\gamma_\beta > 0$ and

$$m^2 - \gamma_\beta^2 = -\beta(\beta-1)(\beta^2 - 5\beta + 5) \ge 0, \qquad m^2 + \gamma_\beta = 2\beta.$$

Put $\beta_0 = 5657/10000$, $T = 93/200$, and $\beta_* = 161/250$. If $\beta m/s \ge \beta_0$, then $\beta m \ge \beta_0$. The function $\phi(\beta) = \beta m$ is strictly increasing, and the exact comparison

$$\beta_0^2 - \beta_*^2 m(\beta_*)^2 = \frac{22782769}{62500000000} > 0$$

therefore gives $\beta > \beta_*$. On $[\beta_*, 1]$ put

$$\chi(\beta) = \frac{107}{200} - Tm - (1-\beta)^2.$$

Since $\chi'' = -3T/(4m^3) - 2 < 0$, the function $\chi$ is concave. Its endpoint values are positive: $\chi(1) = 7/100$, while, for $a_* = 107/200 - (1-\beta_*)^2 = 51033/125000$,

$$a_*^2 - T^2 m(\beta_*)^2 = \frac{1693881}{62500000000} > 0.$$

Thus $\chi > 0$ throughout the interval. The identity

$$(\overline{S}(\beta) - T)(1+m) = \chi(\beta)$$

shows that $\beta m/s \geq \beta_0$ would force $S_0 > T$, a contradiction. Consequently

$$\overline{M}_{c_5^*}(a_5^*) < \frac{5657}{10000} < \frac{7-\sqrt{13}}{6}.$$

The final strict comparison follows by squaring $\sqrt{13} < 18029/5000$, whose squared difference is $44841/25000000 > 0$. Thus the final sum is also below one. The table is exhaustive, proving (93). ☐

**Proposition 4.32** (The $N_+ = 0$ T3-like branch)**.** *Assume that $T_C$ is CE1 or CE2, $N_+ = 0$, no V triangle is Vd1 or Vd2, and at least one V triangle is T3-like. Then the seven triangles do not cover $H$.*

*Proof.* By Proposition 2.12, normalize so that $T_C \cap \{M_0, \ldots, M_5\} = \{M_0\}$, and simultaneously apply the closed-trace domination of Proposition 2.7 to every T3-like triangle. If $T_0$ were not T3-like, a maximal consecutive block of T3-like indices among $1, \ldots, 5$ would have to match its triangles bijectively to the same block of uncovered midpoints. Each match sends $T_i$ only to $M_{i-1}$ or $M_{i+1}$, because a T3-like triangle misses $M_i$ and an open role $U_i$ whose closure $T_i$ is Vd0 cannot contain $M_{i-1}$ or $M_{i+1}$. Its first two triangles therefore form a crossed pair forbidden by Lemma 4.30. Hence $T_0$ is T3-like; reflect so that $\mathcal{H}^1(T_0 \cap r_1) > 0$.

There are at most two T3-like triangles. Indeed, with $N_+ = 0$ and no Vd1/Vd2 triangle, their number $k$ equals $N_{\mathrm{sp}}$. If $k \geq 3$, Theorem 3.14 gives an immediate full-skeleton contradiction. Midpoint locality then forces $T_2, T_4, T_5$ to be Vd0. Consequently $r_1$ has positive support only from $T_C, T_0, T_1$, and $r_5$ only from $T_C, T_5$.

Thus all hypotheses of the support-isolated state in Definition 4.21 hold. The actual forward reaches of $T_1, T_5$ are bounded by the two safe maps in Lemma 4.31; an extra reach lower bound on $r_0$ or $r_2$ for a possible T3-like $T_1$ only shrinks its feasible set. Thus their sum is less than one. V triangles $T_2, T_3, T_4$ must cover more than three units of the four intervening boundary edges, while their three nonsupercritical caps sum to at most three. This contradiction proves the proposition. ☐

4.8.3. *Two reusable high-radial endpoint inequalities.*

**Lemma 4.33** (One-side endpoint loss)**.** *Let $0 < R < 1$, $S = \sqrt{1 - R + R^2}$, and let $s, u, w > 0$ satisfy $s + u + w = 1$ and $0 < u < R$. Put*

$$\delta = \frac{R}{1+S}, \qquad \gamma = u - \delta - \frac{R}{1-R}w,$$

*and assume $\gamma > 0$. Set*

$$c_1 = u/R, \qquad c_5 = 1 - \gamma.$$

*Then*

$$\overline{M}_{c_5}(s) + \overline{M}_{c_1}(u) < 1. \tag{97}$$

*Proof.* Put $\eta = (1-R)\delta = 1-S$ and
$$\widehat{b}_5 = \overline{M}_{c_5}(s), \qquad \widehat{b}_1 = \overline{M}_{c_1}(u).$$
The center identity may be written
$$c_5 = \frac{1-u+\eta-Rs}{1-R}. \tag{98}$$
The hypotheses imply $1/2 < c_1, c_5 < 1$: for example $u > \delta$ gives $c_1 > 1/(1+S) > 1/2$, and $\gamma < u-\delta < R-\delta = RS/(1+S) < 1/2$ gives $c_5 > 1/2$. Thus the four high-radial branches of the four-branch high-radial map in Proposition 4.25 are exhaustive.

*Right low branches.* If the right branch is $Q_-$, its exact frontier equation gives $(u+\widehat{b}_1)^2 = 1-R+R^2 = S^2$, so $\widehat{b}_1 = S-u$. For a right constant branch write $\widehat{b}_1 = kc_1$, $0 \le k \le 1/2$, and $C = 1-k+k^2 = c_1^2$. With $X_R = R+k$, the constant-branch selector factors as
$$q = c_1^2(X_R-1)G_k(X_R), \qquad G_k(X) = CX^3 + CX^2 - k(1-k)X + k^2.$$
For $X \ge k$, $G_k(X) > 0$: when $k > 0$, $G_k'(X) \ge k(1+2k-k^2+3k^3) > 0$ and $G_k(k) = k^2(1+k+k^3) > 0$. Hence $q \le 0$ gives $X_R \le 1$. If $h_R = 1-X_R$, then
$$S^2 - (u+\widehat{b}_1)^2 = h_R(1+2k^2+k(1-k)h_R) \ge 0.$$
Consequently either right low branch satisfies
$$\widehat{b}_1 \le X := S-u. \tag{99}$$
*A left low branch against right high or linear.* For left $Q_-$ put $z = s/c_5 = 1-p$ and $h = \sqrt{1-p+p^2}$. The component selector gives $0 \le p \le 1/2$, the frontier gives $s+\widehat{b}_5 = h$, and its selector gives $s \le (1-p)h$. Equation (98) becomes $c_5(1-Rp) = 1-u+\eta$, whence
$$u \ge \eta + (1-h) + Rph.$$
Put $q_0 = ph$. If $q_0 > R$, then $p(1-p) > R(1-R)$ and the preceding lower bound would give $u > R$. Thus $q_0 \le R$, and
$$\eta + 1 - h > \frac{R(1-R)+p(1-p)}{2} \ge (1-R)q_0.$$
Substitution in the exact formula
$$w = \frac{(1-R)(1-u)p - \eta(1-p)}{1-Rp}$$
now gives $w < 1-h$. A right high output has the form $\omega_\beta - u$, where $\omega_\beta = \sqrt{1-\beta+\beta^2} \le 1$; for linear take $\omega_\beta = 1$. Hence
$$\widehat{b}_5 + \widehat{b}_1 = h + \omega_\beta - (s+u) = 1 + w - (1-h) - (1-\omega_\beta) < 1.$$
For a left constant branch write
$$c_5 = h(x) := \sqrt{1-x+x^2}, \qquad \widehat{b}_5 = xh(x), \qquad 0 \le x \le \frac{1}{2},$$
and, along the center line,
$$s(x) = \frac{R-u+\eta+(1-R)(1-h(x))}{R}.$$

If $y = s(x)/h(x)$, the exact selector factorization is

$$\frac{q}{h(x)^2} = (y - (1-x))P_x(y),$$

where

$$P_x(y) = (1 - x + x^2)y^3 + (1 + 2x - 2x^2 + 3x^3)y^2 \\ + (x + 2x^2 - x^3 + 3x^4)y + (x^2 + x^3 + x^5).$$

Every coefficient is nonnegative on $0 \le x \le 1/2$, and the polynomial is positive at every nondegenerate point. Thus a selected constant-branch point has $s(x) \le (1-x)h(x)$. The two curves meet once in $(0, 1/2)$: at $x = 0$ the left side is smaller, while at $x = 1/2$ it is enough to use

$$u \le \frac{R}{1+R} \quad \text{(Lin)}, \qquad u \le \min\{RS, S/2\} \quad (Q_+).$$

Writing $e_0 = 1 - \sqrt{3}/2$, the corresponding endpoint differences are

$$\frac{(1-R)(R^2 + 2Re_0 + 2e_0)}{2(1+R)} > 0$$

for linear, and respectively $[2e_0(1-R) - R^3]/2 \ge 0$ for $R \le 1/2$ and $(1-R)(3R + 4e_0 - 2)/4 \ge 0$ for $R \ge 1/2$ in the high case. At the unique transition $x_0$, the constant frontier is the $q = 0$ endpoint of the preceding $Q_-$ branch. Since $xh(x)$ increases, every selected left constant-branch output is no larger than that transition output. The preceding strict loss therefore also covers $(\mathrm{Const}, Q_+)$ and $(\mathrm{Const}, \mathrm{Lin})$.

*linear and high exclusions.* Left linear forces $s < 1/2$ and $\gamma \ge s$. Right linear or right high gives $u \le 1/2$, but

$$\gamma - s = 2u - 1 - \delta + \frac{1-2R}{1-R}w < 0:$$

for $R \ge 1/2$ this is immediate, while for $R < 1/2$ use $w < 1 - u$ to bound it by $(u-R)/(1-R) - \delta < 0$. Thus left linear cannot pair with either branch. If the left branch is high and the right one linear, then $s < 1/2$ and $u \le R/(1+R)$, whereas $\gamma > 0$ and $w > 1/2 - u$ force $u > R/2 + \eta > R/(1+R)$; the final inequality is equivalent to $2(1+R) > 1 + S$. This pair is also impossible.

*A right low branch.* Use (99). If $s > X$, the cap gives $\widehat{b}_5 + \widehat{b}_1 < 1$. Suppose $s \le X$ and put

$$c_X = X + 2\delta, \qquad f(c,z) = c^4 - c^2 + zc - z^2.$$

At the limiting value $u = R$, with $\kappa = \delta$, one has

$$X_0 = \frac{1-2\kappa}{1+\kappa}, \qquad c_0 = \frac{1+2\kappa^2}{1+\kappa}.$$

Now $c_0 > \sqrt{3}/2$ because $16\kappa^4 + 13\kappa^2 - 6\kappa + 1 > 0$, and

$$f(c_0, X_0) = \frac{\kappa^2(1-2\kappa)^2(4\kappa^4 + 4\kappa^3 + 10\kappa^2 + 6\kappa + 5)}{(1+\kappa)^4} > 0.$$

Along the $u$-line, $\frac{d}{du} f(c_X, X) = -4c_X^3 + c_X + X < 0$, so the same two signs hold before the limiting endpoint. Therefore, for $\ell(c) = c(1 - \sqrt{4c^2 - 3})/2$,

$$\ell(c_X) < X < c_X - \ell(c_X), \qquad \ell(c_X) < 1 - X.$$

Write $c = h(z) = \sqrt{1 - z + z^2}$ and let $z_X$ satisfy $h(z_X) = c_X$. The actual $c_5(s)$ corresponds to $0 < z \le z_X < 1/2$, and

$$s(z) = s_0 + \frac{1-R}{R}(1 - h(z)), \qquad s_0 = \frac{R - u + \eta}{R} > 0.$$

The transition input $(1 - z)h(z)$ decreases, and the preceding root ordering gives $s(z_X) < (1 - z_X)h(z_X)$, hence the same strict inequality for all $z \le z_X$. To check the order of the two boundary coordinates put

$$\phi(z) = s(z) - zh(z), \qquad k_R = \frac{1-R}{R}.$$

Its endpoint values are $s_0 > 0$ and $X - \ell(c_X) > 0$. Moreover

$$\phi'(z) = \frac{(k_R + z)(1 - 2z)}{2h(z)} - h(z).$$

For $k_R \le 2$ this is nonpositive; for $k_R > 2$ its zero is described by the strictly increasing function $K_0(z) = (2 - 3z + 4z^2)/(1 - 2z)$, so the minimum is again at an endpoint. Thus $zh(z) < s(z)$. Applying the already displayed polynomial $P_z$ gives the strict constant-branch selector, and $s + zh(z) \le X + \ell(c_X) < 1$. Since $\partial_b F_L = c(1 - 2z) > 0$ and each feasible fiber is an interval, the nonsupercritical maximum is

$$\widehat{b}_5 \le zh(z) \le z_X h(z_X) = \ell(c_X) < 1 - X.$$

Together with $\widehat{b}_1 \le X$, this proves the loss for every right low branch.

*High–high.* Let $c_L(z)$ be the unique root in $[\sqrt{3}/2, 1]$ of $f(c, z) = 0$. With $D_c = 4c^3 - 2c + z > 0$, implicit differentiation gives

$$c_L''(z) = \frac{2(c^2 - cz + z^2) + 16c^2 z(c - z)}{D_c^3} > 0.$$

A selected left high branch requires $s < 1/2$ and $c_5(s) \le c_L(s)$. Define

$$s_0 = \frac{R - u + \eta}{R};$$

then $0 < s_0 < s$ and $c_5(s_0) = 1$. The explicitly named function $\psi_L(z) = c_5(z) - c_L(z)$ is concave, with $\psi_L(s_0) > 0$. The right high bound $u \le \min\{RS, S/2\}$ gives $c_5(1/2) > 7/8$. For $R \le 1/2$ this follows after squaring from $17 - 10R + 9R^2 - 64R^3 - 64R^4 > 0$ (a decreasing polynomial with value 9/4 at 1/2); for $R \ge 1/2$ it follows from $(3 + R)^2 - 16S^2 = (1 - R)(15R - 7) > 0$. Finally

$$f(7/8, 1/2) = 33/4096 > 0$$

and $f$ increases in $c$ on this range, so $c_L(1/2) < 7/8$ and $\psi_L(1/2) > 0$. Concavity contradicts $\psi_L(s) \le 0$. Thus high–high is impossible.

If both branches are low, the cap already gives $\widehat{b}_5 + \widehat{b}_1 \le s + u = 1 - w < 1$. The preceding five paragraphs cover every other ordered pair of the four branches, proving (97). □

**Lemma 4.34** (CE2 two-endpoint loss)**.** *Let $[x, U]$ and $[y, v]$ be the two exact CE2 traces in the legacy coordinates of Remark 2.16, where $U$ denotes the endpoint called $u$ there. Put*

$$p = 1 - U, \qquad q = 1 - v, \qquad r = \frac{x}{x+y}, \qquad w = \frac{y}{x+y},$$

*and reflect if necessary so that* $0 < w \le 1/2$ *and* $r = 1 - w$. *Then*

$$\overline{M}_{p/w}(p) + \overline{M}_{q/r}(q) < 1. \tag{100}$$

*Proof.* Put

$$S_0 = x + y, \quad K = \sqrt{1 - wr}, \quad z = \frac{w}{1+K}, \quad \zeta = \frac{r}{1+K}, \quad \alpha = \frac{p}{w}, \quad \beta = \frac{q}{r}.$$

Then $D = S_0 K$, $1 - K = wr/(1+K)$, and the exact midpoint inequalities give $1/2 < \alpha, \beta < 1$. Dividing the coupling equation $(U + v)S_0 - xy = D$ by $S_0$, and using successively $x < U$ and $y < v$, gives the strict endpoint inequalities

$$\beta + p > 1 + z, \qquad \alpha + q > 1 + \zeta. \tag{101}$$

Define the local outgoing-cap functions

$$K_w(\alpha) = \overline{M}_\alpha(w\alpha), \qquad K_r(\beta) = \overline{M}_\beta(r\beta), \qquad h(t) = \sqrt{1 - t + t^2}.$$

Thus the desired sum is $K_w(\alpha) + K_r(\beta)$.

*Selected branch parametrizations.* For a side of weight $\sigma \in \{w, r\}$, every constant branch has

$$c = h(k), \qquad \overline{M}_c(\sigma c) = kh(k), \qquad 0 \le k \le w, \tag{102}$$

on either side. On the large side the range follows from

$$Q_{\mathrm{cell}}/h(k)^2 = (k - w)G_k(r + k),$$

where $G_k(X) = h(k)^2X^3 + h(k)^2X^2 - k(1-k)X + k^2 > 0$ for $X \ge k$. The remaining selected parametrizations are

$$\alpha = \frac{h(a)}{w+a}, \qquad K_w(\alpha) = a\alpha, \qquad r \le a < 1, \tag{103}$$

$$\beta = \frac{h(b)}{r+b}, \qquad K_r(\beta) = b\beta, \qquad r \le b < 1, \tag{104}$$

$$\beta = \frac{K}{r+t}, \qquad K_r(\beta) = t\beta = K - r\beta, \qquad w \le t \le r. \tag{105}$$

For example, the small-high selector factors as

$$h(a)^2 a\left(\frac{w}{(w+a)^2} - (1-a)\right), \qquad w - (1-a)(w+a)^2 = (a - r)(a^2 + wa - w),$$

which gives $a \ge r$; the other two factorizations are $(b-w)(b^2+rb-r)$ and $(t-w)(r^2-wt)$, respectively. These formulas specify the selected geometric components, not discarded algebraic roots.

The small-side $Q_-$ branch is absent for $w < 1/2$. Large-side linear is incompatible with the first inequality in (101). Small-side linear can pair only with the constant branch: either high or $Q_-$ on the large side has $\beta \le K$, which would force $p > (1+r)z > w/(1+w)$, contrary to linear. Thus, up to the symmetric $w = r = 1/2$ tie, the exact list is

$$\begin{gathered}(\mathrm{Lin}, \mathrm{Const}),\ (Q_+, \mathrm{Const}),\ (\mathrm{Const}, \mathrm{Const}),\ (\mathrm{Const}, Q_-),\\ (\mathrm{Const}, Q_+),\ (Q_+, Q_-),\ (Q_+, Q_+).\end{gathered} \tag{106}$$

Low–low sums are at most $K < 1$: a constant-branch output is at most $wK$, while (105) is at most $rK$. For linear–constant, put

$$b_{\mathrm{thr}} = 1 + z - \frac{w}{1+w}, \qquad \kappa = \frac{w}{1 + (1+w)z}.$$

linear and (101) give $\beta > b_{\mathrm{thr}}$ and $(1+\kappa)b_{\mathrm{thr}} = 1+z$. Direct reduction yields

$$b_{\mathrm{thr}}^2 - h(\kappa)^2 = \frac{w^2(A(w)+KC(w))}{D_0(w,K)} > 0,$$

where

$$\begin{aligned} A(w) &= 8 - 5w + 22w^2 + 6w^3 + 5w^4 + 5w^5 - w^6, \\ C(w) &= 8 - w + 14w^2 + 4w^3 - 2w^4 + w^5, \\ D_0(w,K) &= ((1+w)(1+K))^2(1+K+w+w^2)^2. \end{aligned}$$

Both numerator polynomials are positive on $[0,1/2]$. Since $h$ decreases and $(1+k)h(k)$ increases there, $\beta = h(k) > b_{\mathrm{thr}} > h(\kappa)$ implies $(1+k)\beta < 1+z$; combined with (101), this gives $k\beta < p$ and hence the strict linear–constant loss.

It remains to verify the four pairs containing a high branch on the small side.

$(\mathrm{Const}, Q_+)$. Use parameters $0 \le k \le w$ and $r \le t < 1$:

$$\alpha = h(k), \quad K_w = kh(k), \qquad \beta = \frac{h(t)}{r+t}, \quad K_r = \frac{th(t)}{r+t}.$$

The first output and the last output increase in their parameters, while $h(t)/(r+t)$ decreases. If $w \le 2/5$, then

$$K_w + K_r < wK + \frac{1}{2-w} < 1;$$

after using $K \le 1 - wr/2$, the positive cleared gap is $P(w) = w^4 - 3w^3 + 4w^2 - 6w + 2$, which decreases on $[0,2/5]$ and has $P(2/5) = 46/625$. If $w \ge 2/5$, the first endpoint inequality gives $\beta > r + z \ge 3/4$. Since $h(3/4)/(r+3/4) = \sqrt{13}/(7-4w) < 3/4$, monotonicity forces $t < 3/4$. Therefore

$$K_w + K_r < \frac{\sqrt{3}}{4} + \frac{3\sqrt{13}}{20} < 1.$$

$(Q_+, Q_-)$. Use $a, t$ from (103) and (105). Both reach lower bounds are at most $K$. The first endpoint inequality gives

$$\alpha > \frac{1+z-K}{w} = \frac{1+r}{1+K} =: a_0.$$

Thus $K > 1 - w^2$, which forces $w > 1/3$, and comparison at $a = 2/3$ forces $a < 2/3$. The function

$$G(a) = \frac{(1+a)h(a)}{w+a}$$

decreases on $[r, 2/3]$, because its derivative numerator $2a^3 + (4w-1)a^2 + (1-w)a + w - 2$ is at most its value $-7/54$ at $(2/3, 1/2)$. Hence $G(a) \le (1+r)K$. Using the second endpoint inequality and $K_r = K - r\beta$ gives

$$K_w + K_r < (2+r)K - 1 - \zeta < 1.$$

After multiplication by $1+K$, the last nonempty gap is $r(3w - w^2 - K)$; its squared comparison is

$$(3w - w^2)^2 - K^2 = w^4 - 6w^3 + 8w^2 + w - 1 > 0$$

for $w > 1/3$ (value $1/81$ at $1/3$, positive derivative thereafter).

$(Q_+, Q_+)$. Use parameters $a, b$ from (103)– (104). The endpoint inequalities first force $w > 2/5$: for $w \le 2/5$, the asserted contrary inequality $K(w+1/(2r)) \le 1+z$ reduces, after multiplication by $2r(1+K)$, to

$$K(2w^2 - 4w + 1) + 2w^4 - 4w^3 + w^2 - w + 1 > 0,$$

which is immediate according as the coefficient of $K$ is nonnegative or, using $K \le 1$, from the decreasing polynomial $2w^4 - 4w^3 + 3w^2 - 5w + 2 \ge 172/625$.

For the auxiliary function $\mathcal{G}_c(t) = (1+t)h(t)/(c+t)$, the derivative numerator $2t^3 + (4c-1)t^2 + (1-c)t + c - 2$ is increasing on the present rectangle, so $\mathcal{G}_c$ has at most one interior minimum. Endpoint comparison gives

$$K_w + \alpha \le \frac{2}{1+w}, \qquad K_r + \beta \le \frac{2}{1+r}.$$

The nontrivial endpoint signs are exactly

$$4 - (1+r)^2(1+w)^2K^2 = -w^2(w-1)^2(w^2 - w - 3) > 0$$

and

$$16r^2 - (1+r)^4h(r)^2 = (1-r)(r^5 + 4r^4 + 7r^3 + 9r^2 - 4r - 1) > 0;$$

the bracket is increasing on $[1/2, 1]$ and equals $13/32$ at $1/2$. Weighting the two strict endpoint inequalities by $w/K^2$ and $r/K^2$ gives

$$\alpha + \beta > \frac{2K+1}{K(1+K)}.$$

Consequently

$$K_w + K_r < \frac{2}{1+w} + \frac{2}{1+r} - \frac{2K+1}{K(1+K)} < 1.$$

The numerator of the last gap is $3Kwr - Q(w)$, where $Q(w) = w^4 - 2w^3 + 7w^2 - 6w + 2 > 0$. Its squared difference is

$$D(w) = -w^8 + 4w^7 - 9w^6 + 13w^5 - 41w^4 + 65w^3 - 55w^2 + 24w - 4.$$

Here $D'(w) = (1-2w)P_*(w)$ with

$$P_*(w) = 4w^6 - 12w^5 + 21w^4 - 22w^3 + 71w^2 - 62w + 24 > 0$$

on $[2/5, 1/2]$; use $71w^2 - 62w + 24 \ge 743/71$ and the remaining cubic block at least $-11/4$. Thus $D$ increases and $D(2/5) = 4904/390625 > 0$, proving the claim.

$(Q_+, \mathrm{Const})$. Use the fully specified parameters

$$\alpha = \frac{h(t)}{w+t}, \quad p = w\alpha, \quad b = t\alpha, \quad \frac{1}{2} \le t < 1,$$

and

$$\beta = \rho = h(k), \quad \widehat{b}_L = k\rho, \quad 0 \le k \le w.$$

Since $p + b = h(t)$, the first endpoint inequality gives

$$1 - (b + k\rho) > G := 2 + z - h(t) - (1+k)\rho. \tag{107}$$

Put

$$p_0 = 1 + z - K = \frac{w(2-w)}{1+K},$$

let $t_0 \in (1/2, 1)$ be the unique point with $p(t_0) = wh(t_0)/(w+t_0) = p_0$, and put $b_0 = t_0p_0/w$. The exact endpoint lemma is

$$b_0 < 1 - wK. \tag{108}$$

To verify it, set $b_* = 1 - wK$ and $T_* = wb_*/p_0$. One has $0 < T_* < 1$, and

$$p_0^2(w+T_*)^2 - w^2h(T_*)^2 = -\frac{w^3(KE(w)+J(w))}{(1+K)^2(2-w)^2} > 0,$$

where

$$E(w) = 2w^5 + 3w^3 - 20w^2 + 23w - 6,$$
$$J(w) = w^6 + 8w^5 - 22w^4 + 12w^3 + 6w^2 + 2w - 6.$$

Here $J \le -111/64$; if $E > 0$, then $KE + J < E + J < 0$, because $E + J$ is increasing and equals $-139/64$ at $w = 1/2$. Thus $p(T_*) < p(t_0)$. Since $p(t)$ decreases, $t_0 < T_*$ and (108) follows.

Finally put $P = 1 + z - p$. If $P < K$, then $t < t_0$ and $k \le w$, so

$$G > 2 + z - h(t_0) - (1+w)K = 1 - b_0 - wK > 0.$$

If $P \ge K$, then $K \le P \le P_1 := 1 + z - w/(1+w) \le 1$ and there is $a \in [0, w]$ with $h(a) = P$. Since $P < \rho$, one has $k < a$ and $(1+k)\rho < (1+a)P = P + \ell(P)$, where $\ell(P) = P(1 - \sqrt{4P^2 - 3})/2$. Hence

$$G > \mathcal{D}(P) := 1 - b(P) - \ell(P).$$

This function is concave. Indeed $\ell''(P) > 0$, and the high output $b$ is convex in $p = 1+z-P$: with $A_t = 2+w-(1+2w)t > 0$ and $Q_t = 8wt^2+4t^2-8wt-13t-w+4 < 0$,

$$\frac{d^2b}{dp^2} = -\frac{2h(t)(t+w)^3Q_t}{w^2A_t^3} > 0.$$

At $P = K$, (108) gives $\mathcal{D}(K) = 1-b_0-wK > 0$. At $P = P_1$, set $\kappa = w/(1+(1+w)z)$. The linear–constant comparison already proved gives $P_1 > h(\kappa)$, hence $\ell(P_1) < \kappa P_1 = w/(1+w)$, while $b(P_1) = 1/(1+w)$. Thus $\mathcal{D}(P_1) > 0$, and concavity gives $G > 0$ throughout. Equation (107) proves the final hard pair.

The exact list (106) is now exhausted, proving (100). □

4.8.4. *The exactly-one-Vd1/Vd2 placements.* We now assume CE2, $N_+ = 1$, and exactly one V triangle is Vd1 or Vd2. The corner normal form of Proposition 3.6 will be used in the following form: for some $t > 0$ and $d = \sqrt{t^2+t+1}$,

$$x - (t+1)y \le a, \qquad ty - (t+1)x \le tb, \qquad tx + y \le d - a - tb. \tag{109}$$

In particular $a + b < 1/2$ for the Vd1/Vd2 triangle in this corner normal form.

**Lemma 4.35** (Vd1 adjacent rescue)**.** *Assume an original open-triangle skeleton cover in the CE2 branch with $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$, with $T_0$ the unique Vd1 V triangle, $M_1 \in T_0$, $T_1$ uniquely supercritical, $T_2, \dots, T_5$ nonsupercritical Vd0, and*

$$\mathcal{H}^1(T_i \cap r_{i-1}) = \mathcal{H}^1(T_i \cap r_{i+1}) = 0 \qquad (i = 1, \dots, 5).$$

*Then the perimeter together with $r_1$ is not covered.*

*Proof.* Lemma 4.16 proves the two local inequalities in (78). Midpoint and support isolation force the center to cover the $O$-side radial gap before the Vd1 interval. The result is therefore Corollary 4.18. □

**Lemma 4.36** (Vd1-pair replacement interface)**.** *In the remaining adjacent Vd1–supercritical placement, the pair can be replaced by two open nonsupercritical Vd0 triangles while preserving full skeleton coverage.*

*Proof.* This is Proposition 4.38, which uses separate local charts at the two distinguished vertices and proves all reach, triangle, and skeleton-preservation claims explicitly. □

4.8.5. *Final boundary-propagation cases.*

**Proposition 4.37** (Cases excluded by boundary propagation)**.** *Boundary-reach propagation proves the following exclusions:*

(1) *CE1 or CE2, $N_+ = 0$, all V triangles Vd0;*
(2) *CE1 or CE2, $N_+ = 0$, with at least one T3-like triangle and no Vd1/Vd2;*
(3) *CE1 or CE2, $N_+ = 1$, all V triangles Vd0, with at least one boundary gap;*
(4) *CE1 or CE2, $N_+ = 1$, exactly one T3-like triangle and no Vd1/Vd2;*
(5) *the CE2, $N_+ = 1$, exactly-one-Vd1/Vd2 branch with no T3-like triangle.*

*All gap alternatives include singleton gaps.*

*Proof.* The five items are respectively Propositions 4.29, 4.32, 4.27, 4.28, and 4.39. The strictness at open handoffs appears in the weak endpoint bounds, the nonattained supercritical envelope, and the strict replacement margins; no positive-length assumption is made about a boundary gap itself. □

4.9. **The Vd1–supercritical replacement.** The complete four-branch T3 endpoint calculation is already Lemma 4.31; its authenticated source crosswalk is kept in the paper source ledger rather than printed a second time. What remains here is the unique two-chart replacement needed for the Vd branch.

**Proposition 4.38** (Vd1–supercritical two-chart replacement)**.** *Assume the CE2 branch in which an adjacent Vd1 triangle and the unique supercritical Vd0 triangle lie away from the V triangle at the center's unique midpoint. After reflection, write their base vertices as $V_\tau, V_{\tau+1}$. Assume the shared boundary edge is center-free, all other V triangles are nonsupercritical Vd0, and*

$$\mathcal{H}^1(T_j \cap r_\tau) = \mathcal{H}^1(T_j \cap r_{\tau+1}) = 0 \qquad (j \notin \{\tau, \tau+1\}).$$

*Then the pair can be replaced by two open nonsupercritical Vd0 triangles while preserving coverage of the full skeleton.*

*Proof.* After a fresh cyclic renumbering, let the Vd1 triangle be based at $V_0$, the supercritical triangle at $V_1$, and let the Vd1 triangle contain $M_1$. Write $a, b$ for the Vd1 boundary reaches and use the Vd corner parameter $t \geq 1$, with $d = \sqrt{t^2 + t + 1}$. Put

$$c = \frac{d - a - tb}{t+1}, \qquad \lambda = \frac{t(1-b)}{t+1}, \qquad u_{\text{adj}} = \frac{d - a - tb - 1}{t}.$$

The strict midpoint and one-sided-support inequalities give

$$a + c < 1, \qquad a < \lambda \leq \frac{1}{2}, \qquad b < \frac{1}{2}, \qquad u_{\text{adj}} < 1 - a. \tag{1}$$

Let $(A_1, B_1, C_1)$ be the maximal reaches of the supercritical triangle. The center-free shared edge and radial coverage give

$$A_1 \geq 1 - b > \frac{1}{2}, \qquad C_1 \geq \lambda.$$

The selected supercritical admissible cell has $C_1 \leq 1/2$. The half-square lemma

$$(x, y, z) \in \mathcal{A}, \qquad x \geq \frac{1}{2}, \qquad z \leq \frac{1}{2} \quad \Longrightarrow \quad y \leq 1 - z$$

therefore gives $B_1 \le 1 - C_1 \le 1 - \lambda$, and hence

$$a + B_1 < 1. \tag{2}$$

If $c_1^{\mathrm{req}} = 1 - d_1^C$ is the vertex-side beginning of the center trace on $r_1$, open skeleton coverage gives

$$c_1^{\mathrm{req}} < \max\{C_1, u_{\mathrm{adj}}\} \le \max\{1 - B_1, 1 - a\}. \tag{3}$$

Choose $p_2 \in (a, 1 - B_1)$ with

$$\max\{p_2, 1 - p_2\} > c_1^{\mathrm{req}};$$

such a point exists because the supremum of the left side on that interval is $\max\{1 - B_1, 1 - a\}$. Then choose

$$a < p_1 < p_2, \qquad p_1 < \frac{1}{2}, \qquad 1 - p_1 > c.$$

Finally choose $\varepsilon > 0$ smaller than

$$p_1 - a, \quad p_2 - p_1, \quad 1 - B_1 - p_2, \quad 1 - p_1 - c, \quad \max\{p_2, 1 - p_2\} - c_1^{\mathrm{req}}. \tag{4}$$

Use two distinct local charts:

$$X_0(x, y) = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$
$$X_1(x, y) = V_1 + x(V_0 - V_1) + y(V_2 - V_1).$$

For $p \le 1/2$ put

$$\Delta_p^- = \mathrm{conv}\{(0, 1 - p), (1, 1 - p), (0, -p)\},$$

and for $p > 1/2$ put

$$\Delta_p^+ = \mathrm{conv}\{(p, 0), (p, 1), (p - 1, 0)\}.$$

Define

$$U_0' = X_0\big(\mathrm{int}(\Delta_{p_1}^-) + (-\varepsilon, 0)\big)$$

and

$$U_1' = \begin{cases} X_1\big(\mathrm{int}(\Delta_{p_2}^-) + (-\varepsilon, 0)\big), & p_2 \le 1/2, \\ X_1\big(\mathrm{int}(\Delta_{p_2}^+) + (0, -\varepsilon)\big), & p_2 > 1/2. \end{cases}$$

In its own chart the shifted minus template contains the origin and has reach triple

$$(p - \varepsilon, 1 - p, 1 - p),$$

while the shifted plus template contains the origin and has reach triple

$$(p, 1 - p - \varepsilon, p).$$

In the $V_i$ chart, $i = 0, 1$, each closure stays strictly below the two supporting lines of $r_{i-1}$ and $r_{i+1}$; hence

$$\mathcal{H}^1(\overline{U_i'} \cap r_{i-1}) = \mathcal{H}^1(\overline{U_i'} \cap r_{i+1}) = 0.$$

Thus $V_0 \in U_0'$, $V_1 \in U_1'$, both triangles are Vd0, and both boundary sums equal $1 - \varepsilon < 1$.

The first replacement reaches beyond $a$ on $e_{5,0}$ and beyond $c$ on $r_0$. On the shared edge its reach is $1 - p_1$, while the second replacement reaches at least $p_2 - \varepsilon$ from $V_1$; their sum exceeds one by (4). The second replacement reaches beyond $B_1$ on $e_{1,2}$ and has radial reach $\max\{p_2, 1 - p_2\} > c_1^{\mathrm{req}}$ on $r_1$. Hence every boundary and radial component formerly supplied by the special pair remains covered, and all other skeleton components are unchanged.

The modified triangles therefore cover the full skeleton and have six nonsupercritical Vd0 V triangles. Proposition 4.29 gives the contradiction. □

4.10. **Authoritative CE2 One-Vd Placement Assembly.** The following proposition is the sole CE2 exactly-one-Vd placement assembly used by the final proof. It cites only complete placement lemmas and the explicit two-chart replacement in Subsection 4.9.

**Proposition 4.39** (Complete CE2 exactly-one-Vd1/Vd2 assembly)**.** *Assume the center has type CE2, $N_+ = 1$, exactly one V triangle has type Vd1 or Vd2, and no T3-like triangle is present. Then the seven triangles do not cover $H$.*

*Proof.* By Proposition 2.12, normalize so that $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$. If, for an additional V triangle $T_i$, there is some $j \in \{i-1, i+1\}$ with $\mathcal{H}^1(T_i \cap r_j) > 0$, then the unique supercritical V triangle, the Vd1/Vd2 V triangle, and that additional triangle give $N_+ + N_{\mathrm{sp}} \geq 3$; the skeleton contradiction of Theorem 3.14 applies. Hence assume the no-additional-positive-support complement.

First suppose the unique supercritical V triangle is $T_0$. If the Vd1/Vd2 V triangle is adjacent, it is eliminated by Lemma 4.19; for the $T_1$ placement its proof treats $\mathcal{H}^1(T_1 \cap r_0) > 0$ and $\mathcal{H}^1(T_1 \cap r_2) > 0$, as well as the reflected $T_5$ placement. If it is nonadjacent, its index lies in $\{2, 3, 4\}$ after reflection, and Lemma 4.20 gives a strict radial separation.

Next suppose the unique Vd1/Vd2 V triangle is $T_0$. It must rescue the midpoint of the unique supercritical V triangle, so midpoint locality makes that V triangle adjacent. If $T_0$ is Vd2, the $M_1$ or $M_5$ cap Lemma 3.7, combined with the center and supercritical perimeter caps, is Proposition 3.9. If $T_0$ is Vd1, the exact Vd1 local profile and the common center-hiding theorem give Corollary 4.18.

Finally suppose neither special V triangle is $T_0$. Since $T_C \cap \{M_0, \dots, M_5\} = \{M_0\}$, the Vd1/Vd2 V triangle must rescue the midpoint missed by the unique supercritical V triangle. Diameter locality makes the two V triangles adjacent. A Vd2 rescue is again eliminated by the $M_{i-1}/M_{i+1}$ cap. For a Vd1 rescue, the shared edge has no center trace in this placement and the no-additional-support hypotheses are exactly those of Proposition 4.38. Replace the pair by two open nonsupercritical Vd0 triangles while preserving every boundary and radial reach lower bound used by Proposition 4.29. All other vertex triangles were already nonsupercritical Vd0, so that all-Vd0 theorem gives the contradiction.

These alternatives exhaust the relative positions of the center-midpoint V triangle, the unique supercritical V triangle, and the unique Vd1/Vd2 V triangle. □

## 5. Area-Loss Estimate

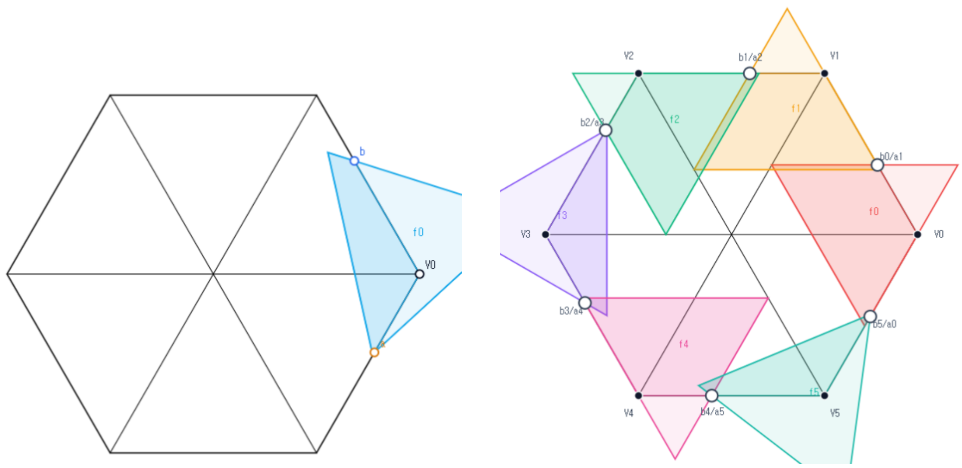
The third mechanism measures the normalized area forced outside $H$ by a V triangle with prescribed boundary reaches. The local square-loss and T3-like estimates are proved first, then summed around the six-cycle to close the two CE0 area branches.

5.1. **Detailed Area-Loss Estimates.** This section proves the local inequalities and cyclic estimates used in the area-loss argument.

Let

$$A_{\triangle} = \frac{\sqrt{3}}{4}$$

be the area of a unit equilateral triangle. Throughout this section areas are divided by $A_\triangle$. Thus a unit triangle has normalized area one and $H$ has normalized area six.




(a) local retained area (b) global retained areas

FIGURE 10. Schematic views of Method 3, not to scale. The local panel shows one V triangle with boundary reaches $a, b$, and the global panel shows the six cyclic triangles. Only the darker portions represent area retained inside $H$; the proof uses the inequalities below, not the picture.

5.1.1. *The local square-loss inequalities.* Fix a vertex $V_i$. Let $P_a$ and $P_b$ be the points at distances $a$ and $b$ from $V_i$ on its incoming and forward boundary edges. For a closed unit equilateral triangle $T$ containing $V_i, P_a, P_b$, put

$$\mathcal{L}(T) = \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

Equivalently, if

$$f(a,b) = \max_T \frac{\operatorname{area}(T \cap H)}{A_\triangle}, \qquad \mathcal{L}(a,b) = 1 - f(a,b),$$

then $\mathcal{L}(a,b)$ is the least possible normalized loss. Reflection in the radial axis interchanges $a$ and $b$, and increasing either reach lower bound can only decrease $f$.

**Lemma 5.1** (Local wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x,y) : x \geq 0,\ y \geq 0\}.$$

*If a closed unit equilateral triangle $T$ contains $V_0$, then*

$$T \cap H = T \cap W.$$

*Normalized Euclidean area is twice the corresponding $(x,y)$-coordinate area.*

*Proof.* The two basis vectors have length one and angle $120°$, so

$$\|(x,y)\|^2 = x^2 + y^2 - xy, \qquad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \le x, y \le 2,\ |x - y| \le 1\} \subset W.$$

Conversely, if $(x, y) \in T \cap W$, then the diameter-one condition gives $x^2 + y^2 - xy \le 1$. Hence

$$|x - y|^2 \le x^2 + y^2 - xy \le 1, \qquad \frac{3}{4}x^2 \le x^2 + y^2 - xy,$$

and the same estimate holds for $y$. Thus $|x - y| \le 1$ and $0 \le x, y \le 2/\sqrt{3} < 2$, proving $(x, y) \in H$. The determinant identity proves the area assertion. □

**Theorem 5.2** (Unconditional local square loss)**.** *For every feasible pair* $0 \le a, b \le 1$ *and every realizing closed unit equilateral triangle* $T$*,*

$$\mathcal{L}(T) \ge \min(a, b)^2. \tag{110}$$

*If* $a + b > 1$*, then*

$$\mathcal{L}(T) \ge \max(a, b)^2. \tag{111}$$

*Consequently the same inequalities hold for* $\mathcal{L}(a, b) = 1 - f(a, b)$*.*

*Proof.* In the coordinates of Lemma 5.1, physical rotation through $60°$ is $\mathsf{R}(x, y) = (x - y, x)$. Put

$$0 \le t \le 1, \qquad D = 1 - t + t^2, \qquad z = \sqrt{D}.$$

After choosing one of the six directed sides, every orientation has one of the following two forms:

$$\begin{array}{c|cc} & d & \mathsf{R}d \\ \hline \mathrm{I} & z^{-1}(1, t) & z^{-1}(1 - t, 1) \\ \mathrm{II} & z^{-1}(t, -(1 - t)) & z^{-1}(1, t). \end{array} \tag{112}$$

Indeed these forms cover, respectively, the physical direction sectors $[120°, 180°]$ and $[60°, 120°]$; their endpoints cover the common boundary orientations.

For Type I introduce

$$U = \alpha + y - tx, \qquad V = \beta + x - (1 - t)y.$$

Then $T$ is the simplex $U, V \ge 0$, $U + V \le z$. Containment of $(0, 0), (0, a), (b, 0)$ gives

$$\alpha \ge tb, \quad \beta \ge (1 - t)a, \quad \alpha + \beta + (1 - t)b \le z, \quad \alpha + \beta + ta \le z. \tag{113}$$

The two inequalities defining $W$ become

$$(1 - t)U + V \ge (1 - t)\alpha + \beta, \qquad U + tV \ge \alpha + t\beta.$$

For fixed $t$, lowering $\alpha$ or $\beta$ therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1 - t)a$. At that placement the portion outside $W$ has vertices, in the $(U, V)$-plane,

$$(0, 0), \quad (a + tb, 0), \quad (tb, (1 - t)a), \quad (0, b + (1 - t)a).$$

Since the Jacobian of $(x, y) \mapsto (U, V)$ has absolute value $D$,

$$\mathcal{L}(T) \ge \mathcal{L}_{\mathrm{I}}(a, b; t) = \frac{(1 - t)a^2 + tb^2 + 2t(1 - t)ab}{D}. \tag{114}$$

If $a \le b$, then

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1 - t)(a^2 + 2ab)\} \ge 0;$$

the reflected factorization proves $\mathcal{L}_{\mathrm{I}} \ge b^2$ when $b \le a$. This proves (110) in Type I.

Suppose now that $a+b>1$ and $a\le b$. Set
$$r=\frac{a}{b},\qquad t_0=\frac{1-r^2}{1+2r}.$$
Since $b>1/(1+r)$, feasibility in (114) and (113) excludes $0\le t<t_0$: for $0<t<t_0$,
$$\left(\frac{1+r(1-t)}{1+r}\right)^2-D=\frac{t(1+2r)(t_0-t)}{(1+r)^2}>0,$$
so $b+(1-t)a>z$, while $t=0$ would give $a+b\le 1$. Hence $t\ge t_0$, and
$$D(\mathcal{L}_{\mathrm{I}}-b^2)=(1-t)b^2\{r^2-1+t(2r+1)\}\ge 0.$$
If $b\le a$, put $r=b/a$ and $t_1=(r^2+2r)/(1+2r)$. The identical reflected calculation excludes $t>t_1$ and gives
$$D(\mathcal{L}_{\mathrm{I}}-a^2)=ta^2\{r^2+2r-t(1+2r)\}\ge 0.$$
Thus (111) holds in Type I.

For Type II put
$$U=\alpha+(1-t)x+ty,\qquad V=\beta+tx-y.$$
Again $T$ is the simplex $U,V\ge 0$, $U+V\le z$. Its exact reaches on the positive $y$- and $x$-axes are
$$A=\beta,\qquad B=z-\alpha-\beta,\qquad A+B\le z\le 1. \tag{115}$$
In particular Type II is impossible when $a+b>1$. Its wedge intersection is the quadrilateral with vertices
$$(0,0),\quad (B,0),\quad \left(\frac{B+(1-t)A}{D},\frac{A+tB}{D}\right),\quad (0,A).$$
Hence
$$\mathcal{L}(T)=\mathcal{L}_{\mathrm{II}}(A,B;t)=1-\frac{(1-t)A^2+tB^2+2AB}{D}. \tag{116}$$
Let $S=A+B$. If $A\le B$, write $A=rS$, $B=(1-r)S$ with $0\le r\le 1/2$. Since $S^2\le D$, and with
$$C=(1-t)r^2+t(1-r)^2+2r(1-r),$$
we have $\mathcal{L}_{\mathrm{II}}\ge 1-C$. The exact factorization
$$1-C-r^2D=(1-t)(1-2r+tr^2)\ge 0$$
gives $\mathcal{L}_{\mathrm{II}}\ge r^2D\ge A^2$. Reflection gives $\mathcal{L}_{\mathrm{II}}\ge B^2$ when $B\le A$. Since $A\ge a$ and $B\ge b$, this proves (110). The two orientation types are exhaustive, so the theorem follows. $\square$

5.1.2. *The direct* T3-like *loss.* The next theorem uses the classification data $(o,n)$ from Section 2.1: $o$ counts triangle vertices outside $H$, and $n=n(T_i)$ is
$$\left|\left\{j\in\{i-1,i+1\}:\mathcal{H}^1(T_i\cap r_j)>0\right\}\right|.$$
Thus a T3-like triangle has $(o,n)=(2,1)$.

**Theorem 5.3** (Direct T3-like loss). *Let $T$ be a closed unit $V_i$-triangle of* T3-like *type, and let $a, b$ be lower bounds for its adjacent boundary reaches. Then*

$$a + b \leq 1. \tag{117}$$

*Moreover, for $0 \leq m \leq 1/2$,*

$$a, b \geq m \implies \mathcal{L}(T) \geq 2m - 4m^2. \tag{118}$$

*Proof.* The first assertion is Lemma 2.9. For the loss bound apply the closed-trace domination in Proposition 2.7. It gives a translated Type-II triangle $\widehat{T}$ of the same area with

$$T \cap H \subset \widehat{T} \cap H, \qquad \mathcal{L}(T) \geq \mathcal{L}(\widehat{T}).$$

In the notation of that proposition, $0 < t < 1$, $D = 1 - t + t^2$, $z = \sqrt{D}$, and after reflection the translated majorant has exact reaches

$$A = \beta, \qquad B = z - \beta$$

and unique wedge vertex

$$Q = \left( \frac{B + (1-t)A}{D}, \frac{A + tB}{D} \right), \qquad Q_x > 1, \quad Q_y \leq 1.$$

Thus $z - tA > D$, so

$$A < L := \frac{z - D}{t} = \frac{z(1-z)}{t}. \tag{119}$$

The exact normalized loss is

$$\mathcal{L}(\widehat{T}) = \mathcal{L}(z, A) = \frac{tA^2 + (1-t)(z-A)^2}{D}.$$

This decreases for $A < L < (1-t)z$, and therefore

$$\mathcal{L}(T) \geq \mathcal{L}(\widehat{T}) > \mathcal{L}(z, L). \tag{120}$$

Set $r = (1-z)/t$. Then $0 < r < 1/2$, and eliminating $t, z$ gives

$$t = \frac{1-2r}{1-r^2}, \qquad z = \frac{r^2 - r + 1}{1 - r^2},$$

$$L = \alpha_{\text{lim}}(r) := \frac{r(r^2 - r + 1)}{1 - r^2}, \qquad z - L = \beta_{\text{lim}}(r) := \frac{r^2 - r + 1}{1 + r}.$$

The limiting normalized inside area and loss are

$$F_0(r) = \frac{(r^2 - r + 1)^2}{1 - r^2}, \qquad \mathcal{L}_0(r) = 1 - F_0(r).$$

Since the exact reach $A$ is at least the required reach lower bound $a \geq m$, (119) gives $\alpha_{\text{lim}}(r) > m$. Direct simplification yields

$$\mathcal{L}_0(r) - \{2\alpha_{\text{lim}}(r) - 4\alpha_{\text{lim}}(r)^2\} = \frac{r^2(5r^4 - 8r^3 + 13r^2 - 8r + 2)}{(1-r^2)^2} \geq 0,$$

because the last polynomial is at least $9r^2 - 8r + 2 \geq 2/9$ on $[0, 1/2]$. Also $\alpha_{\text{lim}}(r) \leq r$, and

$$\mathcal{L}_0(r) - \frac{1}{4} = \frac{(1-2r)(2r^3 - 3r^2 + 6r - 1)}{4(1-r^2)} \geq 0 \quad \text{when } \alpha_{\text{lim}}(r) \geq \frac{1}{4};$$

the cubic is increasing and has value $11/32$ at $r = 1/4$.

Finally $\psi(u) = 2u - 4u^2$ is increasing on $[0, 1/4]$ and never exceeds $1/4$ on $[0, 1/2]$. If $\alpha_{\lim}(r) \le 1/4$, then $\mathcal{L}_0(r) \ge \psi(\alpha_{\lim}(r)) \ge \psi(m)$; if $\alpha_{\lim}(r) \ge 1/4$, then $\mathcal{L}_0(r) \ge 1/4 \ge \psi(m)$. Together with (120), this proves (118). The reflected crossing is identical. □

5.1.3. *The cyclic area certificates.* The selected handoffs of Proposition 2.10 have the form

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1,$$

with cyclic indices. V triangle $i$ is selected supercritical exactly when $\xi_i > \xi_{i-1}$.

**Lemma 5.4** (Cyclic area-loss certificate)**.** *Let the six actual V triangles realize selected handoffs*

$$(a_i, b_i) = (1 - \xi_{i-1}, \xi_i), \qquad 0 < \xi_i < 1.$$

*Their total normalized loss is strictly greater than one if either*

(1) *at least two selected V triangles are supercritical; or*
(2) *exactly one selected V triangle is supercritical, every triangle is* Vd0 *or* T3-like*, and at least one triangle is* T3-like*.*

*Proof.* Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves both alternatives and the total loss. Hence assume $m \le 1 - M$. Every selected V triangle coordinate is then at least $m$, so (110) gives loss at least $m^2$ for every V triangle.

In case (1), choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1 - m)^2$ by (111). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$. The other four V triangles lose at least $m^2$ each, so the total loss is strictly larger than

$$4m^2 + (1 - m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} > 1.$$

In case (2), rotate so that the unique ascent leaves the minimum: $\xi_5 = m < \xi_0 = M$. The supercritical V triangle loses at least $(1 - m)^2$. Lemma 2.9 shows that a T3-like V triangle is different from it, and Theorem 5.3 gives that V triangle loss at least $2m - 4m^2$. The remaining four V triangles lose at least $m^2$ each. Their total loss is at least

$$(1 - m)^2 + (2m - 4m^2) + 4m^2 = 1 + m^2 > 1.$$

□

**Proposition 5.5** (Branches closed by area)**.** *The following branches are impossible:*

(1) $T_C$ *is* CE0*,* $N_+ \ge 2$*, with arbitrary V triangle types;*
(2) $T_C$ *is* CE0*,* $N_+ = 1$*, no triangle is* Vd1 *or* Vd2*, and at least one triangle is* T3-like*.*

*Proof.* In the CE0 class, the C triangle has no positive-length boundary trace. If the six V triangles missed a boundary point, openness of $U_C$ would give such a trace. Hence the V triangles cover $\partial H$, so Proposition 2.10 applies.

If $N_+ \ge 2$, its at-least-two clause selects handoffs with at least two supercritical V triangles. Lemma 5.4 makes the six vertex triangles' total normalized inside area strictly less than five. The center triangle contributes at most one, so the total is strictly less than six.

If $N_+ = 1$, the exact-one clause preserves the unique actual supercritical index for every strict selection. The exhaustive vertex-type hypothesis gives case (2) of Lemma 5.4; the six V triangles contribute less than five normalized area, and the C triangle contributes at most one. Again the sum is strictly below the normalized area six of $H$. $\square$

## 6. Nine-Point Enclosure Obstruction

The fourth mechanism applies when all six V triangles are Vd0, the boundary is covered, and exactly one V triangle is supercritical. Six radial points and three asymmetric frontier points are forced into the center triangle. The detailed construction is followed by the support-arc covering argument and the final enclosure contradiction.

### 6.1. Detailed Nine-Point Enclosure Argument.

This section supplies the exact frontier, witness, sign, and cap-overlap calculations for the direct nine-point obstruction.

This section proves a center-class-independent theorem. Its hypotheses are that all six V triangles are Vd0, that they cover the full boundary, and that exactly one actual boundary V triangle is supercritical. The theorem will therefore apply both to the CE0 branch and to the zero-gap subcases of CE1 and CE2. Six common radial points and three asymmetric points are excluded directly from all six actual V triangles and are therefore forced into the C triangle. Only the unique supercritical V triangle uses an $AB$-union; no nonsupercritical model core is introduced.

6.1.1. *The finite caliper criterion.* We first state the finite caliper criterion used below. It enters the proof through the exact statement and verification argument given here; neither a plot nor an unrecorded numerical convention is used.

Let $\mathsf{R}$ and $\mathsf{J}$ denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For a nonempty finite set $S \subset \mathbb{R}^2$ and a vector $N$, put

$$W_S(N) = \sum_{k=0}^{2} \max_{z\in S} \langle z, \mathsf{R}^k N\rangle. \tag{121}$$

**Theorem 6.1** (Finite caliper certificate)**.** *If $S$ has at least two distinct points, then $S$ is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in S$ and $\varepsilon \in \{-1, 1\}$ for which*

$$W_S\big(\varepsilon \mathsf{J}(q-p)\big) \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{122}$$

*If $S$ consists of one point, it is contained in equilateral triangles of arbitrarily small positive side length.*

*For $S_x = \{O, A, B, x\}$, every clause of* (122) *is a finite polynomial clause of degree at most four in the coordinates of $A, B, x$; at most $12 \cdot 4^3$ clauses are required, and repeated point labels merely delete zero-length pairs.*

*Proof.* Apply the enclosure gauge of Proposition 4.1 to the polygonal hull of $S$. A minimizing normal is perpendicular to a hull edge $[p, q]$; after a cyclic rotation of the three normals it is $\varepsilon\mathsf{J}(q-p)/\|p-q\|$. Homogeneity gives (122). A nondegenerate segment is checked directly with its two normals.

For the polynomial expansion fix $(p, q, \varepsilon)$, put $N = \varepsilon \mathsf{J}(q-p)$ and $N_k = \mathsf{R}^k N$, and choose support labels $z_0, z_1, z_2 \in S_x$. The support cell is

$$\langle z_k, N_k\rangle \geq \langle z, N_k\rangle \quad (z \in S_x,\ k = 0, 1, 2), \tag{123}$$

with $\|p-q\|^2 > 0$. Since $W_S(N) \geq 0$, the remaining test is

$$4\left(\sum_{k=0}^{2}\langle z_k, N_k\rangle\right)^2 \leq 3\|p-q\|^2. \tag{124}$$

The points are affine-linear in the parameters, so (123)–(124) have degree at most four. Six unordered pairs, two signs, and $4^3$ support-label triples give the asserted finite exhaustive list. □

6.1.2. *Corner-clipped $AB$-unions.* Let $e_A, e_B$ be unit vectors with angle 120°, put

$$O = 0, \qquad A = ae_A, \qquad B = be_B, \qquad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$\mathcal{R}(a, b) = C \cap \bigcup\{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}. \tag{125}$$

The finite description in Theorem 6.1 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (125) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

**Theorem 6.2** (Strict supercritical $AB$-union)**.** *Assume*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

*put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define*

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-a + b + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

*Then all four coefficients are positive and*

$$\mathcal{R}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{126}$$

*where*

$$\mathcal{D}_A = \{(u, v) : (u-a)^2 + v^2 - (u-a)v \leq 1\},$$
$$\mathcal{D}_B = \{(u, v) : u^2 + (v-b)^2 - u(v-b) \leq 1\},$$
$$\mathcal{H}_A = \{(u, v) : \alpha(u-a) + \beta v \leq 0\},$$
$$\mathcal{H}_B = \{(u, v) : \gamma u + \delta(v-b) \leq 0\}.$$

*Its non-axis frontier consists, in counterclockwise order, of the $A$-circle, the $A$-line, the $B$-line, and the $B$-circle.*

*Proof.* The identities

$$4\rho - 3(a+b)^2 = (a-b)^2, \qquad (a+b)^2D^2 - (a-b)^2 = 4\rho((a+b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem 6.1. For a point $x = (u, v)$ in the relative interior of $C$ but outside $\triangle OAB$, the four hull vertices occur in the order $O, B, x, A$. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by $h$, are respectively

$$\max\{a, u\} + \max\{b, v - u\}, \qquad \max\{b, v\} + \max\{a, u - v\},$$

and are at least $a + b > 1$. It remains to test the $Ax$ and $Bx$ calipers.

For the $Ax$ caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

(127)
$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \quad N_A = \begin{cases} 0, & p \le 0, \\ ap, & 0 \le p \le v, \\ (a+b)p - bv, & p \ge v. \end{cases}$$

No support label is omitted in these three sectors. If $p \le 0$, $G_A \le h$ is equivalent to

$$r \le 1, \qquad -ap + (a+b)v \le r.$$

The second inequality is the half-plane condition in (126), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \le v$ one has $r \le v$, whereas the numerator in (127) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \ge v$, the inequalities $G_A \le h$ imply

$$r^2 + (a+b)p - bv \le r, \qquad bp + av \le r. \tag{128}$$

Write $x - A = ry(\theta)$, $0 \le \theta \le 60^\circ$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}}\cos(\theta - 30^\circ), \quad w = \frac{2}{\sqrt{3}}\sin\theta.$$

Then (128) gives

$$r \le f(\theta) = 1 - (a+b)t + bw, \qquad S(\theta) = bt + aw \le 1.$$

Let $n_B$ be the unit vector with dual coordinates $(\gamma, \delta)$. The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1 + D), \quad b\delta - a\gamma = \frac{h}{2}(1 - D)$$

show that the unique angle $\theta_B$ of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1 - D)/2$. The function $S$ has at most one critical point, a maximum, so $S(\theta) \le 1$ implies $\theta \le \theta_B$. On the feasible part $f, f' \ge 0$; hence $f(\theta)\cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B\rangle \le a\gamma - b\delta + \frac{h}{2}(1 - D) = 0.$$

The diameter bound also gives $\|x - B\| \le 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges $a, u$ with $b, v$ and supplies the corresponding result for the $Bx$ caliper. This proves (126) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \qquad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2) > 0$ and $b(4-(1-D)^2) > 0$. This proves the asserted four-piece order. □

6.1.3. *Exact-one handoffs and common reach bounds.*

**Lemma 6.3** (Exact-one handoff chain)**.** *Assume the six V triangles cover $\partial H$ and exactly one actual V triangle is supercritical. Rotate its index to* 4*. The strict handoffs supplied by Proposition 2.10 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

*With*

$$a = 1 - \xi_3, \qquad b = \xi_4, \qquad p = 1 - b, \qquad q = 1 - a,$$

*one has*

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1, \tag{129}$$

*and every selected V triangle obeys*

$$a_i \geq p, \qquad b_i \geq q.$$

*Proof.* Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \neq 4$, which is precisely (6.3). In particular $\xi_3$ is the global minimum and $\xi_4$ the global maximum. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (129). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \geq 1 - \xi_4 = p$ and $\xi_i \geq \xi_3 = q$. □

6.1.4. *Six directly forced radial points.* Put $h = \sqrt{3}/2$ and

$$s = p + q, \qquad m = \min(p, q), \qquad M = \max(p, q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (129) gives

$$pq > (1-s)(2-s), \qquad m > 1 - s, \qquad s > 1 - m. \tag{130}$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition 4.23. On the present subcritical domain, equation (81) becomes

$$c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases} \tag{131}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal $s$.

**Lemma 6.4** (Direct radial forcing)**.** *One has*

$$0 < c_* < 1, \qquad c_* > 1 - m. \tag{132}$$

*Assume that every $U_i$ is* Vd0 *and that $U_C, U_0, \dots, U_5$ cover $H$. Then, for every $i$,*

$$D_i = (1 - c_*)V_i \in U_C.$$

*Consequently $U_C$ contains the centered closed disk*

$$\mathcal{D}_\eta = \{x : \|x\| \le \eta\}, \qquad \eta = h(1 - c_*). \tag{133}$$

*Proof.* On the $C_L$ branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \ge \sqrt{3}/2 > s > 1 - m$, while if $s \ge \sqrt{3}/2$ its value at $s$ is $\chi \le 0$, so $c_* \ge s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_* = \frac{2M}{1 - m} - 1,$$

one has

$$A_*^2 - E^2 = \frac{4(1 - s)\{(1 - m)(1 - 2m) + m(2 - m)(1 - s)\}}{(1 - m)^2} > 0.$$

Thus $A_* > E$ and $c_* > 1 - m$. This proves (132).

Fix $i$. If $D_i \in U_i$, openness supplies $\varepsilon > 0$ with $c_* + \varepsilon < 1$ such that $(1 - c_* - \varepsilon)V_i \in U_i$. The same triangle contains its two selected handoff anchors. Since their incoming and forward-reach lower bounds dominate $p$ and $q$, convexity and passage to the closure give a closed unit triangle realizing $(p, q, c_* + \varepsilon)$. This contradicts the definition of $c_* = c_{\max}(p, q)$. Hence $D_i \notin U_i$.

If $D_i$ belonged to $U_{i-1}$ or $U_{i+1}$, a small plane ball inside that open triangle would meet $r_i$ in a positive-length interval. This contradicts the Vd0 definition. For the remaining triangles put $r = 1 - c_* > 0$. Directly,

$$\|rV_i - V_{i\pm 2}\|^2 = 1 + r + r^2 > 1, \qquad \|rV_i - V_{i+3}\| = 1 + r > 1.$$

Since $V_j \in U_j$ and two points of an open unit triangle are at distance strictly below 1, these inequalities exclude $D_i$ from the other three V triangles. Thus no V triangle contains $D_i$. The point lies in $\mathrm{int}(H)$ by (132), so the assumed cover forces $D_i \in U_C$.

The six points $D_i$ form a regular hexagon of circumradius $1 - c_*$. Their convex hull contains its centered incircle of radius $h(1 - c_*)$, and convexity of $U_C$ gives (133). ☐

We next define the three asymmetric witnesses. At $V_4$ use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \qquad \mathsf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is $b$ and the backward-reach lower bound is $a$. Accordingly, set

$$\alpha = \frac{h(b + 2a - bD)}{2\rho}, \qquad \beta = \frac{h(b - a + (a + b)D)}{2\rho},$$

$$\gamma = \frac{h(-b + a + (a + b)D)}{2\rho}, \qquad \delta = \frac{h(2b + a - aD)}{2\rho},$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem 6.2 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$J = \left(\frac{2b + a - aD}{D+3}, \frac{b + 2a - bD}{D+3}\right). \tag{134}$$

Put

$$E_- = (1 - b, 2 - b), \qquad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \qquad 1 - a \le t \le b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \qquad 1 - b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2,$$
$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \tag{135}$$

Let $\lambda_-$ and $\mu_+$ be their first roots; explicitly,

$$\lambda_- = 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6},$$
$$\mu_+ = 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.$$

**Lemma 6.5** (Moving adjacent-triangle signs)**.** *Throughout* (129),

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{136}$$

*Moreover* $J_u, J_v < 1/2$. *The point* $L_+(\mu_+)$ *satisfies*

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

*and the reflected bound holds for* $L_-(\lambda_-)$. *Writing* $F(P,t) = \mathsf{q}(P - Z_5(t)) - 1$, *one has, for every* $t \in [1 - a, b]$,

$$F(L_+(\mu_+), t) \ge 0, \qquad F(J, t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{137}$$

*Under the reflection* $(u, v, a, b) \leftrightarrow (v, u, b, a)$, *the analogous signs hold for every* $y \in [1 - b, a]$: *the first-root point* $L_-(\lambda_-)$ *has nonnegative sign against* $Z_3(y)$, *and the other two witnesses have strictly positive sign.*

*Proof.* Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2 - a, 1 - a)$; reflection gives the calculation for $E_-$. Put

$$F(P,t) = \mathsf{q}(P - (1 + t, t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D + 3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2+U)$, which follows from $D^2 \geq 3U(2+U)$ and $(2+U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v,$$

$F(J, t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1-a, b]$.

On the opposite line $L_-$, the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$\begin{aligned} P_0 = {} & 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ & + 2z^4 + 9z^2 + 9. \end{aligned}$$

The coefficient of $D$ is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$\begin{aligned} P_1 = {} & 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 \\ & - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9. \end{aligned}$$

Its $D$-coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in $t$, so it is positive throughout $[1-a, b]$. Since the restriction of $F$ to $L_-$ is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line $L_+$, the value at its outer endpoint $h^{-1}$ has, after multiplication by a positive factor, numerator

$$A_* = P_2 - 4DU(1 + U + z),$$

where

$$\begin{aligned} P_2 = {} & 3U^4 + 6U^3z + 6U^3 + 4U^2z^2 + 12U^2z + 12U^2 \\ & + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3. \end{aligned}$$

For fixed $U$, replace the odd powers of $z$ by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_* < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the $L_+$ half of (136); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2-(3U-z)^2=6U(1+z-U),\qquad D^2-(3U+z)^2=6U(1-z-U)$$

give $J_u, J_v<1/2$. It remains to control the coordinate sum along $L_+$. For $E=L_+(h^{-1})$, the sign of $3-2a-E_u-E_v$ is the sign of

$$D(6U+2z+6)+B(U,z),$$

where

$$\begin{aligned}B(U,z)=&-9U^3-9U^2z-9U^2-3Uz^2-18Uz+3U\\&-3z^3+3z^2-3z+3.\end{aligned}$$

Here

$$B_z=-3(3z^2+2Uz+6U-2z+3U^2+1)<0,$$

because the quadratic in parentheses has discriminant $-32U^2-80U-8<0$. Hence its minimum on $z^2<\Omega$ is at $z=\sqrt{\Omega}$, where $B(U,\sqrt{\Omega})=6(1-3U-\sqrt{\Omega})>0$ because $(1-3U)^2-\Omega=12U^2$. Thus $E_u+E_v<3-2a$. Also $J_u+J_v<1<3-2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P,t)=1+2t-P_u-P_v.$$

Since $F(L_+(\mu_+),1-a)=0$, the coordinate-sum bound makes its $t$-derivative strictly positive for $t\geq 1-a$. Together with the no-opposite-line result, this proves (137). Reflection gives both the bound on $L_-$ and the complete $Z_3(y)$ statement. □

Define the local and global witnesses by

$$Q_-^{\mathrm{loc}}=L_-(\lambda_-),\qquad Q_0^{\mathrm{loc}}=J,\qquad Q_+^{\mathrm{loc}}=L_+(\mu_+),\tag{138}$$

and by the displayed conversion from $(u,v)$ to the plane.

**Lemma 6.6** (Direct asymmetric forcing)**.** *Assume that $U_C, U_0, \ldots, U_5$ cover $H$. Then*

$$Q_-, Q_0, Q_+\in \operatorname{int}(H)\setminus\bigcup_{i=0}^{5}U_i,$$

*and consequently all three witnesses belong to $U_C$.*

*Proof.* The root order and positivity of the line coordinates give $0<u,v<1$ for all three points. Each belongs to $\mathcal{R}(b,a)$ and hence lies in a closed unit triangle containing $V_4$, so $\mathsf{q}(u,v)\leq 1$; hence $|u-v|<1$ and all three points lie in $\operatorname{int}(H)$.

The closure of $U_4$ is one of the triangles represented by $\mathcal{R}(b,a)$, since it contains $V_4, X_3, X_4$. All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle $U_4$, a sufficiently small plane ball about it would lie both in $U_4$ and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}(b,a)$. This contradiction excludes $U_4$.

For the triangles $U_0, U_1, U_2$, their contained vertices have local coordinates

| $i$ | 0 | 1 | 2 |
|---|---|---|---|
| $V_i$ | $(2,1)$ | $(2,2)$ | $(1,2)$. |

For $P=(u,v)$,

$$\|P-(2,1)\|^2-1=\mathsf{q}(u,v)+2-3u,\quad \|P-(1,2)\|^2-1=\mathsf{q}(u,v)+2-3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from $V_0$ for $Q_-, Q_0$ and from $V_2$ for $Q_0, Q_+$. For the remaining pair, the first-root circle and the sum bound in Lemma 6.5 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude $U_0, U_1, U_2$.

The actual handoff $X_5 = Z_5(\xi_5)$ belongs to $U_5$, and $1 - a < \xi_5 < b$ by (6.3). The three signs in (137) therefore put every witness at distance at least one from $X_5$, excluding $U_5$. Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$, $X_2 = Z_3(y) \in U_3$, and the reflected signs exclude $U_3$. Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into $U_C$. □

The nine directly forced points are $D_0, \ldots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas 6.4 and 6.6 therefore force the compact set

$$K_{\mathrm{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{139}$$

6.1.5. *Tangent reduction and exact mixed overlaps.* For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D + 3)}.$$

Let $g_-, g_+$ be the rescaled forms of (135), and take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= J, \\ C &= \left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

and convert them to global vectors.

**Lemma 6.7** (Newton inner reduction)**.** *The points above have coordinates rational in $a, b, D$ and satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

*Consequently*

$$\widehat{K} = \mathrm{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

*obeys* $\widehat{K} \subseteq \mathrm{conv}(K_{\mathrm{wit}})$.

*Proof.* Each quadratic $g_-, g_+$ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. All displayed coefficients and Newton quotients are rational in $a, b, D$, and convexity gives the asserted containment. $\square$

Put $W = C + RA$. For a disk $\mathbb{D}(X, r)$ let

$$I(X, r) = \{n : \|n\| = 1,\ \langle X, n\rangle + r \geq h\}.$$

6.1.6. *Ray order and four overlaps.*

**Proposition 6.8** (Technical four-overlap verification)**.** *Assume* (129) *and* $c_* > 2/3$. *Then*

$$\|A\|, \|C\| > \eta,$$

*the rays* $A, B, C, W, RA$ *occur in that cyclic order, and the four consecutive cap pairs*

$$I(A, 2\eta) \leftrightarrow I(B, 2\eta) \leftrightarrow I(C, 2\eta) \leftrightarrow I(W, \eta) \leftrightarrow I(RA, 2\eta)$$

*overlap across their intervening short sectors.*

*Proof.* Put $\tau = h - 2\eta = h(2c_* - 1)$. The rays of

$$A,\ B,\ C,\ W,\ RA \tag{140}$$

occur in this cyclic order from $A$ to $RA$. Indeed the two line distances give

$$\frac{\operatorname{cross}(A, B)}{\|B - A\|} = \alpha(1 - b) + \beta > 0, \qquad \frac{\operatorname{cross}(B, C)}{\|C - B\|} = \gamma + \delta(1 - a) > 0.$$

If $A = (-X, -Y)$ and $C = (-P, -Q)$ in the centered cone basis, then $0 < X, Y, P, Q < 1$, $X, Q > 1/2$, and

$$\frac{\operatorname{cross}(C, RA)}{h} = PX + (Q - P)Y > 0.$$

The last sign is immediate unless $Q < P$ and $X < Y$, in which case it is larger than $Q - P/2 > 0$. The ray of $W = C + RA$ lies between those of $C$ and $RA$. The same coordinates give

$$\|A\|, \|C\| > h/2 > \eta. \tag{141}$$

For the two equal-radius overlaps it is enough to prove

$$\frac{\operatorname{cross}(A, B)}{\|B - A\|} \geq \tau, \qquad \frac{\operatorname{cross}(B, C)}{\|C - B\|} \geq \tau. \tag{142}$$

We now give the exact analytic verification. By reflection take $a \leq b$ and set $m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (142) is

$$d = \frac{\mathcal{A} + U(2m - 1) + D(\mathcal{A} + U)}{2\rho}, \qquad \mathcal{A} = 1 - m + m^2. \tag{143}$$

On the $C_L$ sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m - 1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0,1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28-34m+m^2(5m^2-12m+33)>0$. Since $F_L(h)=-(m-h/2)^2\le 0$ and the selected root is unique, $2c_*-1\le\mathcal{A}$. Put

$$X_0=\mathcal{A}+U,\qquad Y_0=2\rho\mathcal{A}-\mathcal{A}-U(2m-1).$$

Then

$$d-\mathcal{A}=\frac{D(\mathcal{A}+U)-Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned}Y_0&=2(m^2-m+1)U^2+(2m^3-2m+3)U\\&\quad+(m^2-m+1)(2m^2-2m+1)>0.\end{aligned}$$

Furthermore

$$D^2X_0^2-Y_0^2=4m\rho\Phi(U,m),$$

where

$$\begin{aligned}\Phi(U,m)&=(1-m)(m^2-m+2)U^2\\&\quad+(2-m^2)(m^2-m+1)U\\&\quad-m(1-m)^2(m^2-m+1).\end{aligned}$$

In these variables

$$\chi=U^4-4U^3+5U^2-(m+2)U+m(1-m).$$

Since $U^2(U^2-4U+5)>0$, the selector $\chi\le 0$ implies $U>m(1-m)/(m+2)$; $\Phi$ is increasing in $U$ and at that lower endpoint is

$$\frac{m^2(1-m)(m^4-2m^3+6m^2-6m+4)}{(m+2)^2}>0.$$

The last quartic equals $(m^2-m)^2+5m^2-6m+4>0$. Thus $d>\mathcal{A}\ge 2c_*-1$.

On the other radial sheet let $E=\sqrt{4s^2-3}$. Then $0\le w<sE$ and $c_*=(s+w)/(1+E)$. At fixed $U$, differentiation of (143) can be checked without an implicit estimate. Namely,

$$d=\frac{1+D}{2}-UB(D,w),\qquad B(D,w)=\frac{(1+U)(3+D)+w(1-D)}{D^2+3}.$$

Here $D^2=w^2+6U+3U^2$, so $0\le D'=w/D\le 1$, and

$$B_w=\frac{1-D}{D^2+3}\ge 0.$$

Writing $N=(1+U)(3+D)+w(1-D)$, direct differentiation gives

$$B_D=\frac{(1+U-w)(D^2+3)-2DN}{(D^2+3)^2}>-\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1+U-w=2a>0$, $N<(7/6)4+1=17/3$, $D<1$, and $D^2+3\ge 3$. If $B_D<0$, then $B_DD'\ge B_D$; otherwise $B_DD'\ge 0$. Hence $B'=B_w+B_DD'>-34/27$, and $U<1/6$ yields

$$d'<\frac{1}{2}+\frac{1}{6}\frac{34}{27}=\frac{115}{162}<1,$$

whereas $(2c_*-1)'=2/(1+E)\ge 1$. Hence their difference decreases with $w$. At the boundary $w=sE$, where $\chi=0$ and $c_*=s$, the preceding sheet applies: this is an admissible point because $h<s<1$ and $D^2=4s^4-12s+9<1$. Indeed

$s^4 - 3s + 2 = (s-1)(s^3 + s^2 + s - 2) < 0$ on this interval; the second factor is increasing and already equals $(7h-5)/4 > 0$ at $s = h$. The preceding sheet gives $d \geq 2s - 1$. Therefore (142) holds on both sheets, and reflection supplies the other line.

It remains to check the two mixed overlaps. The exact rational envelopes, Gram reductions, and global positive-basis identities are recorded once in Appendix A. Theorem A.3 gives precisely the overlaps for $\mathbb{D}(C, 2\eta)$ with $\mathbb{D}(W, \eta)$ and for $\mathbb{D}(W, \eta)$ with $\mathbb{D}(RA, 2\eta)$.

Thus the four consecutive pairs of caps centered on the rays (140) overlap. $\square$

6.2. **Support arcs and the enclosure conclusion.** Let $\mathsf{R}$ denote counterclockwise rotation through $2\pi/3$. Retain the points $A, B, C$, the vector $W = C + \mathsf{R}A$, the forced disk $\mathcal{D}_\eta$, and the exact witness set $K_{\mathrm{wit}}$ constructed above. For uniform notation put

$$P_- = A, \qquad P_0 = B, \qquad P_+ = C, \qquad P_* = W, \qquad r_* = \eta,$$

and

$$\widehat{K} = \operatorname{conv}\bigl(\mathcal{D}_\eta \cup \{A, B, C\}\bigr).$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, write

$$\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\},$$

and define the level-$h$, $h = \sqrt{3}/2$, support arc

$$\operatorname{Cap}(X, r) = \{n \in \mathbb{S}^1 : \langle X, n \rangle + r \geq h\}.$$

**Lemma 6.9** (Cyclic support-arc covering)**.** *Let $K$ be compact and put*

$$\Sigma_K = K + \mathsf{R}K + \mathsf{R}^2 K.$$

*Suppose $\Sigma_K$ contains the full $\mathsf{R}$-orbits of*

$$\mathbb{B}(P_-, 2r_*), \quad \mathbb{B}(P_0, 2r_*), \quad \mathbb{B}(P_+, 2r_*), \quad \mathbb{B}(P_*, r_*).$$

*Assume that the rays*

$$P_-, P_0, P_+, P_*, \mathsf{R}P_-$$

*occur in this cyclic order, that every corresponding support arc is connected with half-width less than $\pi/2$, and that each consecutive pair of arcs intersects across the intervening short angular sector. Then*

$$\Lambda(K) \geq 1.$$

*Proof.* The four consecutive intersections and the cyclic order make the five arcs cover the sector from the ray of $P_-$ to the ray of $\mathsf{R}P_-$. Their $\mathsf{R}$-orbits cover $\mathbb{S}^1$. Hence for every unit vector $n$ one of the displayed disks has support at least $h$ in direction $n$, and therefore

$$h_{\Sigma_K}(n) \geq h.$$

Since

$$h_{\Sigma_K}(n) = \sum_{j=0}^{2} h_K(\mathsf{R}^j n),$$

Proposition 4.1 gives $\Lambda(K) \geq 1$. $\square$

**Theorem 6.10** (Witness enclosure)**.** *For every exact-one-supercritical parameter pair $(a,b)$ constructed above,*

$$\Lambda(K_{\mathrm{wit}}(a,b)) \geq 1.$$

*Proof.* The centered disk alone gives

$$\Lambda(K_{\mathrm{wit}}) \geq 3(1-c_*),$$

so the result follows when $c_* \leq 2/3$. Assume $c_* > 2/3$. Lemma 6.7 gives $\widehat{K} \subseteq \operatorname{conv}(K_{\mathrm{wit}})$. The Minkowski sum $\widehat{K} + \mathsf{R}\widehat{K} + \mathsf{R}^2\widehat{K}$ contains the disk orbits listed in Lemma 6.9. Proposition 6.8 supplies the ray order, radius bounds, and equal-radius intersections; Theorem A.3 supplies the two mixed intersections. Lemma 6.9 therefore gives $\Lambda(\widehat{K}) \geq 1$, and monotonicity yields $\Lambda(K_{\mathrm{wit}}) \geq 1$. ☐

**Theorem 6.11** (Zero-gap obstruction)**.** *There is no cover of $H$ by open unit equilateral triangles $U_C, U_0, \ldots, U_5$ such that every $U_i$ is* Vd0*, the six vertex triangles cover $\partial H$, and exactly one $V$ triangle is supercritical.*

*Proof.* The construction above forces the compact set $K_{\mathrm{wit}}$ into the open C triangle $U_C$, whereas Theorem 6.10 gives $\Lambda(K_{\mathrm{wit}}) \geq 1$. Every compact subset of an open unit equilateral triangle is contained in a closed equilateral triangle of side strictly below one, which would imply $\Lambda(K_{\mathrm{wit}}) < 1$. ☐

**Proposition 6.12** (Cases excluded by the nine-point obstruction)**.** *Theorem 6.11 excludes*

(1) *the* CE0*, $N_+ = 1$, all-*Vd0 *case;*
(2) *the no-gap* CE1 *and* CE2*, $N_+ = 1$, all-*Vd0 *cases.*

*Proof.* In the CE0 case the six V triangles must cover the perimeter, since a missed boundary point would lie in the open C triangle and create a positive center trace. In the CE1 and CE2 cases, absence of a boundary gap means that the two incident vertex traces cover every edge. The unique-supercritical hypothesis is common to all three cases. ☐

## 7. Completion of the Proof

*Proof of Theorem 1.1.* Assume that seven open unit equilateral triangles cover $H$. By Proposition 2.14, the cover lies in exactly one row of Table 1. The rows assigned to trace length are excluded by Proposition 3.10; the rows assigned to area loss by Proposition 5.5; the boundary-reach propagation rows by Proposition 4.37, including the complete CE2 one-Vd1/Vd2 positional assembly in Proposition 4.39; and the no-gap, all-Vd0 rows by Proposition 6.12. Hence every row is impossible, contradicting the assumed cover. ☐

Corollary 1.2 follows from Proposition 2.1.

## Appendix A. Exact Unequal-Radius Support-Arc Intersections

The two mixed cap overlaps in Method 4 are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

A.1. **Certificate manifest.** The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob; the source ledger records their role. They decode to one canonical sparse-polynomial transcript with SHA–256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

A.2. **Rational radial upper envelopes.** Retain the notation of Section 6.1. Thus

$$p = 1 - b, \qquad q = 1 - a, \qquad s = p + q, \qquad m = \min\{p, q\}, \qquad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$mM > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > \frac{2}{3}. \tag{144}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \le 0), \qquad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \ge 0). \tag{145}$$

**Lemma A.1** (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \le \bar{c}_L < 1, \qquad c_* \le \bar{c}_T < 1.$$

*At $\chi = 0$ both upper bounds equal $s = c_*$.*

*Proof.* On the $L$ cell, $F_m(s) = \chi \le 0$ and $F_m(c_*) = 0$. For $x \ge s > 2/3$, $F''_m(x) > 0$, so convexity gives

$$0 \ge F_m(s) + F'_m(s)(c_* - s),$$

and hence $c_* \le \bar{c}_L$. If $U = 1 - s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (144); therefore $\bar{c}_L < 1$.

On the $T$ cell put

$$E = \sqrt{4s^2 - 3}, \qquad m_- = \frac{s(1 - E)}{2}, \qquad m_+ = \frac{s(1 + E)}{2}.$$

Then

$$\chi = (m - m_-)(m_+ - m), \qquad s - c_* = \frac{2(m - m_-)}{1 + E}.$$

Consequently

$$\frac{\chi}{s-c_*}=\frac{1+E}{2}(m_+-m)\le 1-m,$$

which gives $c_*\le \bar{c}_T$. The strict inequality $\chi>0$ gives $\bar{c}_T<s<1$. □

Reflect so that $z=a-b\ge 0$ and put $U=a+b-1$. Then

$$s=1-U,\qquad m=\frac{1-U-z}{2},\qquad D^2=z^2+6U+3U^2=:K(U,z),$$

and

$$\Xi(U,z)=4\chi=1-10U+21U^2-16U^3+4U^4-z^2,$$
$$G(U,z)=2\kappa=5-21U+24U^2-8U^3-z.$$

Thus

$$\bar{c}_L=1-U-\frac{\Xi(U,z)}{2G(U,z)},\qquad \bar{c}_T=1-U-\frac{\Xi(U,z)}{2(1+U+z)}. \tag{146}$$

A.3. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (146), and put

$$\bar{\eta}=h(1-\bar{c}),\qquad e=1-\bar{c},\qquad t=2\bar{c}-1.$$

Lemma A.1 gives $0<\bar{\eta}\le\eta=h(1-c_*)$. It therefore suffices to prove the mixed intersections for the smaller disks with radii $2\bar{\eta},\bar{\eta},2\bar{\eta}$.

Represent a global vector as $(x,hy)$ and put $C'=\mathsf{R}^{-1}C$. Define

$$N_A=\|A\|^2,\qquad N_C=\|C\|^2,\qquad \nu=\langle A,C'\rangle,$$
$$\Delta_0=x_Ay_{C'}-y_Ax_{C'}>0.$$

Thus the Euclidean cross product is $h\Delta_0$, and the Gram identity is

$$h^2\Delta_0^2=N_AN_C-\nu^2. \tag{147}$$

Define

$$Q_A=\Delta_0^2-\|tA-eC'\|^2,\qquad Q_C=\Delta_0^2-\|tC'-eA\|^2. \tag{148}$$

**Lemma A.2** (Residual signs imply the mixed cap intersections)**.** *If $Q_A,Q_C\ge 0$, then*

$$I(C,2\bar{\eta})\cap I(C+\mathsf{R}A,\bar{\eta})\ne\varnothing,$$

*and*

$$I(C+\mathsf{R}A,\bar{\eta})\cap I(\mathsf{R}A,2\bar{\eta})\ne\varnothing.$$

*Consequently the same intersections hold with $\eta$ in place of $\bar{\eta}$.*

*Proof.* Rotate the first pair by $\mathsf{R}^{-1}$. Its centers become $C'$ and $A+C'$, so their difference is $A$. An outer common tangent normal $n_A$ is chosen with

$$\langle A,n_A\rangle=\bar{\eta}.$$

Since $\|A\|>\bar{\eta}$ and $\Delta_0>0$, choose the tangent side for which

$$\langle C',n_A\rangle=\frac{\bar{\eta}\nu+\sqrt{N_A-\bar{\eta}^2}\,h\Delta_0}{N_A}.$$

The common support of the two disks is therefore

$$2\bar{\eta}+\frac{\bar{\eta}\nu+\sqrt{N_A-\bar{\eta}^2}\,h\Delta_0}{N_A}.$$

Let $\bar\tau = h - 2\bar\eta = ht$. Expanding (148) and using (147) gives

$$\bar P_A := (N_A - \bar\eta^2)h^2\Delta_0^2 - (\bar\tau N_A - \bar\eta\nu)^2 = h^2 N_A Q_A. \tag{149}$$

Thus $Q_A \ge 0$ implies

$$\sqrt{N_A - \bar\eta^2}\, h\Delta_0 \ge \bar\tau N_A - \bar\eta\nu,$$

and the common support is at least $2\bar\eta + \bar\tau = h$. Hence the two level-$h$ caps intersect. The same calculation with $A$ and $C'$ exchanged gives

$$\bar P_C := (N_C - \bar\eta^2)h^2\Delta_0^2 - (\bar\tau N_C - \bar\eta\nu)^2 = h^2 N_C Q_C, \tag{150}$$

and proves the second intersection. Enlarging the disk radii from $\bar\eta$ to $\eta$ only enlarges the caps. □

A.4. **Eight exact integer-polynomial signs.** The Newton witnesses $A, C$ are rational functions of $U, z, D$. Substituting (146) into (148), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\operatorname{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\}, \tag{151}$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs $A, C$, both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

A.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \qquad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_\triangle = 1 - \frac{\sqrt{3}}{2}, \qquad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

$$\begin{array}{lll} T: & 0 \le U \le U_\triangle, & 0 \le z^2 \le \mathcal{P}(U), \\ L_0: & 0 \le U \le U_\triangle, & \mathcal{P}(U) \le z^2 \le \mathcal{G}(U), \\ L_1: & U_\triangle \le U \le U_{\max}, & 0 \le z^2 \le \mathcal{G}(U). \end{array}$$

For the $T$ cell use

$$U = \frac{(x-1)(3-x)}{4x}, \qquad z = y\frac{9 - x^4}{8x^2}, \qquad 1 \le x \le \sqrt{3}, \quad 0 \le y \le 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and $x$, the four transformed core polynomials have global tensor Bernstein expansions with data

| | bidegree | coefficient count | zero count |
|---|---|---|---|
| $A_{T,A}$ | $(88, 22)$ | 2047 | 2 |
| $B_{T,A}$ | $(80, 20)$ | 1701 | 0 |
| $A_{T,C}$ | $(88, 22)$ | 2047 | 2 |
| $B_{T,C}$ | $(80, 20)$ | 1701 | 0. |

Every nonzero coefficient is strictly positive.

For an $L$-cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \qquad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}\,|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_\triangle r, \qquad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for $L_0$, and

$$U = U_\triangle + (U_{\max} - U_\triangle)r, \qquad w = s\mathcal{G}(U)$$

for $L_1$. The positivity verifier forms one global Bernstein expansion for each of $E_F, R_F$ on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij} r^i s^j$ has bidegree $(m, n)$, its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

**Theorem A.3** (Exact mixed-overlap certificate)**.** *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in* (151) *are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two mixed cap intersections required by Proposition 6.8 hold.*

*Proof.* The $T$ chart proves the four $T$-cell core signs by nonnegative Bernstein coefficients. On each $L$ chart the positive Bernstein certificates for $E_F$ and $R_F$ prove the four $L$-cell signs. Since $D \geq 0$, (151) gives $Q_A, Q_C \geq 0$ for $z \geq 0$; reflection supplies $z \leq 0$. Lemma A.2 then gives the two cap intersections. □

*Remark* A.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.